Multigrid Self-Gravity in the AthenaK Shearing Box: I. Mesh-Refinement Infrastructure (Phase 0), Shear-Periodic Boundary Conditions (Phase 1), Gravity on Refined Meshes (Phase 2), the Stratified Box with Open-z Boundaries (Phase 3), and Ideal MHD with Self-Gravity (Phase 4)

`athenak-multigrid` tree, branch `multigrid`

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Abstract

This document opens the *multigrid track* of the self-gravity program: the route to gas + dust self-gravity in AthenaK shearing boxes *with mesh refinement*. It records Phase 0 — the shearing-box/refinement infrastructure every later phase rests on: (i) an `orbital_advection` toggle with a fully consistent non-FARGO (full-velocity) mode; (ii) an enforced *interior-x_1* refinement policy replacing AthenaK's blanket shearing-box+refinement prohibition; (iii) orbital advection (FARGO) on refined meshes under an *annular* refinement policy (refined regions span the full azimuthal extent); and (iv) *adaptive* refinement (Phase 0c): the shear-boundary communication state is rebuilt after every AMR update, a graded refinement cap keeps 2:1 balancing from ever refining the shear-periodic boundary columns, and an upstream derefinement-ordering bug that split annular rings is fixed. Validation: a refined-ring run reproduces a reference SMR run bit-for-bit in non-FARGO mode; with FARGO on the ring, mass is conserved exactly and a 20-orbit epicycle test tracks the analytic solution with an amplitude drift of only 0.008% (versus 0.064% on the uniform grid), with the epicyclic amplitude invariant flat to 10^{-4}. Under AMR, a prescribed moving ring that is created and destroyed every orbit for 20 orbits (2296 MeshBlocks created, 2240 destroyed) conserves mass exactly and matches the static-ring run an order of magnitude more closely than either matches the uniform grid; on the vortical-shwave problem the moving-ring solution converges to the static-ring one (4.7×10^{-2} apart at 64^2, 1.8×10^{-4} at 128^2). Phase 1 (§6) then makes the multigrid Poisson solver itself shear-aware: shear-periodicity is imposed purely as a boundary *identification* — a y-remapped x_1 ghost fill applied at every level of the multigrid hierarchy through global boundary planes, with the 7-point operator untouched. Validated against the FFT oracle: solves at shear-periodic instants are bit-identical to plain-periodic solves; the analytic shearing-wave potential is recovered with errors *below* the FFT solver's own (9.5×10^{-8} vs. 1.6×10^{-6} at 128^3); the swing-amplification history matches the FFT run to 2×10^{-7}; and the V-cycle contraction factor — the one open risk named in Phase 0 — degrades only from 0.116 to 0.157 per cycle at a near-worst fractional shift, reaching the same 10^{-11} roundoff floor. Phase 2a (§7) puts the solver on *refined* shearing-box meshes with interior-ring static refinement. The never-exercised octet machinery inherited with Athena++ #776 is first anchored on the isolated binary-potential test of Tomida & Stone §4.2 (region-wise second order against the analytic two-sphere solution); opening the shear machinery to multilevel meshes whose x_1 boundary blocks remain at the root level then requires *no new shear-fill code* — only a relaxed guard. Validation: sheared-instant solves on the refined mesh are bit-identical to the periodic twin; the analytic shearing-wave error converges at second order

in L_2 (first order at the max, which sits on the refinement interface exactly as in the periodic case); the swing test on a level-1 ring matches its uniform twin to the hydro truncation level — the gravity-off control deviates *more* than the gravity-on run — and multi-rank solves reproduce serial to a float32 ulp. Phase 2b (§8) then admits refinement *on* the shear boundary (boundary annuli): the octets tiling the shear faces receive a y-remapped x_1 ghost fill — the Phase-1 plane algorithm at octet granularity, in host code with no MPI seam (octets are rank-replicated) — through scalar per-face twins of the validated remap kernels, and the block shear planes are sized at the boundary blocks' level. Annuli solves at shear-periodic instants are bit-identical to their periodic twins at one and two refinement levels; the generic-phase L_2 equals the interior-ring value to four digits (the meshes differ by a half-box shift); contraction and floors match the periodic twins; and the annuli swing run matches its uniform twin to hydro truncation (again dominated by the gravity-off control) at machine-level phase lock. Phase 2c (§9) admits *adaptive* refinement: the shear planes, offsets, and octet flags rebuild after every remesh inside the existing `PrepareForAMR` path. A prescribed moving ring sweeping through the swing (392 MeshBlocks created, 280 destroyed) matches the uniform run to hydro regridding truncation with the gravity-off control dominating; the fixed-ring limit recovers machine-level phase lock and matches the static-SMR run to the float32 floor; and a 2-rank AMR run reproduces serial to every printed digit. The companion document `fft_selfgravity_validation.pdf` covers the FFT solver used as the machine-precision oracle throughout this program; a Julia notebook (`multigrid_validation_tests.ipynb`) reproduces every test here. Phase 4 (§13) first closes the hydrodynamic case: the Jeans wave, stable and growing ($\lambda/\lambda_J = 0.25$–3), follows $\omega^2 = k^2c_s^2 - 4\pi G\rho_0$ to a relative error in ω of at most 5.9×10^{-3} at 64 cells per wavelength (at most 1.6×10^{-3} against the dispersion relation of the discrete operators), converging at second order, with every statically or adaptively refined run inside the bracket of the uniform runs. It then adds a magnetic field: on the magneto-Jeans wave, ideal MHD (constrained transport, HLLD) with multigrid gravity reproduces the dispersion relation of a magnetized self-gravitating medium at every field angle and on both branches, to a relative error in ω of at most 4.8×10^{-3} at 64 cells per wavelength, converging at second order; a perpendicular field stabilises a hydro-unstable wavelength at the predicted Alfvén speed, and the perturbed field matches the eigenvector to a median 4×10^{-4}. On static and adaptive meshes the magneto-Jeans runs lie between the uniform runs at their root and finest resolution (one 3-D case 5% above), with $\nabla \cdot B$ at round-off through thousands of MeshBlocks created and destroyed. The suite exposed a strong-field instability of AthenaK's *isothermal* HLLD solver (independent of gravity), so the MHD tests are adiabatic. Magnetized swing amplification follows linear theory to a few per cent on the uniform shearing box; on refined shearing boxes integrated without orbital advection gravity is right but the field perturbation carries a 10–20% error that needs the field, the refinement and the shear flow together; a minimal reproducer shows the fine/coarse EMF correction to be exact and places the error in the fine side's interface EMFs.

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1 Plan and scope

1.1 Goal and roadmap

The scientific target is shearing-box simulations combining gas self-gravity, dust with mutual drag (streaming instability, including the background radial pressure gradient), and mesh refinement that zooms onto high dust-density regions in spiral arms launched by gravitational instability (cf. Baehr, Zhu & Yang 2022, ApJ 933, 100, who omitted the pressure gradient). Mesh refinement is the reason AthenaK was chosen, so the Poisson solver must ultimately live on refined meshes — which an FFT cannot. The adopted route therefore makes the *multigrid* solver shear-aware, with the validated FFT solver retained in three roles: machine-precision oracle for the multigrid shear

boundary conditions, production solver for uniform-grid runs, and donor of exact open-z Dirichlet boundary values for stratified multigrid runs.

The phases of the multigrid track:

1. **Phase 0 (this document, done):** shearing box + refinement infrastructure. 0a: an `orbital_advection` on/off toggle with a consistent full-velocity (non-FARGO) mode, and the interior-x_1 refinement policy. 0b: FARGO on refined meshes under the annular refinement policy. 0c: adaptive refinement — dynamic creation and destruction of refined rings, without gravity, isolating the mesh machinery from the solver before Phase 2 couples them. Without 0b, refined shearing boxes pay a timestep penalty of $\sim (c_s + q\Omega L_x/2)/c_s \approx 10$–$30$ for gravito-turbulence boxes.

2. **Phase 1 (this document, §6, done):** shear-periodic boundary conditions in the multigrid driver (`MultigridDriver`), validated against the FFT oracle: exactness at shear-periodic instants, shearing-wave accuracy at generic shear phase, and the swing-amplification test with `solver=multigrid`. The V-cycle convergence rate with remapped boundaries — the one open risk of the program — is measured: the per-cycle contraction factor degrades only from 0.116 to 0.157.

3. **Phase 2:** multigrid gravity on refined shearing-box meshes (SMR first, then AMR), compared against high-resolution uniform FFT references.

4. **Phase 3 (this document, §12, done):** stratification with open-z boundaries — vertical tidal gravity, slab-open (vacuum) multigrid boundary conditions, cooling, and the Shi & Chiang (2014) gravito-turbulence configuration. The original plan had the FFT solver donating open-z Dirichlet values; the implementation instead evaluates the *discrete vacuum Green's function* directly (exact for the multigrid stencil, refinement-compatible, multi-rank), with the FFT solver retained as the independent open-boundary oracle.

5. **Phases 4–5:** merge with the `athenak-dust` particle-mesh dust module (dust density in the Poisson source, gravitational force on particles, pressure-gradient source term for the streaming instability), and the science runs.

1.2 Phase-0 scope and restrictions

What works after Phase 0, all enforced by fatal errors rather than conventions:

- Hydrodynamic shearing boxes with SMR, under the *uniform-boundary-level* rule: both shear-periodic x_1 boundary faces must share one refinement level. Refinement interior in x_1 always satisfies this (faces stay at root); alternatively *both* boundary faces may be refined uniformly (full x_2/x_3 extent, same level) — physically natural for a structure at the boundary, whose shear-periodic images live on both faces.

- Orbital advection on refined meshes, provided every refined region spans the full x_2 (azimuthal) extent — *annular rings* (interior, or boundary annuli as above). Without FARGO (`orbital_advection=false`) arbitrary refinement shapes are allowed subject only to the boundary-level rule.

- Adaptive mesh refinement (Phase 0c). The shear-boundary communication state (`ShearingBox::x1bndry_mbg`
GID lists, MPI request arrays, CC/FC buffers) is rebuilt after every AMR update; AMR flag handling keeps rings intact (refine a ring if any member is flagged, derefine only if all are) and a

graded cap refuses any refinement whose 2:1 balancing would propagate into the shear-periodic boundary columns (§5.8).

Not yet supported/exercised: MHD orbital advection with refinement (the face-centered EMF remap at level interfaces needs a $\nabla\cdot B$ analysis; fatal-guarded), multi-rank (MPI) exercise of the AMR rebuild path (the code is rank-aware but all Phase 0c runs are serial), and a production-size demonstration of the FARGO speedup itself. SMR/AMR runs must write `bin` outputs — the vtk writer does not support refined meshes.

2 Numerical method

2.1 Two frames for the shearing box

With background shear flow $\mathbf{v}_0 = -q\Omega x\,\hat{y}$, AthenaK can now evolve either of two variable sets, selected by `<shearing_box> orbital_advection`:

FARGO frame (`orbital_advection = true`, default). The stored velocity is the *fluctuation* $\mathbf{v}' = \mathbf{v} - \mathbf{v}_0$; the background advection is applied analytically once per step by the orbital-advection shift (Sec. 2.2). The momentum source terms are

$$\dot{M}_x = 2\Omega\,\rho v'_y, \qquad \dot{M}_y = -(2-q)\,\Omega\,\rho v'_x, \tag{1}$$

where the $(2-q)$ coefficient is the sum of Coriolis (-2Ω) and the advection of background shear by the radial fluctuation ($+q\Omega$). The CFL condition sees only $|\mathbf{v}'|$.

Full-velocity frame (`orbital_advection = false`). The stored velocity includes the shear. The source terms are the full Coriolis force plus the tidal force, and (for an ideal EOS) the tidal work:

$$\dot{M}_x = 2\Omega\,\rho v_y + 2q\Omega^2 x\,\rho, \qquad \dot{M}_y = -2\Omega\,\rho v_x, \qquad \dot{E} = 2q\Omega^2 x\,\rho v_x \tag{2}$$

(the Coriolis force does no work). Because the wrapped data at the shear-periodic x_1 faces comes from the opposite side of the box, where the background azimuthal velocity differs by $q\Omega L_x$, the boundary exchange must offset the wrapped azimuthal momentum and energy:

$$M_y \to M_y + \rho\,\Delta v, \qquad E \to E + M_y\,\Delta v + \tfrac{1}{2}\rho\,\Delta v^2, \qquad \Delta v = \pm\, q\Omega L_x \tag{3}$$

($+$ at the inner face, $-$ at the outer). Initial conditions must include $\mathbf{v}_0$. The two frames are the same physics on different variables; comparing them is a strong consistency test (Sec. 5). The cost of the full-velocity frame is the timestep,

$$\frac{\Delta t_{\rm FARGO}}{\Delta t_{\rm full}} \;\approx\; \frac{c_s + q\Omega L_x/2}{c_s + |\mathbf{v}'|_{\max}}, \tag{4}$$

measured directly as 5.7× for the $L_x = 2\pi$ swing-test box (Sec. 5.1) and reaching 10–30× for gravito-turbulence boxes — plus the extra truncation diffusion of advecting structures through the grid at many c_s. Non-FARGO is therefore the correctness baseline, not the production mode.

2.2 Orbital advection and its refinement problem

The orbital-advection step shifts every x-column of every MeshBlock by the background displacement $\delta y(x) = q\Omega\, x\, \Delta t$, as an integer number of cells plus a conservative fractional remap,

$$q_j \;\to\; q_{j-j_{\rm off}} - \big(F_{j+1-j_{\rm off}} - F_{j-j_{\rm off}}\big), \qquad j_{\rm off} = \left\lfloor \frac{\delta y}{\Delta y} \right\rfloor, \tag{5}$$

with upwind remap fluxes F of first (`dc`), second (`plm`), or third (`ppmx`) order — the same kernels as the shear-periodic boundary remap. The required y-ghost data travels in buffers of depth $n_g +$ `maxjshift` exchanged with the two x_2-face neighbors.

The crucial observation for refinement: *the shift kernel is already per-block correct on a multilevel mesh.* Each block computes δy from its own cell centers and applies it in its own Δy; blocks at different levels shift by the same physical displacement expressed in their own cell units, and after the shift the ordinary AMR ghost exchange (with prolongation/restriction at x_1/x_3 level interfaces) restores consistency to interpolation order, since a uniform-in-y translation commutes with the interpolations to that order. What is *not* level-safe is the communication pattern: the buffer exchange assumes both x_2-face neighbors are same-level, same-size blocks, which fails at any coarse–fine interface in y. This is exactly why refinement previously worked only with orbital advection disabled.

2.3 Refinement policies

Rather than generalizing the y-communication to level interfaces, Phase 0 removes the interfaces:

Uniform-boundary-level policy (always). All MeshBlocks on the two shear-periodic x_1 boundary faces must share *one* refinement level. The reason is structural: the unshifted wrap across x_1 goes through the ordinary neighbor machinery (the tree treats `shear_periodic` exactly like `periodic`), and the shear pass then redistributes the wrapped ghost data azimuthally *within each face* — with per-block Δy arithmetic and a level-aware target lookup (`FindTargetMB` shifts lx_2 modulo the block count at the source block's own level). Nothing in that machinery references the root grid; its only real assumption is same-size blocks along each face and a same-level wrap between the faces. Hence the common level may exceed root: refining *both* boundary faces uniformly (full x_2/x_3, same level) is legal, and a mixed-level boundary is a fatal error. Enforced block-by-block at tree build and restart (`Mesh::CheckShearingBoxRefinement`). Truly mixed-level shear boundaries (a refined patch on one face wrapping onto coarse blocks of the other) would require combining the y-shift with prolongation/restriction in the shear comms — deferred; no production code implements it.

Annular policy (when FARGO is on). Every refined region must span the full x_2 extent over its x_1–x_3 footprint: refinement comes in *rings*. Within a ring every x_2-face neighbor is same-level and same-size, so the per-level shift plus the *existing* communication code are valid with no changes. Enforced by ring-completeness counting over (level, lx_1, lx_3); a partial ring with FARGO on is a fatal error naming the offending ring. For AMR, refine/derefine flags are synchronized ring-wise after the global flag exchange — refine the whole ring if any member is flagged, derefine only if every member is — deterministically on all ranks. Scientifically the ring shape is a good match for the target problem: GI spiral arms are azimuthally extended, and a ring at the arm's radius captures the whole arm and every clump in it.

Buffer-depth safety. The integer shift on a refined block at radius x is $q\Omega|x|\Delta t/\Delta y_\ell$ cells, which the construction-time estimate `maxjshift` $= \lfloor \mathrm{cfl} \cdot q\Omega \max|x| \rfloor + 1$ (now including the previously missing $q\Omega$ factor, doubled on multilevel meshes) may underestimate in unusual configurations. The unpack step therefore re-verifies the actual worst-case shift against the communicated buffer depth every step and aborts with a diagnostic rather than silently corrupting data.

3 What was built

Changes live in commits `e7ef37f2` (infrastructure) and `f1468da3` (bin outputs + docs); Phase 0a is ported from the `athenak` branch `test/shwave-smr`, with provenance kept in the input-file comments. Phase 0c adds commits `f0acd383` (derefinement-sort fix), `b5388c02` (AMR rebuild machinery, graded cap, moving-ring test), and `097a97ee` (vortical-shwave AMR test).

File (under `src/`)	Change
`hydro/hydro.cpp`, `mhd/mhd.cpp`	`orbital_advection` toggle; MHD+refinement fatal
`shearing_box/shearing_box.{hpp,cpp}`	store the toggle; `SetX1BndryMBs` / `ReinitAfterMeshUpdate`: rebuild boundary GID lists, MPI requests, and (via virtual `AllocateBuffers`, grow-only) CC/FC buffers after every AMR update
`shearing_box/shearing_box_srcterms.cpp`	dual-frame source terms, Eqs. (1)–(2)
`shearing_box/shearing_box_cc.cpp`	non-FARGO BC momentum/energy offset, Eq. (3)
`shearing_box/orbital_advection.cpp`	`maxjshift` estimate: $q\Omega$ factor, 2× multilevel headroom
`shearing_box/orbital_advection_cc.cpp`	runtime shift-vs-buffer-depth guard (fatal)
`mesh/mesh.cpp`	blanket prohibition → policy warning
`mesh/build_tree.cpp`	`Mesh::CheckShearingBoxRefinement`: interior-x_1 and ring completeness
`mesh/mesh_refinement.{hpp,cpp}`	boundary-block refinement veto + graded cap, Eq. (6); ring-wise AMR flag sync; `ReinitAfterMeshUpdate` calls after each update; derefinement-sort end-iterator fix (upstream bug)
`pgen/tests/shwave.cpp`, `swing.cpp`	background shear in ICs when `orbital_advection=false`; `MovingRingRefine` user AMR criterion (`amr_moving_ring`)
`inputs/shearing_box/` (repo root)	`*_smr*.athinput`, `epicycle*.athinput`, `*_amr_ring*.athinput` test inputs (`bin` outputs)

Runtime options (`<shearing_box>` block): `orbital_advection = true|false` (default `true`). Everything else is policy, enforced automatically. The gravity solver selection of the companion document (`<gravity> solver = multigrid|fft`) is unaffected by any of this; Phase 1 will add `shear_periodic` boundary support to the multigrid driver behind the same input block.

4 Building and compiling

Phase 0 needs no external dependencies beyond AthenaK's vendored Kokkos:

```
cd athenak-multigrid
git submodule update --init kokkos     # once
mkdir build && cd build
cmake ..                               # add -D Athena_ENABLE_FFT=ON for the FFT
make -j8                               #   oracle / swing+gravity cross-checks
```

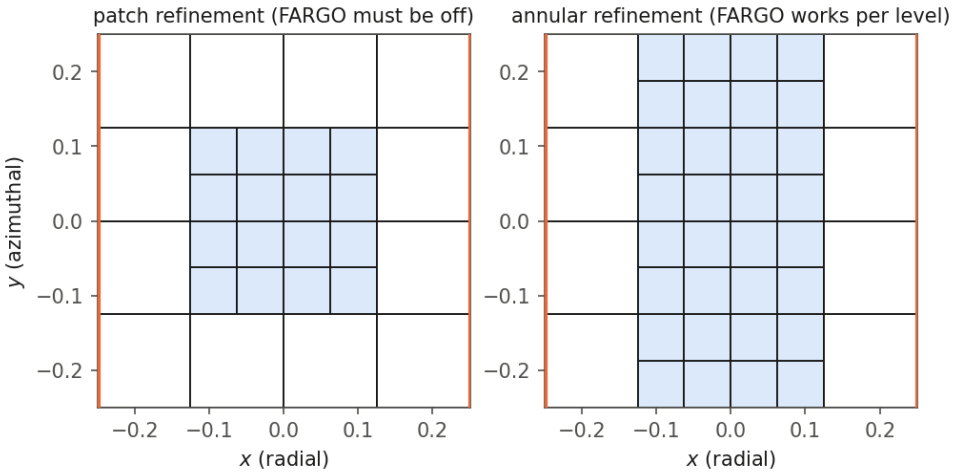


Figure 1: The two refinement layouts, drawn from the `bin` output block structure of actual runs (orange lines: shear-periodic x_1 boundaries; shaded: level-1 blocks). Left: a patch interior in both x_1 and x_2 — legal only with `orbital_advection=false`, because the coarse–fine interfaces in y break the shift communication. Right: the annular layout — the ring has no level jump in y, so FARGO runs per level with unmodified communication.

Notes: SMR/AMR runs need `<mesh> nghost` ≥ 4 (even, for prolongation); the FFT flag is only required for the tests that put gravity in the loop (it fetches kokkos-fft and needs `brew install fftw` on macOS, as detailed in the companion document). All test inputs referenced below are under `inputs/shearing_box/` and `inputs/tests/`; runs in this document were executed from `validation/run/` and `validation/run_epicycle/`.

5 Tests and results

5.1 Frame consistency: FARGO vs. full-velocity, uniform grid

```
ATH=build/src/athena
$ATH -i inputs/shearing_box/hydro_incompress_shwave.athinput job/basename=shw_fargo
$ATH -i inputs/shearing_box/hydro_incompress_shwave.athinput job/basename=shw_nofargo \
     shearing_box/orbital_advection=false
# with gravity in the loop (requires FFT build):
$ATH -i inputs/tests/swing_selfgrav.athinput job/basename=SwingNoFargo \
     shearing_box/orbital_advection=false
```

The two frames agree on the frame-independent diagnostics of the JG05 vortical shearing wave — mass exactly, x-kinetic-energy history to 3.3% (the truncation gap between two different discretizations of the same problem). With self-gravity, the non-FARGO swing test tracks the linearized shearing-sheet solution with $L_2(\delta) = 1.2 \times 10^{-2}$ (FARGO: 0.9×10^{-2}) and phase-locking of the density to the instantaneous shearing wavevector at the 10^{-12} level; it costs 5.7× more cycles, exactly Eq. (4).

5.2 Refinement without FARGO (Phase 0a)

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
     shearing_box/orbital_advection=false
```

An interior level-1 patch (Fig. 1, left) in the JG05 vortical-shwave problem. The run reproduces the reference run from the original `test/shwave-smr` branch **bit-for-bit** in every history column — the port changed nothing.

5.3 Guard checks (must abort)

```
# refined region snapping to a shear-boundary block: interior-x1 policy
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
     refined_region1/x1min=-0.24 refined_region1/x1max=-0.13
# partial-x2 ring with FARGO on: annular policy
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
     shearing_box/orbital_advection=true
```

Both abort with explanatory fatal errors; the second names the offending ring ("ring at level=3 lx1=2 lx3=0 has 4 of 8 MeshBlocks").

5.4 FARGO on an annular ring (Phase 0b)

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_ring.athinput \
     job/basename=ring_fargo
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_ring.athinput \
     job/basename=ring_nofargo shearing_box/orbital_advection=false
```

The full-x_2 level-1 ring of Fig. 1 (right). With FARGO active on the refined mesh: mass conserved exactly (drift 0.0); x-kinetic-energy history agrees with the non-FARGO twin to 3.8% — the same truncation-level gap the two frames show on uniform grids — and with the uniform-grid FARGO run to 4.1%.

5.5 Refined x_1 boundary faces

The uniform-boundary-level rule allows refinement *at* the shear-periodic boundaries when both faces are refined uniformly. The test input refines the leftmost and rightmost root columns to level 1 over the full x_2/x_3 extent (boundary annuli), interior at root, orbital advection on:

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput \
     job/basename=epi_bndry problem/ipert=1
# guard (must abort): refine only ONE face -> mixed levels across the wrap
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput \
     refined_region2/x1min=0.0 refined_region2/x1max=0.12
```

Results: the epicycle matches the analytic solution to 5.0×10^{-6} — *identical to the last digit* with the uniform-grid and interior-ring values, so the level-1 shear-periodic wrap and the level-1 azimuthal redistribution preserve uniform fields exactly. The 20-orbit endurance run on this mesh (`epicycle_smr_bndry.athinput`) is *bit-identical* to the interior-ring endurance run over all 202 history outputs: for a uniform field the refined shear-boundary wrap contributes exactly nothing, the strongest available statement for this configuration. The vortical shwave conserves mass exactly and agrees with the uniform run to 4.9% in x-KE (interior ring: 4.1%) and with the

interior-ring run to 0.9%; a three-panel density movie of uniform | interior ring | boundary annuli (`make_bin_movie_compare.jl`) shows the sheared wave crossing the refined boundary with no visible artifacts. The single-face guard aborts with a message naming the level span.

5.6 Two-level refinement (levels 1 and 2)

Both policies recurse in level, so nothing new is required for deeper hierarchies — verified with two configurations at levels 1+2 (8 root + 32 level-1 + 256 level-2 blocks each, FARGO on): nested interior rings (`hydro_incompress_shwave_smr_ring2.athinput`) and level-2 boundary annuli with level-1 buffer rings inboard (`hydro_incompress_shwave_smr_bndry2.athinput`). The epicycle matches the analytic solution to 5.0×10^{-6} on both meshes and the two runs are *bit-identical* to each other; the vortical shwave conserves mass exactly and agrees with the uniform run at truncation level (4.5% and 5.2% in x-KE; 0.45% between the two-level and one-level rings). The three-panel density movie (uniform | nested rings | two-level boundary annuli) shows no artifacts at any level interface or at the refined shear boundary.

5.7 The endurance test: epicycles for 20 orbits

The `ipert=1` epicycle has spatially uniform $v_x(t)$, $v_y'(t)$: an exact nonlinear solution that any correct y-shift must preserve *exactly*, making it the sharpest probe of the per-level shift and its ring-boundary communication — any indexing or buffer defect appears as $O(1)$ corruption. Two long runs ($L = 20$, $128^2 \times 4$ effective resolution, ideal EOS with $c_s = \sqrt{5/3}$, $v_x(0) = 0.1 = 0.077\, c_s$, 20 orbits = 20 epicyclic periods since $\kappa = \Omega$ at $q = 3/2$):

```
$ATH -i inputs/shearing_box/epicycle.athinput             # uniform 128x128x4
$ATH -i inputs/shearing_box/epicycle_smr_ring.athinput    # level-1 ring + FARGO
```

run	v_x range / amp	amplitude drift, 20 orbits	max $\lvert v_x - \text{analytic}\rvert$ / amp
uniform 128^2	$[-1.0006, +1.0005]$	+0.064%	2.4%
annular ring + FARGO	$[-1.0000, +1.0001]$	+0.008%	0.63%
boundary annuli + FARGO	*bit-identical to the annular ring, all 202 outputs*		

Both runs hold the ± 1 envelope; the amplitude invariant is flat to a few $\times 10^{-4}$ (uniform) and 10^{-4} (ring) — a slow secular *growth* characteristic of RK2 on an oscillator, not shearing-box or refinement physics. The 3.3% peak difference between the runs is pure phase drift: the ring run takes smaller steps (finer cells) and accumulates *less* RK2 phase error, which is why the refined-mesh FARGO run is the one closer to the analytic solution. A defect in the ring communication would instead show as amplitude noise — none is present at the 10^{-5} level.

5.8 AMR in the shearing box (Phase 0c)

AMR renumbers MeshBlock GIDs and re-assigns blocks to ranks on every mesh update, so the shear-boundary communication state built at construction — the `x1bndry_mbgid` GID lists, the MPI request arrays sized to them, and the send/recv buffers — goes stale. `ShearingBox::ReinitAfterMeshUpdate` now rebuilds all of it from the current MeshBlockPack, called from `AdaptiveMeshRefinement` after the post-update policy check and before any boundary exchange on the new mesh; the startup fatal guard is gone. Orbital advection needs no rebuild (its buffers are neighbor-indexed and sized to `max_nmb_per_rank` from the start).

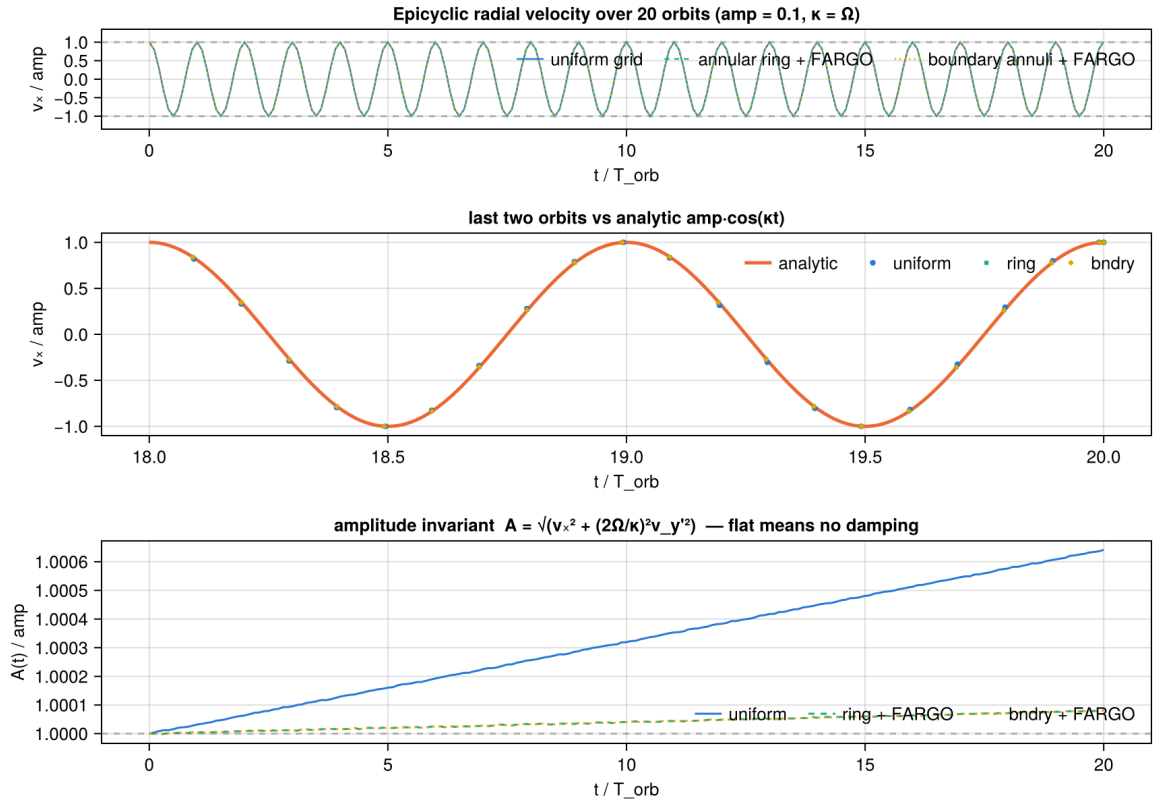


Figure 2: Volume-averaged radial velocity of the epicycle test over 20 orbits, normalized to the initial amplitude. Top: both runs oscillate between ± 1 with no visible decay. Middle: the final two orbits against the analytic $v_x = A\cos\kappa t$. Bottom: the epicyclic amplitude invariant $A = [v_x^2 + (2\Omega/\kappa)^2 v_y'^2]^{1/2}$, which isolates numerical damping/growth from phase error.

Two defects surfaced by the multi-level tests, both now fixed, are worth recording. First, vetoing refinement of the boundary blocks themselves is not enough: a criterion refining blocks *near* the boundary can force refinement *of* the boundary columns through 2:1 balancing. The veto is therefore extended to a *graded cap*: a block at level $s \geq 1$ above the (uniform) boundary level may be refined only if there is room for the intermediate buffer columns between it and the boundary, which in integer logical coordinates at the block's own level reads

$$2^{s+1} - 1 \;\leq\; l_{x1} \;\leq\; n_{\mathrm{mb},x1} - 2^{s+1}. \tag{6}$$

Any post-update policy violation is therefore a genuine bug rather than a criterion asking for too much. Second, an upstream AthenaK bug: the derefinement list was sorted with end iterator `&cllderef[ctnd-1]`, which *excludes the last element*. When a mixed-level derefinement left a fine-level parent in the unsorted last slot, coarser parents merging before it were vetoed by the tree's finer-neighbor check order-dependently — some parents of an annulus merged and some did not, splitting what must be a ring-complete derefinement (the policy check caught it as a partial ring, e.g. 4 of 8 members). Outside the shearing box the bug merely delays some derefinements by one AMR interval, which is why it went unnoticed upstream.

5.8.1 Dynamic ring creation and destruction: the moving-ring criterion

The dynamic exercise uses a deterministic user AMR criterion (`MovingRingRefine` in the `shwave` pgen): refine every MeshBlock whose x_1 extent overlaps $[x_c(t) - w,\, x_c(t) + w]$ with $x_c(t) = x_{c0} + A\sin(ft)$, derefine all others. The criterion depends only on x_1, so it naturally flags complete rings; the ring-sync logic and the vetoes above do the rest. With A chosen so the interval sweeps past the (vetoed) boundary columns, *all* refined rings are destroyed near each turning point and re-created on the way back — one full create/destroy cycle per $2\pi/f$.

```
$ATH -i inputs/shearing_box/epicycle_amr_ring.athinput          # 20 orbits, 1 level
$ATH -i inputs/shearing_box/epicycle_amr_ring2.athinput         # 2 orbits, 2 levels
$ATH -i inputs/shearing_box/epicycle_amr_ring_nofargo.athinput  # 2 orbits, non-FARGO
```

The 20-orbit epicycle run ($x_c = 7\sin t$, $w = 1.9$, one sweep per orbit) creates 2296 and destroys 2240 MeshBlocks (Fig. 3, left); the counts and all history columns are bit-reproducible across reruns. Mass drift is exactly zero. The epicyclic amplitude invariant of §5.7 drifts by +0.015% over 20 orbits, between the static ring (+0.008%) and the uniform grid (+0.064%), and the time-matched x-KE history differs from the *static*-ring run by at most 2.6×10^{-3} of the peak — an order of magnitude closer than either run is to the uniform grid (2.4×10^{-2}): the dynamic regridding adds essentially nothing beyond the static refined-boundary truncation already characterized in §5.7. The two-level variant (nested moving rings, 16−296 blocks, graded per Eq. (6)) drifts by 0 to six digits over 2 orbits; the non-FARGO variant exercises the momentum-offset boundary path with exact mass conservation. Two-panel movies with per-frame mesh outlines (`dens_compare_epi_unif_vs_epi_amr{,2}.mp4`) show the rings sweeping, collapsing, and re-forming with no visible artifacts.

5.8.2 Wave structure through a moving refinement boundary

The epicycle is spatially uniform, so it never exercises prolongation/restriction on real structure. The vortical-shwave AMR test (`hydro_incompress_shwave_amr_ring.athinput`) runs the JG05 problem of §5.1 under a moving ring ($x_c = 0.2\sin t$, $w = 0.05$), so wave structure is prolonged, restricted, and shear-remapped across every regrid event, including full ring destruction near the turning points at 64^2:

resolution	peak x-KE err. (static ring)	peak x-KE err. (moving ring)	AMR vs. static ring	mass drift
64^2	−4.05%	−8.55%	4.7×10^{-2}	0.0
128^2	−0.29%	−0.27%	1.8×10^{-4}	0.0

(errors vs. the same-resolution uniform run; curve maxima relative to the peak x-KE; the 128^2 runs use CLI overrides `mesh/nx1=128 mesh/nx2=128`). This test is in fact the code-verification setup of Johnson & Gammie (2005, ApJ 635, 149, their Fig. 1, left): the vortical-shwave amplitude obeys their eq. (9),

$$\delta v_x(t) = \delta v_{x0}\,\frac{1+\tau_0^2}{1+\tau^2}, \qquad \tau = q\Omega t + k_{x0}/k_y, \tag{7}$$

with $\tau_0 = n_{wx}/n_{wy} = -4$ here, so the swing through $\tau = 0$ at $\Omega t = 8/3$ amplifies δv_x by $1+\tau_0^2 = 17$. At 128^2 all three runs track Eq. (7) to better than 0.5% of the peak amplitude (uniform 0.39%, static ring 0.40%, moving-ring AMR 0.44%). At 64^2 the moving ring roughly doubles the static-ring truncation — there the ring covers only ~ 1.6 wavelengths of the $|n_{wx}| = 8$ mode and full ring destruction repeatedly restricts fine data mid-swing. At 128^2 the moving-ring and static-ring solutions are indistinguishable (1.8×10^{-4}, Fig. 3, right): the regridding cost converges away *at*

least as fast as the static refined-boundary error, with no floor. (The apparent order is inflated by geometry — at 128^2 the same physical criterion keeps a relatively tighter ring that never fully vanishes — so this is a convergence statement, not an order measurement.)

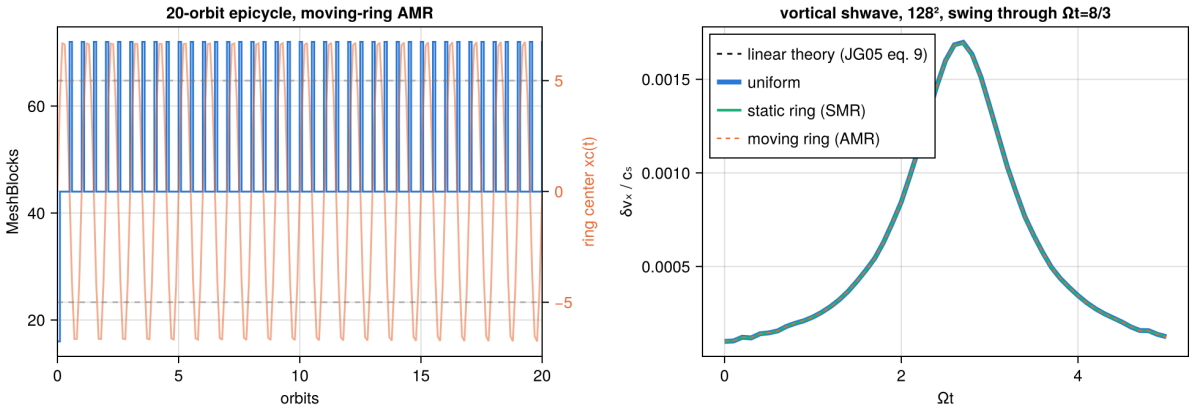


Figure 3: Phase 0c. Left: MeshBlock count (blue, left axis) in the 20-orbit moving-ring epicycle AMR run, with the prescribed ring center $x_c(t) = 7\sin t$ (orange, right axis; dashed lines mark the vetoed boundary columns beyond $|x_1| = 5$). Rings are destroyed near every turning point and re-created on the way back, 2296 blocks created / 2240 destroyed in total. Right: radial-velocity perturbation amplitude $\delta v_x/c_s$ of the 128^2 vortical shwave through the swing, with the linear-theory amplitude of Eq. (7) (Johnson & Gammie 2005, eq. 9) overplotted — uniform, static SMR ring, and moving-ring AMR are visually indistinguishable from each other (moving vs. static ring: 1.8×10^{-4} of peak) and track the theory to $< 0.5\%$ of the peak.

5.9 Summary

Check	Metric	Result
SMR no-FARGO vs `test/shwave-smr` reference	all history columns	bit-identical
Uniform: FARGO vs non-FARGO	mass / x-KE	exact / 3.3%
Swing + gravity, non-FARGO vs linear theory	$L_2(\delta)$ / phase	1.2×10^{-2} / 10^{-12}
Non-FARGO cost (L_x=2π box)	cycles vs FARGO	5.7×, as predicted
Ring + FARGO: mass conservation	max drift	exact (0.0)
Shwave: ring-FARGO vs ring-non-FARGO	x-KE	3.8%
Epicycle, 20 orbits, ring + FARGO	amplitude drift / invariant	0.008% / flat to 10^{-4}
Epicycle, 20 orbits, boundary annuli	vs ring run, all columns	bit-identical (0.0)
Two-level (1+2) rings / bndry annuli	epicycle vs analytic; ring2 vs bndry2	5×10^{-6}; bit-identical
Two-level shwave (both configs)	mass / x-KE vs uniform	exact / 4.5–5.2%
Boundary annuli + FARGO: epicycle	max rel. vs analytic	5×10^{-6} (= uniform)
Boundary annuli + FARGO: shwave	mass / x-KE vs uniform	exact / 4.9%
Mixed-level x_1 boundary	—	fatal, names the level span
Partial-x_2 ring with FARGO	—	fatal, names the ring
AMR epicycle, 20 orbits (2296 MBs created)	mass / invariant drift	exact / +0.015%
AMR epicycle vs static ring	time-matched x-KE / peak	2.6×10^{-3} (vs 2.4×10^{-2} to unifo
AMR epicycle, two-level nested rings	amp. drift, 2 orbits	0 to six digits
AMR epicycle, non-FARGO	mass	exact
AMR shwave vs static ring, $64^2 \to 128^2$	curve max rel. diff	$4.7 \times 10^{-2} \to 1.8 \times 10^{-4}$
AMR determinism	reruns, all counts/columns	bit-reproducible

6 Phase 1: shear-periodic multigrid boundary conditions

6.1 Method: an identification, not an operator change

Shear-periodicity identifies the two radial faces with a time-dependent azimuthal offset,

$$u(x,\, y,\, z) \;=\; u(x + L_x,\; y - q\Omega L_x\, t_s,\; z), \qquad t_s = t \bmod T_{\rm shear}, \quad T_{\rm shear} = \frac{L_y}{q\Omega L_x}, \tag{8}$$

and nothing more: the interior 7-point Laplacian, the red–black smoother, restriction, and prolongation are all untouched. Only the x_1 ghost fill changes — the ghost column is the opposite boundary column with its content shifted by $\Delta y = \pm\, q\Omega t_s L_x$ (henceforth $\pm\,\mathrm{qomt}\, L_x$; the sign convention, validated in the FFT track, is: inner ghosts shift by +, outer by −). The shift is generally a non-integer number of cells *at every multigrid level*: the same Δy is 35.52 fine cells but 4.44 cells four levels down. The integer part is index arithmetic with periodic wrap; the fractional part ϵ is applied with the same conservative remap kernels (`DC/PLM/PPMX_RemapFlx`) shared by orbital advection, the hydro shear boundary, and the FFT solver (order selected by the new `<gravity> mg_remap`, default `plm`).

Two structural facts make this cheap. First, the fixed point of the V-cycle (and of FAS) satisfies the *finest-level* discrete system, so the remap quality on coarse levels affects only the convergence *rate*, never the converged answer — one code path serves every level with no accuracy tax. Second, because each boundary plane is remapped as a full-period global row, the wrap supplies all guard cells and the `ngh=1` multigrid ghost width imposes no stencil restriction even for PPMX.

The shear phase is frozen once per solve as $\mathrm{qomt} = q\Omega\,(t \bmod T_{\rm shear})$ evaluated at `pmesh->time` — the FFT solver's convention exactly, so both solvers see identical boundary identifications at any instant (hydro's shear BC uses $t + \mathrm{d}t$ on the final RK stage; that $O(\mathrm{d}t)$ offset is a pre-existing, FFT-validated convention).

6.2 The global boundary-plane algorithm

At each MeshBlock level the driver keeps two global boundary planes (full y–z extent of that level, `ngh` deep in x). One fill is three stream-ordered kernels: *gather* the interior columns adjacent to each x_1 face of the boundary blocks into the planes; *remap* each plane row by $\pm\,\mathrm{qomt}\,L_x$ in place; *scatter* into the x_1 ghost slabs of the opposite-face blocks — over the *entire* slab $k, j \in [0, n{+}2\,\mathtt{ngh})$, with periodic wrap in y and z supplying the slab's edge and corner ghosts (trilinear prolongation reads the full 3^3 stencil, so corners matter). The fill is invoked at the top of `MultigridDriver::PhysicalBoundary`, which already runs after every boundary exchange in all three MG task lists including the final finest-level exchange — so the one ghost layer of Φ handed to the `SelfGravity` source term is also sheared, and no task-graph surgery was needed. It overwrites the unsheared periodic copy the neighbor communication just wrote (the tree wraps `shear_periodic` x_1 exactly like `periodic`); at $\mathrm{qomt} = 0$ the fill is skipped since that copy is already exact.

The root grid is a single replicated array, so `MGRootBoundary` gains a shear branch (`RootShearBoundaryX1`) that remaps the interior-row ghost columns in place; the existing x_2/x_3 passes then wrap the corner rows, and the 1-cell block level inherits correct ghosts through `SetFromRootGrid` unchanged. Mean-source subtraction — required for solvability — is re-enabled whenever every face is `periodic` or `shear_periodic` (it was silently disabled for any not-strictly-periodic mesh before, one of three defects that made shearing-box multigrid gravity *silently wrong* rather than unsupported: the ghosts were also overwritten with a $\Phi = 0$ Dirichlet wall after the periodic copy).

Multi-rank support is one collective at one seam: each rank gathers only its own boundary blocks' columns into a zero-filled plane, and a single `MPI_Allreduce(SUM)` then reconstructs the global plane on every rank before the (rank-redundant) remap. Every plane cell has exactly one writer, so summing the zeros is exact and independent of reduction order — the multi-rank solve is *bit-identical* to the serial one. The root-grid fill needs no communication (the root grid is already replicated on every rank). Remaining scope is enforced by fatal guards (uniform grid, periodic x_2/x_3, no `mg_bc` override, `mg_nghost=1`, device root grid).

A byproduct worth noting: MG task lists are now executed through a driver-independent runner, so `pgrav->Solve(nullptr, 1)` from a problem generator works with `solver=multigrid` — static Poisson tests (below) run the multigrid solver through the identical code path as evolution.

6.3 Tests and results

All tests use the shared inputs of the FFT track with the single override `gravity/solver=multigrid` (the inputs now carry the MG parameters: `threshold=0.0` stagnation mode, FMG with one V-cycle per level, `mg_remap=ppmx`); the FFT solver runs the same input as the oracle. Multigrid requires cubic MeshBlocks, so the swing runs override `meshblock/nx*=16`.

Shear-periodic instants (plumbing arbiter). `fft_poisson_shear.athinput` with `solver=multigrid` at $t_0 = 0$ *and* at $t_0 = T_{\mathrm{shear}} = 2/3$ (exercising the phase wrap) produces Φ **bit-identical** to the same solve with plain-periodic x_1 and no shearing box. Any plumbing or indexing error shows up here first.

Generic phase: residual, and what it does and does not prove. At $t_0 = 0.37$ ($\Delta y = 0.555\,L_y$) the discrete residual $L[\Phi] - 4\pi G(\rho - \bar{\rho})$ evaluated with the solver-filled ghosts is 5.9×10^{-15} / 5.5×10^{-14} / 2.7×10^{-13} at $32^3/64^3/128^3$: the V-cycles converge to machine precision with no floor from the sheared boundary. But for multigrid this residual is a *self-consistency* check only — a converged solve with a wrong boundary identification would also drive it to zero (the FFT solver's

$\sim 10^{-3}$ values on the same metric measure its remap truncation against the *exact* identification instead). The accuracy arbiter is the next test.

Analytic shearing wave (the sign arbiter). `profile=shwave` initializes a single rolled-frame Fourier mode whose discrete potential is known in closed form; the pointwise Φ error is $O(1)$ under any sign or offset error. Both solvers converge; the multigrid errors are *smaller* than the FFT solver's own, because MG remaps only the boundary ghosts while the FFT rolls and unrolls the entire field:

max rel. Φ error	32^3 (ϵ=0.76)	64^3 (ϵ=0.52)	128^3 (ϵ=0.04)
multigrid (`mg_remap=ppmx`)	3.2×10^{-5}	4.9×10^{-6}	9.5×10^{-8}
FFT (`fft_remap=ppmx`)	1.1×10^{-4}	1.3×10^{-5}	1.6×10^{-6}

(ϵ = fractional part of the cell shift at that resolution; it varies with N, so these are not clean order measurements — the FFT track's fixed-ϵ study established the remap orders.)

V-cycle convergence rate (the named open risk). Iterative mode (`full_multigrid=false`, warm start off a zero guess) at 64^3, $t_0 = 0.37$ — a near-worst fractional shift $\epsilon = 0.52$ — against the identical solve at shear phase zero:

	contraction/cycle	cycles to floor	floor
periodic phase (qomt = 0)	0.116	15	1.28×10^{-11}
sheared ($\epsilon = 0.52$)	0.157	17	1.27×10^{-11}

The remapped boundary costs two extra V-cycles and reaches the *same* roundoff floor — no stall, no loss of the solution. The open risk is retired.

Dynamic oracle A/B: swing amplification. `swing_selfgrav.athinput` (the Phase-0/FFT-track configuration: $128^2 \times 16$, self-gravity at $4\pi G = 1.9$, two swing times) run twice, changing only the solver. Peak amplification -3.511418 in **both** runs (relative difference 5.7×10^{-8}); the full 401-point amplification histories differ by at most 2.0×10^{-7} (rms 8.6×10^{-8}); phase locking (sine projection) holds at 4×10^{-14} (MG) vs. 8×10^{-14} (FFT). (§11.4 derives why the $d_{\cos}$ projection equals the wave amplitude exactly and why the sine channel is empty.) At matched settings the two solvers are interchangeable.

Multi-rank (the Allreduce seam). With `-D Athena_ENABLE_MPI=ON`, the static tests across 1–8 ranks reproduce the serial results: residuals at machine precision, and the shwave error *identical to all printed digits* (the plane reconstruction is exact, so the multi-rank solve is bit-identical to serial); the qomt = 0 shear-vs-periodic bit-identity also holds at 2 ranks. The 2-rank swing run matches the serial multigrid history to 10^{-11} — the MPI summation order of the history reductions, not the shear fill. (The rank-local diagnostics of the `fft_poisson` pgen were made MPI-aware in the process.)

Periodic regression. With no `<shearing_box>` block the code path is unchanged: the Jeans-wave evolution (`jeans_wave` vs. `jeans_fft`) reproduces the documented post-merge comparison — final density bit-identical between MG and FFT, velocities within 4×10^{-14} — and the static periodic Poisson test now also runs under `solver=multigrid` (residual 5.9×10^{-14}).

6.4 Consistency with the published Athena++ solver (Tomida & Stone 2023)

Before Phase 2 builds on this solver, we anchor it to the published results of the Athena++ multigrid method paper it descends from (Tomida & Stone 2023, ApJS 266, 7; `docs/Tomida_2023_ApJS_266_7.pdf`), reproducing their uniform-grid tests of §4.1 and §4.3 on our tree (notebook §10–11; inputs `tests/ts41_sin3.athinput` `tests/ts43_jeans.athinput`). Their §4.2 (binary potential on refined meshes) is the planned acceptance test for Phase 2.

§4.1: static sinusoidal waves. Triple-sine density ($A = 1$, $\rho_0 = 2$, $G = 1$) in a fully periodic unit box, with the analytic potential of their eq. (29). The `fft_poisson` pgen gains `profile=sin3` and a per-iteration study mode (`<problem> conv_niter`): in MGI mode successive `Solve()` calls warm-start from `pgrav->phi`, so N calls with `niteration=1` reproduce N V-cycles from the zero initial guess, and ϵ (vs. analytic, their eq. 6), δ (vs. the fully converged discrete solution, eq. 5, built by a bit-identical reference pass so $\delta_{20} \equiv 0$), and the defect d (eq. 7) are measured after every iteration inside one process; in FMG mode iteration 1 is one full FMG sweep. Results at $h = 1/16 \ldots 1/256$ (their 1/512–1/1024 points exceed laptop memory):

- convergence factor per V-cycle 0.125 at 64^3 (rising to 0.127 at 256^3, 0.088 at 16^3) in both δ and d, for both MGI and FMG — their reported ∼0.125 (Yavneh 1996). The 0.116 measured in §6.3 on the modes+blob source is source-dependent variation, not a discrepancy;
- converged ϵ exactly second order (order 2.008 → 2.000), **identical to the FFT solver's ϵ to every printed digit** at all resolutions (both solve the same discrete system);
- the first FMG sweep alone already *below* the converged discretization error ($\epsilon_{\rm sweep}/\epsilon_{\rm conv}$ = 0.39–0.60), their "remarkable" finding.

§4.3: Jeans-wave dynamical test. Periodic $1 \times 0.5 \times 0.5$ box; the `jeans_wave` pgen's oblique-wave machinery reproduces their $\mathbf{k} = 2\pi(1/L_x, 1/L_y, 1/L_z)$ exactly ($\lambda = 1/3$), and its `n_jeans` is their $\nu_{\rm Jeans}$ ($\lambda_J = \sqrt{\pi c_s^2/(G\rho_0)}$; stable case $\nu = 0.5$, $A = 10^{-6}$, $\Gamma = 5/3$, PLM + HLLC + `rk2` in place of their VL2). Two additions were needed: their *isothermal* pressure perturbation (`<problem> perturb=isothermal`; $\delta p = p_0 A \sin kx$), and their error metric (`ts_errors=true`): the rms of the L1 errors of the five conserved variables against the analytic linear solution. The isothermal perturbation is not an eigenmode — it excites a *static* self-gravitating mode ($\delta p = c_s^2 \nu^2 \delta\rho$ balancing gravity, fraction $f_{\rm st} = (1-1/\Gamma)/(1-\nu^2) = 8/15$ of the density perturbation) plus a standing gravito-acoustic wave at the dispersion-relation frequency — and the metric compares against this two-mode solution. (Its correctness is validated by the convergence itself: a wrong decomposition floors the error at $\mathcal{O}(10^{-7})$.) At $t = 1.5$ (the time of their Fig. 11; Fig. 10's is unstated), $N_x = 16 \ldots 128$:

rms of L1 errors	N_x=16	32	64	128
fully converged	3.60×10^{-7}	1.92×10^{-7}	3.98×10^{-8}	8.12×10^{-9}
one FMG sweep ($N_{\rm iter}$=0)	3.99×10^{-5}	6.00×10^{-6}	1.37×10^{-6}	3.13×10^{-7}
FFT (exact potential)	3.60×10^{-7}	1.92×10^{-7}	3.98×10^{-8}	7.86×10^{-9}

Both of their findings reproduce: the converged runs are roughly second order with a slight deviation (order 2.27–2.29; N_x=16 is pre-asymptotic at ∼5 cells per wavelength), while the single-sweep error is *precisely* second order (2.13) but 30–40× larger — gravity-dominated, so "use of the fully converged solutions improves the accuracy." The FFT solver (exact discrete potential every stage) bounds the converged curve to six digits.

6.5 Summary

Check	Metric	Result
Shear-periodic instants ($t_0 = 0$ and $T_{\rm shear}$)	Φ vs. periodic solve	bit-identical
Generic phase 32^3–128^3	discrete residual	6×10^{-15}–3×10^{-13}
Analytic shwave, 128^3	max rel. Φ error (MG / FFT)	9.5×10^{-8} / 1.6×10^{-6}
V-cycle contraction, $\epsilon = 0.52$	per-cycle factor (vs. periodic)	0.157 (vs. 0.116)
Swing amplification vs. FFT	peak / trajectory / phase	5.7×10^{-8} / 2×10^{-7} / 10^{-14}
Multi-rank, 1–8 ranks	shwave digits / qomt=0 identity	identical / bit-identical
Jeans periodic regression	final density MG vs. FFT	bit-identical
T&S §4.1: CF / ϵ order / FMG sweep	vs. published	0.125 / 2.00 / sub-truncation
T&S §4.1: converged ϵ vs. FFT	all N, all digits	identical
T&S §4.3: converged / one-sweep orders	vs. published	2.3 / 2.13, sweep 30–40× larger

Next: Phase 2 — multigrid gravity on refined shearing-box meshes; delivered for interior rings in §7.

7 Phase 2a: multigrid gravity on refined shearing-box meshes — interior rings

7.1 Scope and the incremental decomposition

Phase 2 combines the two halves validated so far — refinement in the shearing box (Phase 0) and the shear-aware multigrid solver (Phase 1) — in three steps that track *where refined blocks sit relative to the shear boundary*:

- **2a (this section): interior rings.** Under the Phase-0 interior-x_1 policy, refined blocks never touch the shear-periodic x_1 faces, so the boundary blocks — the only participants in the Phase-1 shear-plane fill — all remain at the root level. No octet ever needs a sheared ghost.
- **2b: boundary annuli.** Uniformly refined x_1 boundary faces (allowed by the Phase-0c-lite hydro policy) put octets *at* the shear faces and need a sheared octet ghost fill — the host-side analogue of §6.2.
- **2c: AMR.** The shear planes and offset tables are sized once at construction; adaptive refinement adds a rebuild-after-remesh step (the analogue of `ShearingBox::ReinitAfterMeshUpdate`).

The `MGGravityDriver` constructor now enforces exactly this decomposition: with shear-periodic x_1 it aborts if refinement is *adaptive* (2c), or if any x_1 boundary MeshBlock sits above the root level (2b) — a check stricter than the hydro-side `CheckShearingBoxRefinement`, which also admits uniformly refined boundary faces. Static interior rings pass.

7.2 Step 0: the binary-potential octet control (no shear)

Upstream #776 ships the Athena++ binary-potential test (`pgen/tests/binary_gravity.cpp`, `tests/binary_grav` — precisely the mesh-refinement acceptance test of Tomida & Stone §4.2: two uniform spheres (M_1=2 at x=6/1024, M_2=1 at x= − 12/1024, R = 6/1024, 10^3-subcell surface averaging) under four nested static refinement levels. Since none of our Phase-0/1 work had ever exercised the octet

machinery (every earlier multigrid run was uniform), we run this control *first*, so that any later failure on sheared refined meshes indicts the shear coupling and not the inherited octet code.

Two caveats about the shipped input. First, its periodic boundaries are *not comparable* to the isolated two-sphere analytic solution the pgen's error norms use (the errors come out $O(1)$ by construction); the meaningful configuration is the paper's: outflow mesh boundaries with `gravity/mg_bc=multipole`, `mporder=4` (hexadecapole), now present as overridable defaults in the input. Second, the pgen's printed "Potential L2" is $\sqrt{\langle|\delta\Phi/\Phi|\rangle}$ (Athena++ convention), and the `<refined_region>` blocks are *absolute physical* volumes: doubling the root resolution leaves every region at its absolute size, so between the 64^3- and 128^3-root runs only the innermost box gains resolution and the global mean is frozen *by construction* — region-wise norms are the meaningful metric.

With 40 V-cycles (FMG), roots $32^3/64^3/128^3$ + 4 levels (232/232/288 blocks; at 64^3 the level-1 region force-refines all root blocks, so the far field lands at h=1/128 in both larger runs):

rms rel. Φ error per region	R_0 (far)	R_1	R_2	R_3	R_4 (inner)
32^3 root	1.26×10^{-4}	1.58×10^{-4}	2.56×10^{-4}	3.43×10^{-4}	9.57×10^{-4}
64^3 root	3.19×10^{-5}	4.23×10^{-5}	6.55×10^{-5}	8.68×10^{-5}	2.44×10^{-4}
ratio	3.94	3.73	3.91	3.95	3.92

Second order in every region (the h of each region halves exactly between these two runs). From 64^3 to 128^3 only R_4 refines ($h = 1/1024 \to 1/2048$) and its error drops by 2.37 — saturated by the unimproved surrounding regions, whose inherited error it cannot undercut; the same-resolution regions agree between the two runs to four digits (a strong bit-level reproducibility check through the float32 snapshots). On the Tomida & Stone normalized-rms metric (their eq. 31) the 64^3 run gives 7.5×10^{-5} against their Fig. 7(d) value of $\sim 1.3 \times 10^{-4}$ at h=1/64 — consistent, ours slightly lower because of the force-refined far field. The late-stage convergence factor is $\sim$0.5 per V-cycle, their reported refined-mesh value (§6.3 measured 0.12–0.16 on uniform grids: refinement, not shear, is what costs convergence rate). The defect reaches its 10^{-12} relative floor. The octet machinery works as published.

7.3 What 2a required — and what it did not

The entire code change is the constructor guard described above. The Phase-1 shear-plane fill needed *nothing*:

- `FillShearGhostPack` gathers and scatters only blocks whose own `mb_bcs` carry `shear_periodic` on an x_1 face. Interior fine blocks have internal (`BoundaryFlag::block`) faces there, so they are skipped automatically — their (root-unit, hence meaningless) entries in the per-block offset table are never read. No level test was needed.
- The plane geometry (`nmmbx2_` × block cells per MG level) is expressed in root-grid units, and under the 2a guard every participating block *is* a root-level block, so the planes tile exactly as on the uniform grid.
- Octet x_1 neighbor tables already wrap shear-periodic like periodic (Phase-1 groundwork); under the 2a guard no octet abuts the x_1 faces, so the unsheared wrap is never consulted. (It becomes live — and wrong — only in Phase 2b, which is why boundary-level refinement stays fatal.)

SMR additionally requires `nghost=4` (AthenaK wants even ghosts with refinement) and, as always for this solver, cubic MeshBlocks; the test inputs use 16^3 blocks, giving four x_1 block columns of which the middle two are refined.

7.4 Static tests: bit-identity, the analytic arbiter on SMR, MPI

Input `tests/mg_poisson_shear_smr.athinput`: the Phase-1 shear test (§6.3) with a level-1 ring over $x_1 \in [-1/4, 1/4]$ (full x_2/x_3).

Shear-periodic instants. At $t_0 = 0$ the refined-mesh solve with shear-periodic x_1 is **bit-identical** (float64 comparison of every cell of every block) to the same mesh with plain-periodic x_1, for both the multi-mode and shwave sources. (Compare the bin *data*: the file header echoes the input, so whole-file comparison always differs.)

Analytic shearing wave. The reference potential $\Phi_{\rm ref}$ is the *exact solution of the discrete system on the uniform root grid* — not the continuum potential. The shwave profile initializes a single rolled-frame Fourier mode,

$$\rho = \rho_0 + A\cos(k_x x + k_y y + k_z z), \qquad k_x = \frac{2\pi}{L_x}\Big(n_1 + \mathrm{qomt}\,\frac{L_x}{L_y}\,n_2\Big), \quad k_y = \frac{2\pi}{L_y}n_2, \quad k_z = \frac{2\pi}{L_z}n_3, \tag{9}$$

a plane wave tilted by the current shear phase and therefore exactly shear-periodic across the x_1 wrap. Acting on a cosine, the 7-point Laplacian has the exact (no-expansion) eigenvalue

$$\lambda_h = \frac{2\cos(k_x\Delta x) - 2}{\Delta x^2} + \frac{2\cos(k_y\Delta y) - 2}{\Delta y^2} + \frac{2\cos(k_z\Delta z) - 2}{\Delta z^2}, \tag{10}$$

so $L_h\Phi = 4\pi G(\rho - \bar\rho)$ is solved exactly by $\Phi_{\rm ref} = [4\pi GA/\lambda_h]\cos(k_x x + k_y y + k_z z)$ at cell centers, with the Δ's at the *root* spacing and both fields mean-subtracted (the periodic gauge). This is the closed form behind the pgen's `SHWAVE ERROR` line, inherited from the FFT track — the rolled k-space kernel *is* $1/\lambda_h$, which is why FFT solves it to machine precision. The choice fixes the attribution: on a *uniform* grid $\Phi_{\rm ref}$ is exact, so a correct solver deviates only through its ghost-fill remap (Fig. 4, third panel: 5×10^{-6}, and the roundoff floor at qomt = 0); against the *continuum* $-1/k^2$ instead, both regions would show their own $O(h^2)$ discretization error ($\sim (k\Delta x)^2/12$ per direction, 8×10^{-4} at the root spacing) and the refinement structure would blur. On the refined mesh the composite operator is *not* the root-grid operator: the fine region's eigenvalue is closer to the continuum, differing per direction by $(kh)^2/12$ terms that sum to 6.0×10^{-4} relative at 64^3 — and the measured fine-region plateau is 6.7×10^{-4}: the fine grid deviating from $\Phi_{\rm ref}$ by *being more accurate than it*. The pointwise maximum sits in the first fine cell at the refinement interface ($x = \pm 0.246$ at 64^3), symmetric about the box center, decaying harmonically into the coarse region, with *no* feature at the shear faces $x = \pm 1/2$; at the generic phase it is within 3% of the qomt=0 value at 64^3 and 0.6% at 128^3 — the interface, not the shear, owns the error:

	qomt = 0		generic phase $t_0 = 0.37$	
	max rel.	L_2 rel.	max rel.	L_2 rel.
32^3	1.18×10^{-2}	5.05×10^{-3}	1.44×10^{-2}	8.14×10^{-3}
64^3	5.51×10^{-3}	1.28×10^{-3}	5.66×10^{-3}	2.06×10^{-3}
128^3	2.66×10^{-3}	3.25×10^{-4}	2.68×10^{-3}	5.18×10^{-4}
L_2 ratios		3.94, 3.94		3.96, 3.97

L_2 is cleanly second order at both phases — the same interface behavior Tomida & Stone report (their rms metric) — while the interface *peak* converges at first order, which we took for the known local property of the mass-conserving level-boundary stencil. **Correction (November 2026):** it was mostly a bug in the fine ghost cells (a halved transverse slope); with the fix of §10 the

interface peak converges at second order (3.35×10^{-3}, 8.19×10^{-4}, 2.03×10^{-4} at qomt = 0); with `mg_fc_interp = quadratic`, §10.5, 2.64×10^{-3}, 6.67×10^{-4}, 1.67×10^{-4}). The shear remap contributes only the small, second-order-converging difference between the two L_2 columns. Figure 4 shows all of this at once: the error concentrates in the fine region (the reference uses the root-grid eigenvalue) and peaks on the interfaces, while the shear faces are featureless — and the same metric on the uniform Phase-1 run is three decades smaller, so the entire visible structure is refinement, not shear. One warning: the `profile=modes` residual norms print $O(1)$ on any refined mesh because the pgen evaluates a naive per-block Laplacian, which is *not* the composite multigrid operator at level interfaces; on SMR the shwave comparison is the valid metric.

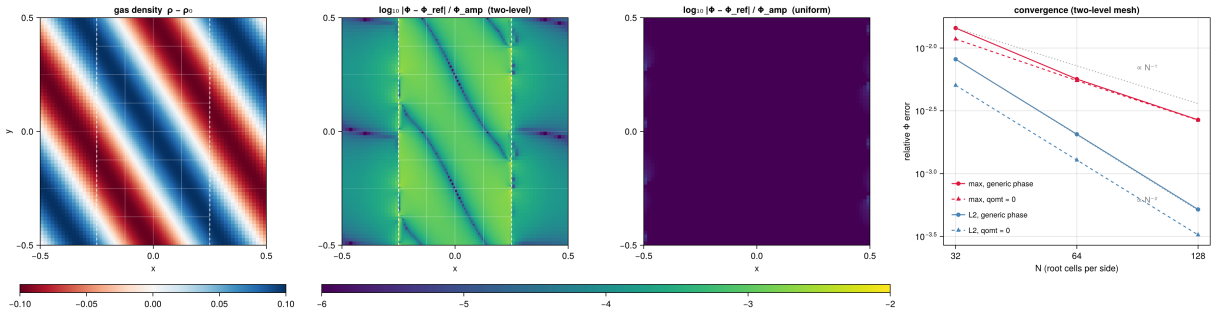


Figure 4: The static shwave test at the generic shear phase (64^3 root, level-1 ring over $|x| < 1/4$, block outlines in white). First panel: the sheared density mode. Second: $\log_{10}$ relative potential error against the root-grid closed form (eq. 10) on the two-level mesh — the error lives in the *fine* region (root–fine eigenvalue mismatch, the 6×10^{-4} plateau of §7.4), peaks in the first fine-cell layer on the refinement interfaces (dashed; bright cells modulated by the mode phase, pinched at its zero-crossing curves), and shows no feature at the shear-periodic faces $x = \pm 1/2$. Third: the *same* metric on the uniform Phase-1 run — pure remap truncation at 5×10^{-6}, three decades below the interface and eigenvalue structure. Fourth: error vs. resolution at both shear phases — L_2 second order, interface peak first order, the two phases nearly coincident. (Before the fine-ghost fix of §10; after it the interface peak is second order too.)

Contraction and MPI. Static FMG solves contract at 0.18/cycle on this two-level mesh (identically for periodic and sheared boundaries; uniform grids gave 0.12–0.16). Runs at 1/2/4 ranks print identical doubles and their float32 snapshots agree to one ulp (5.8×10^{-11}, np=2 ≡ np=4).

7.5 Dynamic test: swing amplification on a ring, with control

Input `tests/swing_selfgrav_smr.athinput`: the §6.3 swing configuration ($128^2 \times 16$, $4\pi G = 1.9$, two swing times) with a level-1 ring nominally over the middle half in x_1 — in fact over the middle *six* of eight block columns: the region edges ($\pm\pi/2$ as a decimal literal) sit 3×10^{-15} inside the neighboring columns and pull them in, a wholesome warning about `<refined_region>` edges at non-representable coordinates; the uniform twin is the single override `mesh_refinement/refinement=none`, the oracle twin additionally `gravity/solver=fft`.

	peak $d_{\cos}/A$	peak time	max $\lvert d_{\sin}\rvert/A$
uniform, FFT	−3.511418	7.804	2.8×10^{-13}
uniform, multigrid	−3.511418	7.804	0.7×10^{-13}
ring SMR, multigrid	−3.583481	7.804	1.3×10^{-13}

The uniform twins reproduce the Phase-1 A/B exactly (the 16^3-block relayout and `nghost=4` change nothing). The SMR peak differs from the uniform one by 2.05% — *toward* the converged answer (linear theory 3.605; the 256^2 FFT-track run gave 3.62), as refining most of the domain should move it, at machine-level phase lock. The decisive attribution is the **gravity-off control** (`hydro_srcterms/self_gravity=false`, both meshes): the hydro-only SMR-vs-uniform deviation is *larger* than the self-gravitating one — peak shift +2.78% vs. +2.05%, normalized trajectory deviation 7.7×10^{-2} vs. 5.6×10^{-2} — so multigrid gravity on the refined shearing mesh contributes *no* deviation of its own beyond hydro truncation. Figure 5 shows the three amplification histories (indistinguishable at plot scale) with the pairwise differences: the SMR-vs-uniform gap rides at 10^{-2}–10^{-1} while the MG-vs-FFT solver seam sits five decades below it; Figure 6 compares the final density fields.

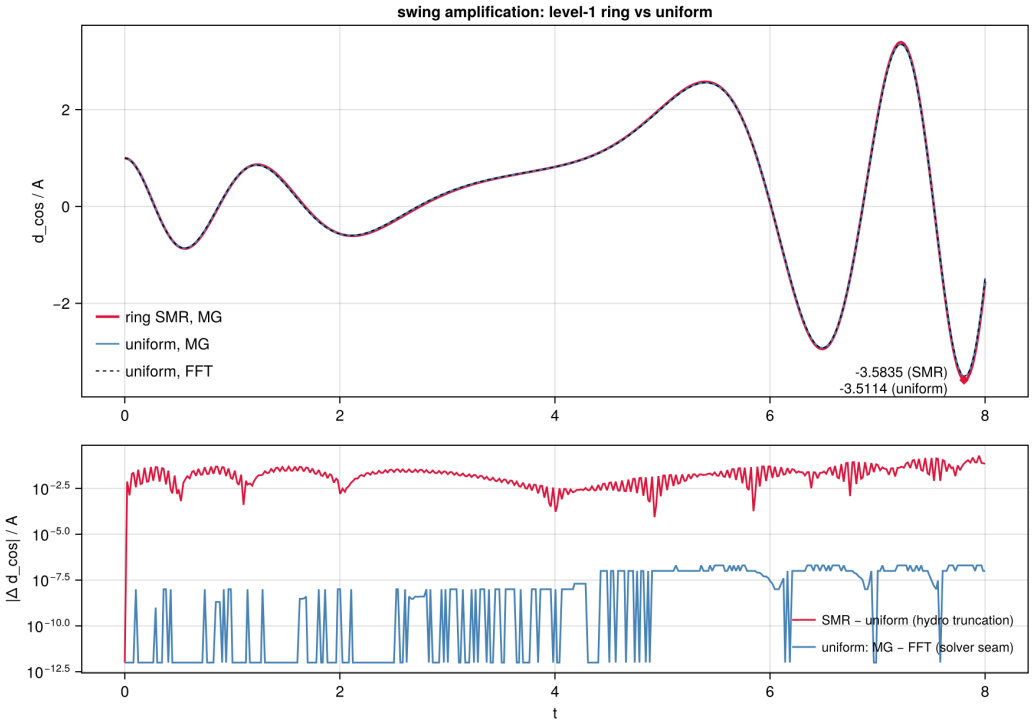


Figure 5: Swing amplification histories $d_{\cos}/A$ for the three runs (top; the curves overlap to the line width, peaks annotated) and the pairwise trajectory differences (bottom, log scale): SMR-vs-uniform is the hydro-truncation resolution blend; uniform MG-vs-FFT is the solver seam, floored by the history-file print precision.

7.6 The stopping rule on refined meshes (practical note)

The first full-length SMR swing run stopped at $t = 5.5$: `threshold=0.0` (stagnation-exit mode) reached the driver's 40-V-cycle cap — *while still converging healthily* (defect 2.9×10^{-12}) — and the cap gracefully ends the run via `nlim`. The mechanism: near the swing peak the solution is dominated by the wound-up $k_x \simeq 0$, $k_y = 1$ mode, whose coarse-grid correction must pass through the sheared x_1 wrap where the y-remap acts on 8-cell rows; the warm-started contraction degrades to ∼0.55/cycle (static solves on the same mesh: 0.19), so the defect ratio never exceeds the 0.9 stagnation trigger within 40 cycles. This is a stopping-rule interaction, not a solver failure — and it

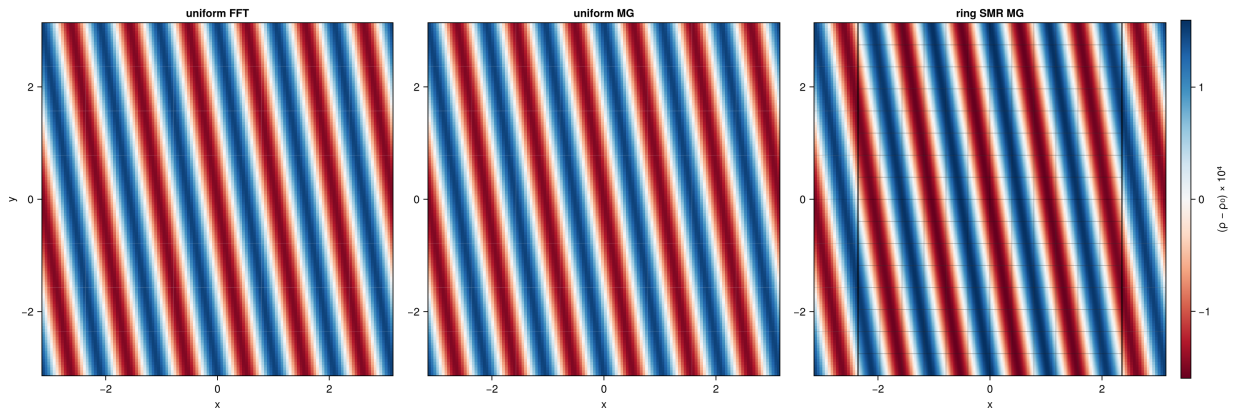


Figure 6: Final ($t = 8 = 2\,t_{\rm swing}$) density deviation $(\rho - \rho_0) \times 10^4$ at the z mid-plane for the three swing runs: uniform FFT, uniform multigrid, and the level-1 interior-ring SMR run (block outlines drawn; the heavier lines mark the refinement-level boundary). The two uniform panels are indistinguishable ($\leq 2 \times 10^{-7}$ everywhere); the SMR panel carries the same trailing wave with the amplitude difference of §7.5.

is exactly why Tomida & Stone run production refined-mesh gravity as FMG with a *fixed* iteration count. The SMR inputs therefore default to `threshold=-1`, `niteration=10`; on the uniform twin, fixed-10 and stagnation mode agree to 10^{-11}, and the full-length SMR run completes with no MG warnings.

7.7 Summary

Check	Metric	Result
Binary potential (octet control)	region-wise order $32^3 \to 64^3$	3.7–3.95
Binary potential vs. T&S Fig. 7(d)	$\epsilon_{\rm rms}$ at h=1/64 root	7.5×10^{-5} vs. $\sim 1.3 \times 10^{-4}$
SMR shear instants ($t_0 = 0$)	Φ vs. periodic twin	bit-identical
SMR analytic shwave	L_2 order (qomt=0 / 0.37)	3.94 / 3.96–3.97
SMR contraction (static)	per-cycle, periodic = sheared	0.18
Multi-rank 1/2/4	printed digits / f32 snapshots	identical / 1 ulp
Swing: uniform twins	peak, vs. Phase-1	-3.511418, exact
Swing: ring SMR	peak shift, direction	+2.05%, toward theory
Swing: gravity-off control	hydro-only shift (must dominate)	+2.78%
Swing: phase lock	$\max \lvert d_{\sin}\rvert / A$, all runs	$\sim 10^{-13}$

Next: Phase 2b — sheared octet ghosts, so refined blocks may sit *on* the shear boundary (boundary annuli); delivered in §8.

8 Phase 2b: sheared octet ghosts — refinement on the shear boundary

8.1 Where the shear enters the octet machinery

With boundary annuli — both x_1 faces uniformly refined over the full x_2/x_3 extent, the configuration Phase 0c-lite validated for hydro — the multigrid hierarchy places *octets* (per-cell 2^3 refinement

units bridging the block levels down to the root grid) directly on the shear-periodic faces. Phase 1 had already wrapped the octet x_1 neighbor tables like periodic and exempted the shear faces from the physical-boundary reflection, so octet x_1 ghosts received *unsheared* periodic values — correct only at qomt = 0, and the reason 2b was fatal-guarded. Every consumer of those ghosts (octet smoothing, the FAS right-hand side, restriction, FMG and V-cycle prolongation, and the octet→block transfer `SetFromRootGrid`) funnels through exactly two chokepoints, `SetBoundariesOctets` (per level, inside the V-cycle) and `SetOctetBoundariesBeforeTransfer` (before the root→block transfer), which makes the fix local.

Two structural facts shape the implementation. First, octets are *host-side* flat buffers, and they are *globally replicated* on every rank (the block→root transfer runs an `MPI_Allgatherv` and every rank populates every octet), so the sheared fill is plain serial host code with **no MPI seam at all**. Second, under the uniform-boundary-level policy the boundary octets tile the full y–z surface at every octet level — precisely the precondition for the Phase-1 global-plane algorithm to transplant.

8.2 The fill

`FillShearOctetGhosts(lev, folddata)` is the Phase-1 algorithm at octet granularity: gather the interior cell columns adjacent to each x_1 face from the boundary octets into a per-level (y, z) plane (level resolution 2 (`nrbx2` $\ll$ lev) cells), remap each row by $\pm\,\mathrm{qomt}\,L_x$, and scatter into the x_1 ghost slabs of the opposite-face octets — the whole slab including its y/z edge and corner ghosts (trilinear and flux-conserving prolongation read those), overwriting the unsheared values the neighbor-table exchange just wrote, with the same sign convention as `FillShearGhostPack`. The conservative remap runs through new *scalar per-face twins* of the team kernels (`DC/PLM/PPMX_RemapFlxFace` in `remap_fluxes.hpp`, host-callable, kept in documented lockstep with the validated device versions — the device path is untouched, preserving every Phase-1/2a result bit-for-bit). As in Phase 1, octet ghost quality can only affect the convergence *rate*: the V-cycle fixed point is set by the finest block level, whose ghosts come from the (generalized) block planes.

On the block side, the only change is plane geometry: the participating blocks — the only ones whose `mb_bcs` carry `shear_periodic` — now sit at a uniform level L above root, so the planes are sized at that level (`shear_reflev_` $= L -$ root); the per-block offsets were already expressed at each block's own level. The constructor guard weakens accordingly: x_1-face blocks must share *one* uniform level (the hydro invariant), AMR remains fatal (Phase 2c), and the deferred host-root shear fill stays guarded.

8.3 Static validation

Input `tests/mg_poisson_shear_bndry.athinput`: the §7.4 problem with the level-1 ring replaced by two annuli hugging the x_1 faces ($|x| \in [1/4, 1/2]$, full x_2/x_3); a two-level variant (level-2 annuli of width 1/16 plus the automatically graded level-1 buffers) runs off the same input via CLI overrides. The Phase-2a interior-ring results reproduce digit-exact with the new code in place (the octet fill is inert without boundary octets).

- **Shear-periodic instants.** At $t_0 = 0$ the annuli solves are **bit-identical** to their periodic twins — at one *and* two refinement levels — and their norms equal the interior-ring values to all printed digits, as they must: the two meshes differ by a half-box shift in x, under which the periodic mode's norms are invariant.

- **Generic phase** ($t_0 = 0.37$): $L_2 = 2.057 \times 10^{-3}$, *equal to the interior-ring value to four digits* (the same half-shift symmetry, now with the remap active across the refined wrap);

max $= 5.95 \times 10^{-3}$ at the annulus' interior interface — 5% above the ring's peak, and still no feature *on* the shear faces themselves, now covered by fine blocks. Two-level: 6.03×10^{-3} / 4.88×10^{-3} (the larger L_2 is the root-eigenvalue reference artifact growing with off-root volume, §7.4).

- **Contraction.** Static FMG: 0.18–0.20/cycle at both one and two levels, with the *sheared* runs reaching floors slightly below their periodic twins (2.0×10^{-10} vs. 8.2×10^{-10} at one level) — the sheared octet coarse correction costs nothing.
- **Multi-rank.** np = 2/4 reproduce the serial digits exactly (the octet fill is replicated host work; the block planes keep their Phase-1 Allreduce).

8.4 Dynamic validation: swing on boundary annuli

Input `tests/swing_selfgrav_bndry.athinput`: the §7.5 swing configuration with the level-1 interior ring replaced by two annuli hugging the x_1 faces (4 of 8 block columns refined — two per face, the same edge-representation effect as §7.5 — vs. the ring's 6). The full-length run completes with no multigrid warnings:

	peak $d_{\cos}/A$	shift vs. uniform	max $\lvert d_{\sin}\rvert/A$
uniform, multigrid	−3.511418	—	0.7×10^{-13}
annuli, multigrid	−3.550196	+1.10%	1.4×10^{-13}
ring (§7.5)	−3.583481	+2.05%	1.3×10^{-13}

All three peak at the same $t = 7.804$ at machine-level phase lock. The annuli shift is again *toward* the converged answer (theory 3.605), and its size scales with the refined volume — two-thirds of the ring's refined columns, about half its shift. The gravity-off control closes the attribution exactly as in §7.5: hydro-only annuli-vs-uniform deviates *more* than the self-gravitating run (peak +1.70% vs. +1.10%; normalized trajectory deviation 6.4×10^{-2} vs. 4.3×10^{-2}) — the sheared octet machinery and level-1 shear planes add no deviation of their own beyond hydro truncation.

8.5 Summary

Check	Metric	Result
2a regression	interior-ring shwave digits	exact
Annuli shear instants, 1 and 2 levels	Φ vs. periodic twin	bit-identical
Annuli vs. ring, qomt = 0	norms (half-shift symmetry)	all digits
Annuli generic phase	L_2 vs. ring L_2	equal to 4 digits
Contraction, 1 and 2 levels	per-cycle, sheared vs. periodic	0.18–0.20, floor below
Multi-rank np = 2/4	printed digits	identical
Swing: annuli peak	shift vs. uniform, direction	+1.10%, toward theory
Swing: gravity-off control	hydro-only shift (must dominate)	+1.70%
Swing: phase lock	max $\lvert d_{\sin}\rvert/A$	1.4×10^{-13}

Next: Phase 2c — AMR in the shearing box with multigrid gravity; delivered in §9.

9 Phase 2c: adaptive refinement with multigrid gravity in the shearing box

9.1 What the remesh invalidates, and where it gets rebuilt

Phases 2a–2b sized every shear structure once, at construction: the per-MG-level block boundary planes and their per-block (y, z) offsets, the boundary-level factor `shear_reflev_`, and the per-octet-level face flags `oct_shear_face_`. Adaptive refinement invalidates all of them — blocks appear, disappear, and migrate between ranks; the octet population changes; in principle even the boundary level can change. The fix rides an existing vehicle: `MultigridDriver::PrepareForAMR` already detects any remesh (via the mesh's AMR/load-balance sequence counter) before every solve and rebuilds the block arrays and octets. Phase 2c makes `AllocateShearPlanes` idempotent (planes are `realloc`'d, offsets recomputed from the current logical locations — load balancing can reassign blocks without changing this rank's block count) and calls it from the same rebuild branch; the octet face flags were already recomputed inside `InitializeOctets`. The constructor's adaptive fatal is removed: the uniform-boundary-level invariant is re-enforced after *every* AMR update by the hydro-side `Mesh::CheckShearingBoxRefinement` (Phase 0c), whose graded-refinement cap also keeps 2:1 balancing from ever refining the boundary columns.

9.2 Dynamic validation: swing with a prescribed moving ring

The swing pgen gains the shwave pgen's prescribed moving-ring criterion (`amr_moving_ring`: refine MeshBlocks overlapping $[x_c(t) - h_w,\, x_c(t) + h_w]$, $x_c(t) = x_{\rm amp}\sin(\omega_r t)$, derefine all others); input `tests/swing_selfgrav_amr.athinput` sweeps a thin ring ($x_{\rm amp} = \pi/4$, $h_w = \pi/16$: one to two block columns) through the swing. The full-length run creates 392 and destroys 280 MeshBlocks — a remesh roughly every fifth cycle, each one exercising the shear-plane/octet rebuild — with no multigrid warnings:

	peak $d_{\cos}/A$	vs. static uniform	norm. traj. dev.
moving-ring AMR, gravity on	-3.510989	-0.01%	2.9×10^{-2}
moving-ring AMR, gravity off	$+1.7403$ (peak $\lvert d\rvert$)	$+0.38\%$	5.1×10^{-2}

The thin transient ring adds negligible net resolution, so the gravity-on peak lands on the uniform value; the gravity-off control again *dominates* both the peak shift and the trajectory deviation — several hundred remesh events add nothing beyond hydro regridding truncation.

One metric changes character under AMR and deserves its attribution: the phase lock $\max|d_{\sin}|/A$ is 9.9×10^{-5} (gravity on) — nine decades above the 10^{-13} of every static-mesh run. This is *regridding* truncation, not a shear or gravity error: each remesh prolongates the wave onto newly created fine blocks, and second-order interpolation of a cosine injects a sine component at truncation level. The gravity-off control shows the same signature at 3.7×10^{-5} — the ratio matching the amplified wave's larger amplitude — so the sheared multigrid machinery preserves phase exactly as well as the hydro does.

9.3 The fixed-ring limit

With $x_{\rm amp} = 0$ and $h_w = \pi/2$ the criterion reproduces the static-SMR ring footprint and never remeshes after the initial refinement — but *bit*-identity with the §7.5 run is structurally impossible: user-criterion AMR starts on the unrefined root mesh and refines within the first `refinement_interval` cycles, so the first few steps run at the coarse CFL. After that transient the two runs are the same

computation: the meshes are identical block-for-block, the final density fields agree to 4.8×10^{-7} (the float32 snapshot floor is 10^{-7}; the wave amplitude 3.5×10^{-4}), the peaks differ by 1.2×10^{-4} relative, and — decisively — the phase lock returns to 4×10^{-14}: with no mid-run remesh the machine-level $d_{\sin}$ of the static runs is recovered, closing the regridding attribution of §9.2 from the other side.

9.4 Multi-rank

The 2-rank moving-ring run reproduces the serial one *to every printed digit*: identical MeshBlock-cycle count (identical mesh evolution), identical amplification history (maximum trajectory difference exactly zero at history precision), identical peak and $d_{\sin}$. The remesh path rebuilds the planes and offsets on every rank from replicated mesh metadata, the octet fill is replicated host work, and the block-plane Allreduce is exact by construction (Phase 1) — so rank count drops out of the entire AMR + shear + gravity loop. This also exercises the hydro-side AMR shear-state rebuild (`ShearingBox::ReinitAfterMeshUpdate`) under MPI, which Phase 0c had written but never tested multi-rank.

9.5 Summary

Check	Metric	Result
Moving-ring AMR swing	MBs created/destroyed, MG warnings	392/280, none
Peak vs. static uniform	gravity on	-0.01%
Gravity-off control	peak shift / traj. dev. (must dominate)	$+0.38\%$ / 5.1×10^{-2}
Phase $d_{\sin}$ under AMR	regridding truncation (off-control ratio)	amplitude-scaled
Fixed-ring limit vs. static SMR	final ρ / peak / $d_{\sin}$	5×10^{-7} / 1.2×10^{-4} / 4×10^{-14}
Multi-rank np = 2 AMR	history vs. serial	identical, all digits

Next: §11 pushes the machinery into the gravitationally unstable regime; the Gammie (2001) gravitoturbulence program (cooling, Q/α diagnostics) follows.

10 The fine-ghost fix: a converging force at level jumps (November 2026)

This section concerns the multigrid solver on *any* refined mesh, shearing box or not. Until November 2026 the gravitational force did not converge within a few cells of a level jump; the cause was a halved transverse slope in the fine ghost cells, and the fix makes the potential second-order and the force first-order accurate there. An input option, `mg_fc_interp = quadratic`, then makes the force second order too (§10.5). Every refined-mesh potential in Phases 2–4 of this document was computed before the fix; §10.4 measures how much that matters.

10.1 The trigger: a solver that converges where ours did not

R. Li's multigrid on the `amrSG_dev` branch of `athenak-dust` (a composite finite-volume operator solved by FGMRES) passes his `self_gravity_wave` Poisson test — $\rho = 1 + 0.1\, S \cos(2\pi(2x + y) + 0.37)$ on the periodic unit square, $4\pi G = 1$, a level-1 square over $[0.25, 0.75]^2$, 4-cell blocks, root $N = 16$–256, S the cell average of the cosine — with the maximum force error in the cells touching a level jump falling from 2.7×10^{-4} to 1.2×10^{-6}, about $4\times$ per doubling (4% to 0.02% of the peak

force $0.1/|k| = 7.1 \times 10^{-3}$). We had measured the opposite in September 2026 in a 3-D test (a $\sim$9% relative error flat from 32^3 to 128^3), and had attributed it to the conservative level-boundary stencil, as §7.4 did for the potential.

The same test on our solver. The problem generator `mg_interface_wave` (`src/pgen/tests/mg_interface_wave.cpp`, on `fargo-amr-mhd`) is a port of his `poisson` mode: the same density, exact potential $-4\pi G\rho_0 A\cos\theta/k^2$, gauge and definition of interface cells (the first or last active cell of a block on a face whose neighbor is on another level). Our multigrid needs logically cubic 3-D blocks, so the 2-D problem is run as a periodic slab one 4^3 block thick ($x_3 \in [0, 4/N]$; the solution does not depend on x_3). The force is the one the gas feels, the centred difference of Φ through the ghost zones the solver fills (the routine `AddSelfGravity` of `SourceTerms`). The solves are converged (30 and 80 fixed V-cycles agree to 7 digits). Before the fix the interface force error was 1.0×10^{-3}, 8.1×10^{-4}, 7.4×10^{-4}, 7.1×10^{-4}, 7.0×10^{-4} for $N = 16$–256: about 10% of the peak force and not converging, so that the refined mesh as a whole was *worse* than the uniform root grid (table below).

10.2 The cause

At a face whose neighbor is coarser, the routine `ProlongateFCMG` of `MultigridBoundaryValues`, in `src/multigrid/multigrid_bvals.cpp`, fills each fine ghost cell from the coarse value c across the face and the first fine cell u_f,

$$u_g = \tfrac{1}{3}\left[2\left(c \pm g_t \pm g_s\right) + u_f\right], \tag{11}$$

where g_t, g_s are the offsets of the fine ghost's centre from the coarse centre in the two transverse directions: a quarter of a coarse cell times the transverse slope, $g_t = \frac{1}{8}(c_{t+1} - c_{t-1})$ for a centred difference. The code clamped that difference one-sided at *both* ends of the coarse patch facing the fine block ($c_{t-1} \to c_t$ at the low end, $c_{t+1} \to c_t$ at the high end) and kept the centred weight $\frac{1}{8}$, so at every clamped cell the offset was half of what it should be. The resulting ghost error, $\frac{2}{3}\cdot\frac{h_f}{4}\,\partial_t\Phi = h_f\,\partial_t\Phi/6$, is $O(h)$, and the one-cell difference of the force turns it into an $O(1)$ error. The $\pm$ pairs cancel in the face flux, so the operator stayed conservative and the defect still converged to round-off; the error showed up only as a sign-alternating flux perturbation along the interface, an $O(h)$ potential error decaying over a few cells, which is everything measured in September. With 4-cell blocks every ghost cell at a jump is clamped; with 16-cell blocks only the edges of each coarse patch are.

The fix. The clamp threw data away: the coarse buffer received from the coarser neighbor already holds `ngh` cells beyond the patch on the side of the parent's interior (the `icoar` receive extents grow by `ngh` toward that side, chosen by the fine block's child parity, which is also the slot index (f_1, f_2) of a coarser neighbor). The slope is now centred wherever that halo exists and one-sided only at the genuine outer edge, with the correct one-cell weight:

$$g_t = \tfrac{1}{4}\,\frac{c_{t^+} - c_{t^-}}{t^+ - t^-}, \qquad t^- = t{-}1 \text{ unless } (t = t_{\rm lo}\ \&\ f = 0), \quad t^+ = t{+}1 \text{ unless } (t = t_{\rm hi}\ \&\ f = 1). \tag{12}$$

About fifteen lines in the three face branches of `ProlongateFCMG`; uniform meshes do not take this path and are unchanged. Commits `d86c16ac` (the fix) and `57a70d8d` (the test problem) on branch `fargo-amr-mhd`; the fix is on `dust-multigrid` as `e1e0a072` (cherry-picked; the Phase-2a test above gives the same numbers to every printed digit there). Implementation details: `multigrid_sb_implementation.pdf`, §"Phase 2d".

10.3 Results

Li's Poisson test. Maximum force error in the interface cells, its fraction of the peak force, the volume-weighted relative L_2 error of the force over the whole mesh, and the same L_2 error on the uniform root grid:

	interface max, before		interface max, after		global L_2 (relative)		
N	abs.	of peak	abs.	of peak	before	after	uniform
16	1.0×10^{-3}	14%	6.0×10^{-4}	8.5%	7.4×10^{-2}	7.1×10^{-2}	8.0×10^{-2}
32	8.1×10^{-4}	11%	2.6×10^{-4}	3.7%	2.2×10^{-2}	1.8×10^{-2}	2.0×10^{-2}
64	7.4×10^{-4}	10%	1.2×10^{-4}	1.8%	9.3×10^{-3}	4.6×10^{-3}	5.1×10^{-3}
128	7.1×10^{-4}	10%	6.1×10^{-5}	0.86%	5.7×10^{-3}	1.2×10^{-3}	1.3×10^{-3}
256	7.0×10^{-4}	9.9%	3.0×10^{-5}	0.43%	3.9×10^{-3}	3.1×10^{-4}	3.2×10^{-4}

After the fix the interface force converges at first order (2.0–2.3× per doubling) and the potential at second order (maximum error $1.5 \times 10^{-5} \to 7.1 \times 10^{-8}$, 4× per doubling; before, first order), and the refined mesh is as accurate as or more accurate than the uniform root grid everywhere. With 8^3 blocks ($x_3 \in [0, 8/N]$) the interface error is 1.9×10^{-4}, 1.2×10^{-4}, 6.0×10^{-5}, 3.0×10^{-5} for $N = 32$–256 (the same 0.43% at 256). Li's solver is still 25× better at $N = 256$ (1.2×10^{-6}): it is second order at the interface. Runs, logs and commands: `validation/run/mg_interface/` of the `fargo-amr` worktree.

This document's own interface test. The Phase-2a static shearing-wave test (§7.4; 8^3, 16^3, 32^3 blocks at 32^3, 64^3, 128^3, fixed 30 V-cycles), rerun with the code before the fix reproduces that table to every printed digit; after the fix (relative to the root-grid closed form, eq. 10):

	qomt = 0				generic phase $t_0 = 0.37$			
	max rel.		L_2 rel.		max rel.		L_2 rel.	
	before	after	before	after	before	after	before	after
32^3	1.18×10^{-2}	3.35×10^{-3}	5.05×10^{-3}	3.73×10^{-3}	1.44×10^{-2}	6.57×10^{-3}	8.14×10^{-3}	7.13×10^{-3}
64^3	5.51×10^{-3}	8.19×10^{-4}	1.28×10^{-3}	9.27×10^{-4}	5.66×10^{-3}	1.61×10^{-3}	2.06×10^{-3}	1.78×10^{-3}
128^3	2.66×10^{-3}	2.03×10^{-4}	3.25×10^{-4}	2.31×10^{-4}	2.68×10^{-3}	4.02×10^{-4}	5.18×10^{-4}	4.45×10^{-4}

The interface peak, first order before (2.1× and 2.1× per doubling), is now second order (4.1×, 4.0× at qomt = 0) and no longer stands above the L_2 error: what remains is mostly the fine region being more accurate than the root-grid reference (the 6×10^{-4} eigenvalue plateau at 64^3). The "known local property of the mass-conserving level-boundary stencil" of §7.4 was, for the most part, this bug.

10.4 Regression

Each case run with the committed code before the fix and with the fix (binaries `athena_before`, `athena_after`; runs in `validation/run/mg_fixA_regress/` of the `fargo-amr` worktree):

case	before	after
uniform swing, 128^2 (`swing_selfgrav_smr`, refinement off)	history bit-identical	
uniform Poisson, $N = 64$ (`mg_interface_wave`)	identical to all printed digits	
swing, static ring: peak $d_{\cos}/A$ (vs uniform 256^2)	-3.583481 (-0.976%)	-3.584414 (-0.950%)
swing, static patch	-3.532422 (-2.387%)	-3.533085 (-2.369%)
swing, moving-ring AMR	-3.510989 (-2.979%)	-3.511909 (-2.954%)
MHD swing, static ring, $B_0 = 1$: c_y vs theory, $t \le 6$	2.66%	2.65%
slab sheet on SMR (§12.3): max rel. error	8.7269×10^{-2}	8.7270×10^{-2}
refined Poisson $N = 64$, 1 vs 4 ranks	—	identical to all printed digits
swing ring, 4 vs 1 rank: $\max\lvert\Delta d_{\cos}\rvert / \max\lvert d_{\cos}\rvert$	1.0×10^{-12}	4.8×10^{-13}

(The rank dependence of the dynamic swing run is round-off that predates the fix; the static solves are rank-independent.) The refined swing runs move toward the uniform 256^2 answer by 0.02–0.03% of the peak (the swing is smooth on the scale of a block, where the error was smallest); the phase lock ($\max|d_{\sin}|/A \approx 10^{-13}$ on the static meshes) is unchanged. The conclusions of Phases 2–4 stand; numbers that measure the interface itself (§7.4, the Phase-2b annulus maximum) were set by the bug.

10.5 Second order: quadratic level-jump ghosts (`mg_fc_interp = quadratic`)

With the slopes fixed, the interface force is first order because both ghosts are linear interpolations in the direction normal to the jump: the fine ghost $\frac{1}{3}(2c+u_f)$ and the coarse-side ghost $\frac{1}{3}(4\bar{u}-u)$. The input option `<gravity> mg_fc_interp = quadratic` replaces them with a quadratic pair (default `linear`, bit-identical to the code above; implementation: `multigrid_sb_implementation.pdf`, §"Phase 2e"). Normal to the jump (fine cells of width h at $h/2$ and $3h/2$, coarse cell of width $2h$ at $-h$), the fine ghost at $-h/2$ is the quadratic through the coarse value and the first two fine cells,

$$g_f = \tfrac{8}{15}\,\tilde{c} + \tfrac{2}{3}\,f_1 - \tfrac{1}{5}\,f_2, \tag{13}$$

with $\tilde{c}$ the coarse layer evaluated at the fine ghost's transverse position from a quadratic fit on 3×3 coarse cells (slopes, second differences δ^2 and the mixed term, which averages out over the four fine ghosts but is needed pointwise). Flux matching (the coarse face flux equals the sum of the four fine fluxes) then fixes the coarse ghost,

$$g_c = -\tfrac{1}{15}\,c_1 + \tfrac{2}{3}\,A_1 + \tfrac{2}{5}\,A_2 - \tfrac{1}{30}\,(\delta_a^2 + \delta_b^2), \tag{14}$$

with $A_{1,2}$ the 2×2 averages of the first and second fine layers; it is also the quadratic interpolant at the coarse ghost's centre, and the δ^2 term removes the $O(h^2)$ transverse bias of the face averages. Both sides use the same transverse stencils (one-sided only at the coarse neighbor's block edges), so the operator stays exactly conservative. The coarse side needs A_1 and A_2 only as $\frac{2}{3}A_1 + \frac{2}{5}A_2$, which is packed into the slot that carried A_1: no buffer, index or MPI-message change. The quadratic pair is used on the finest multigrid level only; the coarser block levels, octets and levels with fewer than 4 cells per block keep the linear pair (they set the convergence rate, not the answer). Commits: `aad3deb7`, `917b93bc`, `5808b2fd` on `dust-multigrid`; `ddae4ca6`, `64a9e22e`, `c9eb45e2` on `fargo-amr-mhd`, with the test inputs and notes (`validation/mg_fc_quadratic_notes.md` there).

Li's Poisson test. Maximum force error in the interface cells (4 ranks, 40 FMG V-cycles; numbers in parentheses: convergence order from the previous row):

N	slab, 4^3 blocks linear	slab, 4^3 blocks quadratic	slab, 8^3 blocks quadratic	refined cube, 8^3 blocks linear	refined cube, 8^3 blocks quadratic
16	6.0×10^{-4}	5.77×10^{-4}	—	—	—
32	2.6×10^{-4}	1.52×10^{-4} (1.92)	1.51×10^{-4}	4.57×10^{-4}	1.43×10^{-4}
64	1.2×10^{-4}	3.93×10^{-5} (1.95)	3.90×10^{-5} (1.96)	2.23×10^{-4}	3.60×10^{-5} (1.99)
128	6.1×10^{-5}	9.87×10^{-6} (1.99)	9.76×10^{-6} (2.00)	1.10×10^{-4}	9.10×10^{-6} (1.98)
256	3.0×10^{-5}	2.49×10^{-6} (1.98)	2.47×10^{-6} (1.98)	—	—

The interface force now converges at second order (0.035% of the peak force at $N = 256$, from 0.43% with the linear pair), as in Li's solver ($2.7 \times 10^{-4} \to 1.2 \times 10^{-6}$; ours $5.8 \times 10^{-4} \to 2.5 \times 10^{-6}$, the same factor of $\sim$230 over four doublings; our interface L_2 at 256, 1.37×10^{-6}, is close to his maximum). The refined cube (`inputs/tests/mg_interface_wave_cube.athinput` on `fargo-amr-mhd`: $[0.25, 0.75]^3$ in a periodic unit cube, wave $(2, 1, 1)$) exercises all six faces and the mixed transverse term; 16^3 blocks and a nested level-2 cube give the same numbers. What remains at the interface is the ordinary second-order error of the coarse level: the refined mesh's global force error is now below the uniform root grid's at every N (2.80×10^{-4} vs 3.20×10^{-4} relative L_2 at 256), and the potential is second order everywhere.

The other interface tests. The Phase-2a shearing-wave test (§7.4; reference = the root-grid discrete solution, so the fine region's own accuracy is not measured): maximum relative error 3.35×10^{-3}, 8.19×10^{-4}, 2.03×10^{-4} (linear) $\to$ 2.64×10^{-3}, 6.67×10^{-4}, 1.67×10^{-4} (quadratic) at 32^3–128^3, qomt = 0; 24–26% lower at the generic phase. The slab-open sheet on SMR (§12.3): 1.84×10^{-1}, 8.73×10^{-2}, $4.24 \times 10^{-2} \to 1.64 \times 10^{-2}$, 4.56×10^{-3}, 1.20×10^{-3} — the quadratic ghosts are exact for the background's $\Phi \propto z^2$, so the constant $k_\perp = 0$ offset $\frac{3}{4}(4\pi G\rho_0)h_f^2$ vanishes and the error equals the pure-mode sector's (19$\times$ lower at 64^3, second order).

Convergence and regressions. The composite operator is nonsymmetric in quadratic mode. With the quadratic pair on the finest level only, the red-black V-cycle contracts as with the linear pair (defect reduction per V-cycle, quadratic vs linear: 0.19 vs 0.19 on the 64^3 cube, 0.23 vs 0.23 at 128^3, 0.19 vs 0.18 on the sheared shearing-wave test) and reaches round-off in the same number of cycles; quadratic ghosts on every level (`mg_fc_interp = quadratic_all_levels`, kept as a diagnostic) give the same answer but an alternating contraction up to 0.45. FMG, plain V-cycles and the stagnation exit agree to 10 digits; 1 vs 4 ranks agree to the last printed digit. Linear mode is bit-identical to the code before the option (all ladder, cube and shearing-wave outputs; the refined swing history byte for byte); uniform meshes are identical in both modes. Dynamic runs barely move: refined swing peaks $-3.584414 \to -3.584413$ (ring) and $-3.533085 \to -3.533080$ (patch), the AMR moving ring 4×10^{-7} of the amplitude, the MHD ring swing 1.9×10^{-6} with $\nabla \cdot B$ unchanged — there the refined-vs-uniform difference is set by the hydrodynamics. The benefit is in the force at the jumps themselves and in the $k_\perp = 0$ sector. On `dust-multigrid` (`5808b2fd`) the shearing-wave test at 64^3 reproduces the same numbers (linear identical to `e1e0a072`, quadratic 6.667×10^{-4}). Not yet tested: GPU builds, restarts, dust self-gravity runs. **Recommendation:** use `mg_fc_interp = quadratic` where the self-gravity force at level jumps matters (collapse and fragmentation with AMR, dust self-gravity on refined meshes); it costs no extra V-cycles or communication. The option must be written in the input file (a command-line override cannot add a new parameter).

10.6 What remains

- *The default is still linear* (first order at the interface), for bit-for-bit continuity of existing runs; switching it is a one-line change once production setups have been compared.
- *Nonsymmetric operator in quadratic mode.* Fine for the red-black FAS V-cycle here, but any future symmetric Krylov wrapper (CG) would need care.
- *Edge and corner neighbors* at a coarser level still take simple injection; they do not enter the face flux and did not show in any test here.
- The unused `FillFineCoarseMGGhosts` carries the old clamp; it is not in any task list.

11 Into the unstable regime: the swing-run compendium and first collapse

11.1 All swing runs at a glance

Every swing-amplification run of this program shares the base configuration: $128^2\times16$ cells (16^3 MeshBlocks), $L_x{=}L_y{=}2\pi$, $L_z{=}\pi/4$, isothermal $c_s{=}1$, $\Omega{=}1$, $q{=}3/2$ ($\kappa{=}1$), $\rho_0{=}1$, single leading wave $(n_{wx}, n_{wy}) = (-6, 1)$, FARGO on, multigrid gravity with FMG + `niteration=10` unless noted. The gravest box modes obey $\omega^2 = \kappa^2 + c_s^2 - 4\pi G\rho_0 = 2 - 4\pi G$, so $4\pi G = 2$ is the box-scale Jeans threshold.

run	mesh	$4\pi G$	amp	$t_{\rm end}$	outcome	peak
validation ladder (linear, stable): §6.3, §7.5, §8.4, §9.2						
P1 uniform (MG ≡ FFT)	uniform	1.9	10^{-4}	8	swing	−3.511418
P2a ring SMR (+twins, off)	6/8-col ring	1.9	10^{-4}	8	swing	−3.583481
P2b annuli (+off)	4-col annuli	1.9	10^{-4}	8	swing	−3.550196
P2c AMR ring / fixed (+off, np2)	moving/static	1.9	10^{-4}	8	swing	−3.5110/−3.5839
unstable window (linear)						
window super-swing	uniform	2.1	10^{-4}	8	swing, $\omega^2{<}0$ in $\lvert k_x\rvert{<}0.32$	−4.4569
nonlinear ladder (max ρ/ρ_0 in the peak column)						
bounce I	uniform	3.0	0.05	10	crest bounces	1.37
bounce II	uniform	3.0	0.10	16	crest bounces	1.73
marginal miss	uniform	5.0	0.05	10	crest $\lambda_J{\approx}$ width	2.70
collapse	uniform	**6.0**	**0.20**	10	**bound clump**	**203**
collapse, AMR zoom	density rings	6.0	0.20	10	§11.3	§11.3

Figure 7 completes the linear grid: uniform and moving-ring AMR at each of $4\pi G = 1.9/2.1/3.0$ (amp $= 10^{-4}$) against the linearized shearing-sheet ODE. Peaks: $-3.51/-3.51$ (theory -3.61), $-4.46/|4.21|$ (theory -4.52), and $-8.53/-8.26$ (theory -8.02) — the amplification nearly doubles between 1.9 and 3.0 as the $\omega^2 < 0$ window widens, the AMR runs track their uniform twins to 0.1–0.4 in $d_{\rm cos}/A$ (hydro-truncation level), and at $4\pi G = 3.0$ the growing spread between simulation and continuum theory is the 128^2 resolution limit amplified through the window, with the AMR curve between the two.

11.2 What it takes to clump

The ladder teaches three lessons (Fig. 8). First, $4\pi G > 2$ alone does not clump: the winding wave only *transits* its unstable window and bounces, and its quadratic self-coupling deposits only

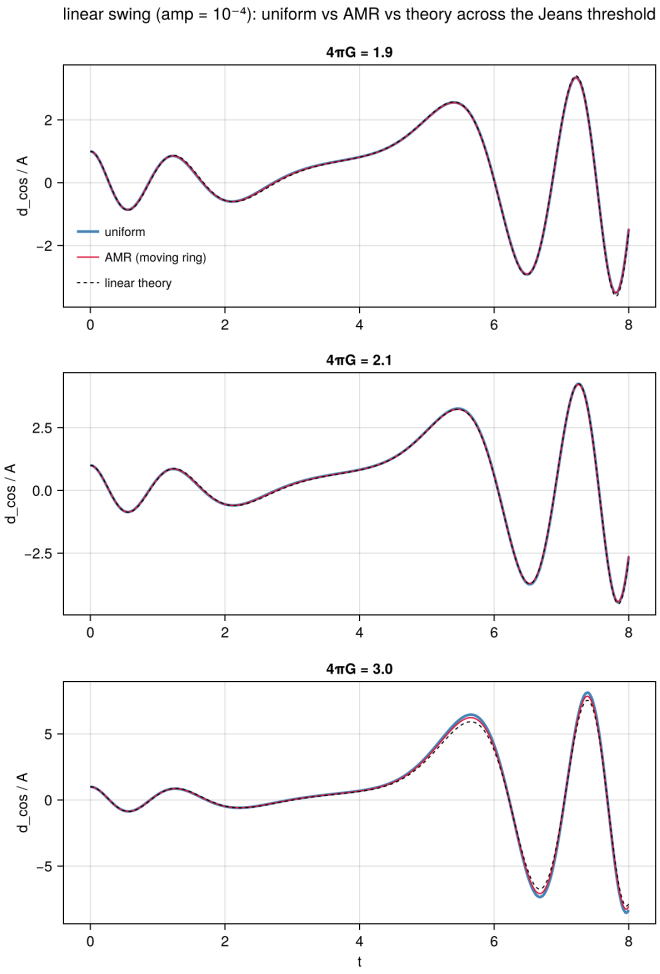


Figure 7: The linear swing grid: uniform (blue) vs. moving-ring AMR (red) vs. the linearized ODE (dashed) at $4\pi G = 1.9$, 2.1, 3.0, all at amp $= 10^{-4}$.

a whisper on the non-winding axisymmetric $(k_x, k_y) = (\pm 1, 0)$ mode (the $k_y = 0$ beat carries $k_x = 2k_x(t)$, which sweeps past the discrete mode instead of parking on it) — the $4\pi G = 3$ runs stay bounded even for 16 time units. Second, collapse requires the *crest itself* to become Jeans-unstable during the window: the local $\lambda_J = 2\pi c_s/\sqrt{4\pi G\rho}$ must drop below the crest width. At $4\pi G = 5$, amp $= 0.05$ this misses by $\sim$15% (peak $2.70\rho_0$, re-expands); at $4\pi G = 6$, amp $= 0.2$ it tips over: fragmentation at $t \approx 4.5$, runaway to $\rho = 203\rho_0$, and a bound clump that persists to the end of the run. Third, resolution: the uniform-grid Truelove ceiling ($\lambda_J \geq 4\Delta x$) at $4\pi G = 6$ is $\rho \approx 171\rho_0$ — the collapsed clump *exceeds* it, so the uniform run is marginally under-resolved precisely where it matters.

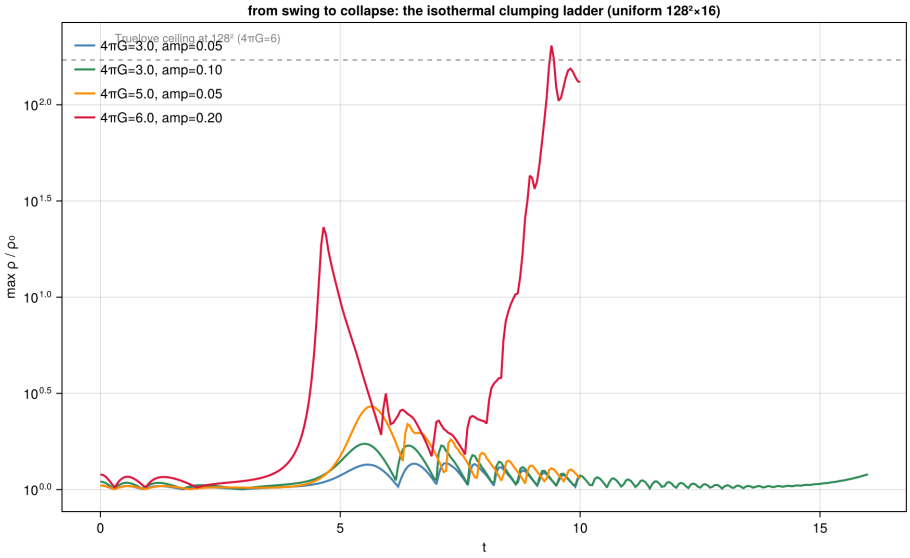


Figure 8: Maximum density vs. time for the nonlinear ladder: two bounces, a marginal miss, and the collapse, with the 128^2 Truelove ceiling marked.

11.3 Zooming on the clump under FARGO: rings, not patches

The clump is local, but with orbital advection on, the Phase-0 annular policy requires refinement in complete x_2 rings — so the AMR "zoom" refines the clump's x_1 columns at all y: a density criterion (`min_max` on `hydro_u_d`, `value_max=2`) flags the clump's blocks and the ring-sync completes the rings automatically. This is cheap here (one ring = 16 root blocks $\rightarrow$ 128 fine) and is what lets FARGO tolerate the clump's azimuthal drift. True local patches are legal only with `orbital_advection=false` at the measured $\sim$5.7$\times$ timestep cost; per-ring bookkeeping for partially refined rings would be the Phase-3 alternative if deep zooms demand it. Input `tests/swing_selfgrav_clump_amr.athinput` runs the collapse with one refinement level chasing the crest and clump (level-1 Truelove ceiling 4$\times$ higher, $\approx 684\rho_0$); the notebook (§15) compares the final density fields and mesh block structure of the uniform and AMR runs. The full run creates 1344 and destroys 1232 MeshBlocks — the rings form on the crest at $t \approx 4$, follow the fragmentation, and contract to a single ring tracking the clump (176 blocks at the end, 128 at level 1) — with zero multigrid warnings through the entire collapse. The density comparison carries the resolution lesson (Fig. 10): the uniform run peaks at $203\rho_0$, *above* its own Truelove ceiling ($171\rho_0$), while the AMR run peaks at $184\rho_0$ at $t = 9.9$ — comfortably inside its level-1 ceiling ($684\rho_0$) — so the uniform

excess is under-resolution overshoot and only the AMR number is trustworthy. Both runs agree on formation time, location, and bulk structure of the *primary* clump. One open observation: the AMR run forms a *second* clump straddling the periodic y boundary (peak at $y = -3.105 \equiv +\pi$), absent in the uniform run — at the same x and separated by $\Delta y = \pi$, the wavelength of the k_y=2 second harmonic, which at $4\pi G = 6$ is itself Jeans-unstable ($\omega^2 = -1$) and quadratically seeded by the primary. This points to real fragmentation that the refined run resolves and the uniform grid merges, not a boundary artifact (the y wrap is plain periodic and translation-invariant). The 256^2 uniform discriminator run **settles it**: at matched fine resolution the same two-site structure appears — and an **FFT-solver twin** (single-rank; the FFT solver's scope) reproduces it *across solvers*: linear-phase $d_{\cos}$ agreement at 5×10^{-8}, and at t=10 the same two fragments within a third of a cell in position (boundary knot: FFT 41.9ρ_0 at $(-0.01, -2.86)$ vs. multigrid 40.4ρ_0 at $(-0.01, -2.88)$; companion 23.7 vs. 23.4) after five nonlinear dynamical times. Extended to t=14, the FFT twin also goes winner-takes-all at the same boundary site with the companion's site evacuated, and its runaway *clock* matches multigrid to 0.05 time units (the ρ=100 crossings); the two runs then part ways only *beyond* the Truelove ceiling — FFT saturates near it ($\sim$600ρ_0, a grid-scale bounce) while multigrid shoots to $2.6 \times 10^5 \rho_0$ — a clean, solver-based demonstration that past-ceiling densities are resolution artifacts in either code (Fig. 11) — — the dominant knot straddling $y = \pm\pi$ at $x \approx 0$ plus a center-side companion — confirming the k_y=2-harmonic fragmentation as physical and the 128^2 single clump as the under-resolved outlier. The properly resolved runs also collapse *later* (peak 40ρ_0 at t=10, still contracting, vs. the 128^2 run's overshot 131): under-resolution both merges fragments and accelerates collapse. Extending the 256^2 run to t=14 at 64 ranks (the run that exposed and then validated the shear-exchange consistency fix) supplies the endpoint: the dominant knot undergoes full isothermal runaway to $2.6 \times 10^5 \rho_0$ — 375× past its Truelove ceiling, deep into the unresolved regime — while consuming the center companion entirely (its site is a 0.4ρ_0 void at the end). Absent sink particles or a floor, that runaway is the physical endpoint of isothermal collapse and the natural stopping criterion for uniform-grid studies; the multigrid solver held machine-clean through the entire quarter-million density contrast, ending exactly on a shear-periodic instant with zero warnings. The production lesson stands: chase fragmentation with AMR, and stop (or sink) at the Truelove ceiling. Figure 9 shows the three runs' winding-wave amplitudes $d_{\cos}/A$ — they share the linear swing and part ways at fragmentation ($t \approx 4.5$), where collapse hands energy to other harmonics — and Figure 10 their final states with the MeshBlock structure.

11.4 Why $d_{\cos} = \delta$: the projection diagnostic

The history diagnostic used throughout the swing program,

$$d_{\cos}(t) = \frac{2}{V} \int \left(\frac{\rho}{\rho_0} - 1 \right) \cos\big(\mathbf{k}(t)\cdot\mathbf{x}\big)\, dV, \qquad \mathbf{k}(t) = \big(k_{x0} + q\Omega k_y t,\ k_y,\ 0\big), \tag{15}$$

equals the wave's density amplitude $\delta(t)$ exactly in the linear regime. Three steps:

(i) Linear disturbances wind. Linearizing about $\mathbf{v}_0 = -q\Omega x\, \hat{y}$ and passing to sheared coordinates $y' = y + q\Omega t\, x$ removes the background's explicit x-dependence, so plane waves in (x, y') remain plane waves for all time; mapped back, $e^{i(k_{x0}x + k_y y')} = e^{i\,\mathbf{k}(t)\cdot\mathbf{x}}$ with the winding $k_x(t)$ above. A single shwave is therefore exactly $\rho/\rho_0 = 1 + \delta(t)\cos\theta + B(t)\sin\theta$, $\theta \equiv \mathbf{k}(t)\cdot\mathbf{x}$.

(ii) The cosine channel is closed, and $B \equiv 0$. Differentiating $\delta\cos\theta$ at fixed $\mathbf{x}$ produces an awkward term $-\delta\,(q\Omega k_y x)\sin\theta$ from the winding of k_x; the background advection $\mathbf{v}_0\cdot\nabla\rho_1 = +\,q\Omega k_y x\,\delta\sin\theta$ cancels it *exactly* — the winding wavevector is precisely the choice that lets the

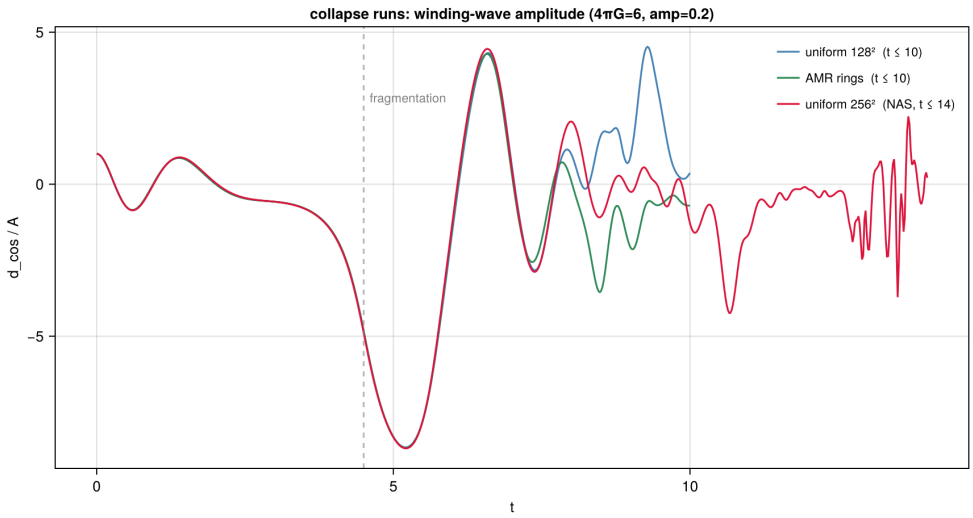


Figure 9: Winding-wave amplitude $d_{\cos}/A$ for the three collapse runs. The projection tracks only the $k(t)$ wave component, so the departure from the common curve is a clock for when fragmentation takes over.

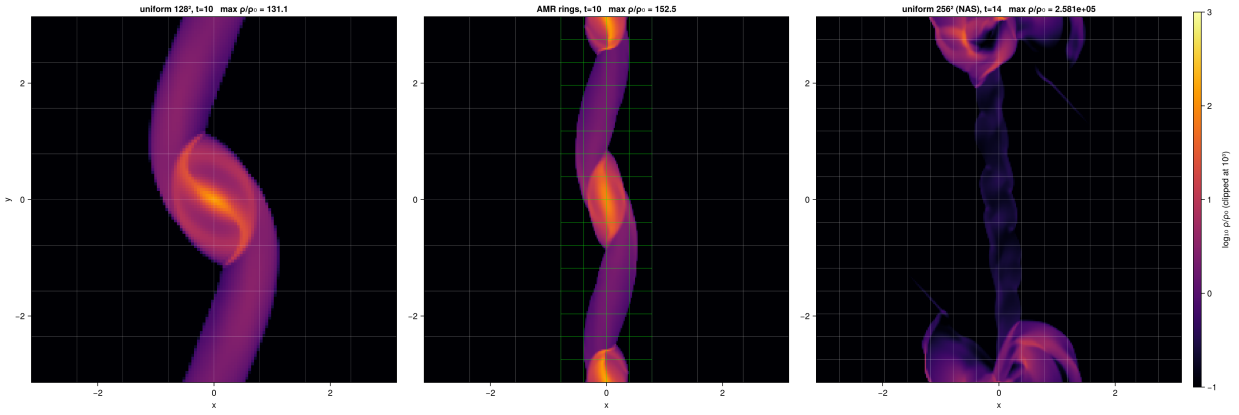


Figure 10: Final states of the three collapse runs ($\log_{10}\rho$, clipped at 10^3; MeshBlock outlines drawn, green = level 1): uniform 128^2 and the AMR run at $t = 10$, and the 256^2 NAS run at $t = 14$, deep in the winner-takes-all runaway with the companion's site evacuated.

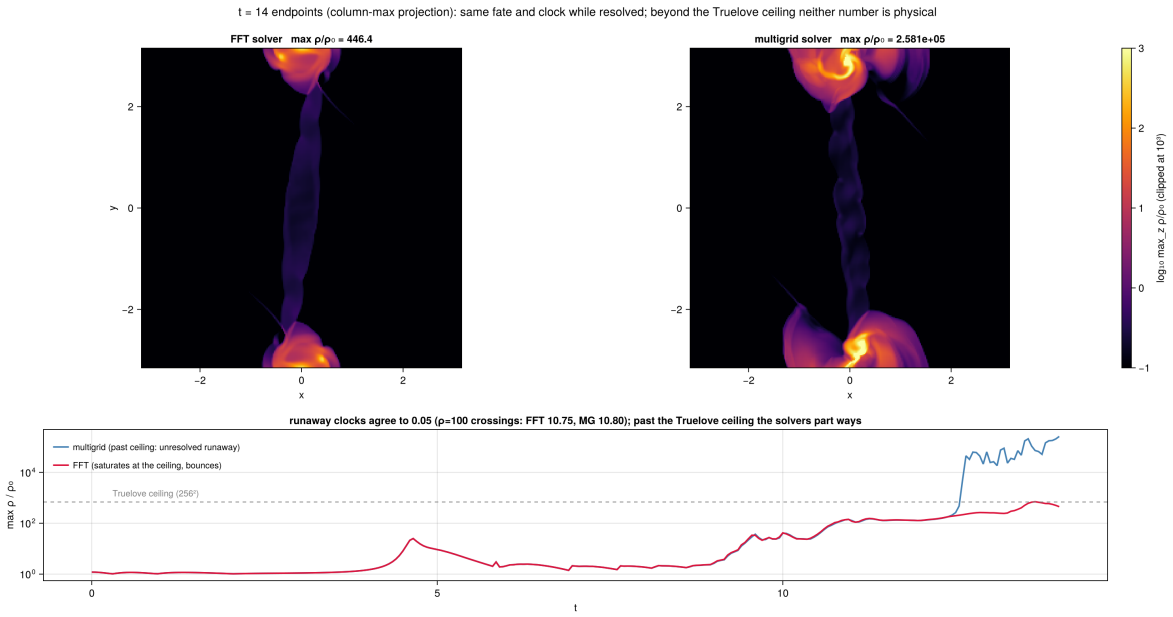


Figure 11: $t = 14$ endpoints of the 256^2 collapse under the two solvers (density panels are column-maximum projections over z — a mid-plane slice hides the multigrid core, which sits off-plane at $z = -0.21$): same fate (boundary knot wins, companion consumed) and runaway clocks agreeing to 0.05 while resolved; beyond the Truelove ceiling (dashed) the solvers diverge qualitatively — neither number is physical there.

background shear advect the phase. What survives is the algebraic system quoted in the pgen header,

$$\dot{\delta} = -(k_x w_x + k_y w_y), \quad \dot{w}_x = 2\Omega w_y + k_x C\delta, \quad \dot{w}_y = -(2-q)\Omega w_x + k_y C\delta, \quad C = c_s^2 + \frac{4\pi G\rho_0}{D}, \tag{16}$$

for the phase assignment $\rho_1 \propto \cos\theta$, $(v_x, v_y) \propto \sin\theta$ (D the discrete Poisson eigenvalue, $\to -k^2$ in the continuum). This set is *closed*: cosine density couples only to sine velocities and back. The initial data ($\rho_1 = A\cos\theta$, $\mathbf{v}_1 = 0$) lies inside it, so the orthogonal channel is never excited and $B(t) = 0$ identically — which is why $\max|d_{\sin}|/A \sim 10^{-13}$ is such a sharp phase diagnostic: theory keeps that channel exactly empty.

(iii) The projection is exact, including on the grid. Inserting the ansatz, $d_{\cos} = (2\delta/V)\int\cos^2\theta\, dV = (\delta/V)\int(1+\cos 2\theta)\, dV$. The $\cos 2\theta$ term dies by the y-integral alone: 2θ contains $2k_y y$ with k_y an integer box mode, so $\int_0^{L_y}\cos(\cdot + 2k_y y)\, dy = 0$ *regardless of the instantaneous, generally non-resonant value of* $k_x(t)$ — the subtle point; y-periodicity suffices. The same kills $\int\cos\theta\sin\theta\, dV$, making $d_{\cos}$ and $d_{\sin}$ independent. Hence $d_{\cos} = \delta(t)$, and normalizing by A gives the amplification factor. On a uniform grid the y-sum is $\sum_j e^{4\pi i n_y j/N_y} = 0$ exactly whenever $2n_y \not\equiv 0 \pmod{N_y}$, so the projection is exact on the grid too — the phase lock can reach machine precision rather than truncation level. Once a run goes nonlinear the field is no longer a single shwave and $d_{\cos}$ reverts to the amplitude of the $\mathbf{k}(t)$ Fourier component alone (the departure at fragmentation in Fig. 9).

12 Phase 3: the stratified box — slab-open multigrid boundaries and the Shi & Chiang configuration

Phases 0–2 lived in fully (shear-)periodic boxes: mean subtraction was mandatory and the mass sector of the potential was projected out by construction. Phase 3 adds vertical stratification — the configuration of Shi & Chiang (2014, ApJ 789, 34; SC14): a 3D shearing box with linearized vertical tidal gravity $-\rho\Omega^2 z$, self-gravity with *open (vacuum)* vertical boundaries, β-cooling, and outflow-type hydro boundaries in z. Implementation details live in the companion document (`multigrid_shear_implementation.pdf`, Phase 3 section); this section records the method in brief and the test results.

12.1 Method in brief

For each horizontal Fourier mode of the rolled frame the 7-point stencil has, in vacuum, the exact decaying solution $\mu^{|\Delta n|}$ in z, with $\mu = 2/(\alpha + \sqrt{\alpha^2 - 4})$, $\alpha = 2 - \lambda_\perp \Delta x_3^2$, and $\lambda_\perp$ the discrete horizontal eigenvalue at the shear-shifted radial wavenumber. Summing the corresponding infinite-vacuum discrete Green's function ($G = -\Delta x_3^2 \mu^{|\Delta n|}/\sqrt{\alpha^2 - 4}$ for $k_\perp \neq 0$; $G = +\Delta x_3^2 |\Delta n|/2$, the discrete isolated-slab potential, for the mass sector) against the density gives exact Dirichlet *face-value planes* on both x_3 faces, applied multipole-style ($u_{\rm ghost} = 2\Phi_{\rm face} - u_{\rm interior}$) at every multigrid level through a restricted plane pyramid with shear-remapped ghost rings. The converged interior solution is then exactly the restriction of the infinite-vacuum discrete solution — no boundary truncation error, and no free constant. Enabled by `<gravity> mg_bc = slab`; requires an FFT build (kokkos-fft) and root-level MeshBlocks on the x_3 faces (enforced, and honored automatically by the AMR criterion veto).

12.2 Static tests: the sheet as absolute arbiter

A new `profile = sheet` in the `fft_poisson` pgen places a uniform background ρ_0 plus one horizontal mode on a single z-layer; the discrete infinite-vacuum solution is closed-form in *both* sectors (μ-decay for the mode, the discrete parabola for the background), so the comparison is made **without mean subtraction** — constant included, no gauge freedom. On 64^3 with 16^3 MeshBlocks (`tests/mg_poisson_slab.athinput` and the `_shear` variant):

Configuration	max relative error	interpretation
periodic x_1, $q\Omega t_s$=0	1.2×10^{-12}	solver tolerance: planes, pyramid, block/root BC application all exact
MG vs FFT-KO09, same density	3.0×10^{-13}	independent open-boundary methods agree
shear-periodic, t_0=0	1.2×10^{-12}	identical to the periodic twin
shear-periodic, t_0=0.37	6.6×10^{-6}	PPMX roll/unroll remap order
MG vs FFT at t_0=0.37	8.8×10^{-8}	both roll; different pipeline points
4 MPI ranks, all three configs	bit-identical	the zero-fill + Allreduce gather

The sech^2-slab physics test (analytic $4\pi G\rho_0 H^2 \ln\cosh(z/H)$ potential) converges at second order: 5.8×10^{-3} / 1.5×10^{-3} / 3.7×10^{-4} at $32/64/128^3$, with MG-slab and FFT-open agreeing to every printed digit at $n = 64$.

12.3 Refined meshes, and what the mass sector revealed

With a level-1 slab spanning the middle half of z (`tests/mg_poisson_slab_smr.athinput`), the screened mode sector converges at second order (1.64×10^{-2} / 4.57×10^{-3} / 1.20×10^{-3} relative at $32/64/128^3$). The unscreened $k_\perp = 0$ sector — physical for the first time in this program — exposes a property of the Tomida & Stone conservative level interface that no periodic test could see: a *constant, horizontally uniform, force-free* potential offset of exactly $\frac{3}{4}(4\pi G\rho_0)h_f^2$ filling the enclosed refined region. The interface flux error is continuous across the boundary (zero monopole — the "conservative" in the formulation), so the residual truncation is a surface *dipole*, whose far field is a constant jump: $O(h^2)$, second-order convergent, and dynamically inert. Diagnosing it cost a detour worth recording: the sheet metric originally evaluated its reference at integer root-layer indices, adding an $O(\partial_z\Phi\,\Delta z)$ *reference* error on fine cells that masqueraded as a first-order solver error; the metric now samples the closed form at the true fine-cell centers.

12.4 Hydro-side pieces, and why the vertical boundary must be a diode

Phase 3a closed two hydro gaps: the vertical tidal force's *energy work term* $-\rho\Omega^2 z\, v_z$ (missing from all four shearing-box source branches; invisible in isothermal runs), and the SC14 cooling laws (`beta_cooling`: $\dot{U} = -U\Omega/\beta$; `thermal_cooling`: $t_{\rm cool} = b(\rho/P)^3$, verified to drain at the mass-weighted rate implied by the $(\rho/P)^3$ law).

The hydrostatic-hold test (SC14 initial state, no cooling, no perturbations) exposed a boundary trap: with plain `outflow` z-boundaries the halo density grew two orders of magnitude in $10\,\Omega^{-1}$ and the accreting fountain destroyed the disk by $t \sim 8$ — ghost-copy outflow happily feeds *inflow* at whatever velocity gravity establishes. With `diode` boundaries the same test holds: midplane profile unchanged over $10\,\Omega^{-1}$, disk-body velocities $\lesssim 2\%$ of $c_s(0)$, mass constant to 0.25% (floor injection in the halo). All Phase-3 inputs use `diode`.

Addendum (September 15, 2026): the diode leaks too, and `hse_outflow` closes it. The high-β SC14 runs later died of a slower version of the same fountain (`HANDOFF_ZBC.md`): once the halo lifts off the density floor, `diode` lets mass *enter* the box — the ghost is a flat reservoir at the edge density with $dP/dz = 0$ across the face, so the vertical gravity accelerates the edge cell inward and the Riemann solver draws mass in from a ghost that is regenerated every step; the clamp constrains the ghost *state*, not the *flux*. Booth & Clarke (2019) avoid it by extrapolating the ghost density along the hydrostatic profile at the edge cell's temperature, which is now the boundary flag `hse_outflow` (`src/bvals/physics/hydro_bcs.cpp`, x_3 faces): $\rho(z) = \rho_e \exp[-\Omega^2(z^2 - z_e^2)/(2a^2)]$ with $a^2 = P_e/\rho_e$, velocities copied with the normal one zeroed if inward, the energy rebuilt from a^2 and the ghost velocity, scalars scaled; it reduces to `diode` without a stratified shearing box and is applied to the fine and coarse arrays alike. In the hand-off's test box ($32 \times 32 \times 24$, $8 \times 8 \times 6$ H, $\beta = 80$, $150\,\Omega^{-1}$ from scratch, data `run/zbc/`) the diode box gains 15.8% in mass and is still gaining at 0.4% per Ω^{-1}, $\langle c_s \rangle$ doubles, and the face density sits at 6×10^{-2}; with `hse_outflow` the mass is constant to 0.1% (peak 1.9%, then a slow *outflow* of -0.07% per Ω^{-1}, as an evaporating halo should), $\langle c_s \rangle$ saturates at $1.10\times$ its initial value, and the face density is 10^{-2}, falling from 2×10^{-2} two cells in. At $\beta = 10$ the two boundaries agree to every digit for as long as the halo stays on the floor. The new inputs (`gravito_turb_bc16.athinput`) use `hse_outflow`; the SC14 and Baehr inputs keep `diode` so their archived runs stay reproducible.

12.5 Dynamic validation: the evolution twin and adaptive refinement

Evolution twin. The same small β=10 box evolved to $t = 5\,\Omega^{-1}$ through both gravity solvers: after removing the constant gauge offset between the two open-boundary conventions, the potentials agree to 8×10^{-5} rms and density/thermal energy to 10^{-3}, while the tiny random-seeded perturbation velocities decorrelate chaotically — the expected separation of two correct solvers differing at roundoff-method level, with the gravity demonstrably not driving it.

AMR. A $16H$-wide β=10 box with a $\rho > 2\rho_0$ criterion (`shearing_box/gravito_turb_amr_test.athinput`) refined onto the cooling collapse clumps at $t \sim 15$ ($\rho_{\max} = 38$ at t=20, 64 fine blocks), tracked them, and derefined as they sheared apart (32 fine blocks by t=25) — multiple remeshes in both directions, slab planes rebuilt each time, zero guard trips, fine blocks confined to $|z| \leq 2H$ throughout by the automatic slab-z policy. One policy interaction is worth remembering: a box with only two x_1 MeshBlock columns cannot refine *anywhere* (both columns sit on the protected shear faces); the test provides four.

12.6 The SC14 reproduction: calibration, the full-width run, and the box-width lesson

The `gravito_turb` pgen integrates SC14's self-gravitating vertical hydrostatic balance; with their published constants ($c_s(0) = 2.12625$, $4\pi G = 4.25231$, $\gamma = 5/3$) the computed half-thickness $\int_0^{z_{\max}} \rho\, dz/\rho_0 = 1.0005$ — an independent confirmation of their $Q_0 = 1$, $H = 1$ calibration.

Two production runs of the standard $\beta = 10$ configuration were carried to $t = 300\,\Omega^{-1}$: the **full-width** SC14 box ($[-32, 32]^2 \times [-6, 6]H$, $256 \times 256 \times 48$, 64 ranks on NAS Athena, restart-continued across two jobs) and a **half-width** twin ($[-16, 16]^2$, $128^2 \times 48$, 4 laptop ranks, 6.8 h). Both reproduce SC14's Figure-2 phenomenology end to end (Fig. 12): violent cooling collapse peaking at $t \approx 20$–25, vertical rebound with mass loss, then $250\,\Omega^{-1}$ of self-regulated gravito-turbulence. Averages over their window ($t = 200$–300):

	full box ($64H$)	half box ($32H$)	SC14 tc=10	full vs SC14
Q	1.29 ± 0.06	1.11 ± 0.08	1.33	−3%
$\alpha_{\rm grav}$	0.0379 ± 0.010	0.034 ± 0.017	0.0387	**−2%**
$\alpha_{\rm Reyn}$	0.0139 ± 0.043	0.044 ± 0.113	0.0188	−26%[1]
α (total, eq. 19)	0.052 ± 0.040	0.080	0.0576	−10%
α' (eq. 20)	0.0382 ± 0.016	0.040 ± 0.052	0.0424	−10%; eq. 21 gives 0.0400
rms δv $[H\Omega]$	1.71 ± 0.17	1.68	1.79	−4%
$\langle c_s \rangle_\rho$ $[H\Omega]$	2.31 ± 0.11	2.19	2.18	+6%
M/M_0 at t=300	0.83	0.92	0.8–0.9	consistent

The full-width box lands on the paper across the board, with $\alpha_{\rm grav}$ — the channel the new gravity machinery computes — the most accurate and steadiest number, and α' within 5% of the eq.-21 energy-balance prediction (a nontrivial audit: it requires the imposed cooling to be balanced by stress heating with the correct normalization). One post-processing trap is worth recording: eq. 20's prefactor is $|d \ln \Omega / d \ln R|^{-1} = 2/3$ against $\int \rho c_s^2$; carrying the paper's equivalent $2/(3\gamma)$-against-$\int P$ form while dividing by $\int \rho c_s^2$ double-counts γ and produced a phantom factor-1.67 "discrepancy" in α' until caught.

[1] The most fluctuation- and resolution-sensitive channel: SC14's own resolution doubling moves it 0.0188 → 0.0170, and its instantaneous value alternates in sign.

The half-box column is the box-width lesson: a $32H$ box supports a *single* dominant box-scale swing mode that recharges quasi-periodically (period ~ 35–$40\,\Omega^{-1}$), inflating the Reynolds channel in bursts (variance $\times 2.6$ the full box's), depressing the mean Q, and blurring the heating–cooling balance. SC14's own width test only ever *doubled* the box (changes $\lesssim 10\%$); halving it is evidently the harsher direction — worth knowing before shrinking boxes to save cycles in the Phase-4/5 science runs.

12.7 The slab planes on a GPU, and the memory fault they hid

The distributed rewrite above made the slab boundary affordable on a CPU, but on a GPU it was latency-bound rather than work-bound: the per-plane form issued about seven kernels per root k-plane, ~ 3600 launches per solve at 512 planes, and the phase timer added under `show_defect` ≥ 1 put 0.113 s of a 1.0 s solve in the slab phase alone. Batching the gather and the forward transforms over a stack of B planes (commits `73204c3a`, `bf1a4446`) cut that to 0.015 s and took the $16H$ box from 4.9×10^7 to 9.2×10^7 cell-updates per second, the defect norms unchanged.

That batching then caused every large-rank failure of the dust stage-2 runs, and it is worth recording how well it hid. Two faults compounded. The y transform ran along axis 1 of the (B, n_y, n_x) stack rather than its last axis, and kokkos-fft answers a non-trailing transform axis by transposing into work arrays that it allocates and frees *inside every* `execute` call — two arrays the size of the whole stack, once per batch, per solve. And B was a compile-time 32, so the stack grew with the box without limit. At the Baehr root grid, $512 \times 512 \times 256$, one complex plane is 4 MiB and the stack 128 MiB: each rank held 320 MiB of slab buffers and churned a further 256 MiB through the allocator on every batch. With 128 ranks to a Vista node that is 72 GiB of slab buffers on one node.

The failures it produced looked like three unrelated cluster problems. At 128 ranks the job died of signal 9 with no `MaxRSS` recorded. At 512 ranks it died inside `MPI_Test` with a UCX *Transport retry count exceeded* on an `RDMA_READ`, reproducibly, on disjoint node sets, always on a rank holding no particles and always before the first cycle line — because a large allocate/free cycle is exactly what invalidates the memory-registration cache UCX serves RDMA from. Disabling that cache (`UCX_RCACHE_ENABLE=n`) as a control test removed the disguise rather than the fault: the run then failed with `Kokkos ERROR: Host memory space failed to allocate 128 MiB (label="out_T")` on 86 of the 128 ranks of one node, `out_T` being kokkos-fft's transpose scratch and 128 MiB being the stack size to the byte.

The fix (`cb3b4b8a`) is two changes. Every transform now runs along the last axis of its buffers, the y transforms taking x-major copies; the kernel that applies the shear phase between the two transforms rewrites the stack in any case, so it carries the index swap at no cost, and kokkos-fft never transposes and never allocates — which also removes the two per-call allocations the original *per-plane* path had been making all along. And B is sized from a memory budget rather than fixed: 48 MiB per rank on a host build, where the ranks of a node share its memory and batching buys nothing because kernel launches are free, and 2 GiB on a device build, where one rank owns the GPU and launch latency is the binding constraint. The Baehr box falls from 576 MiB to 36 MiB per rank, 72 GiB to 4.5 GiB per node.

B is a pure performance knob and the physics does not see it: batches of 1, 3 and 32 give bit-identical solutions. Against the previous code the three slab tests agree to the last digit — `mg_poisson_slab` sheet $\max_{\mathrm{rel}} = 1.203474 \times 10^{-12}$, `mg_poisson_slab_shear` MG-vs-FFT max $= 8.764760 \times 10^{-8}$, `mg_poisson_slab_smr` $\max_{\mathrm{rel,all}} = 4.851125 \times 10^{-12}$ — and the shear test on 4 MPI ranks gives 6.591364×10^{-6} against 6.591365×10^{-6}, the last digit moving with the Allreduce order.

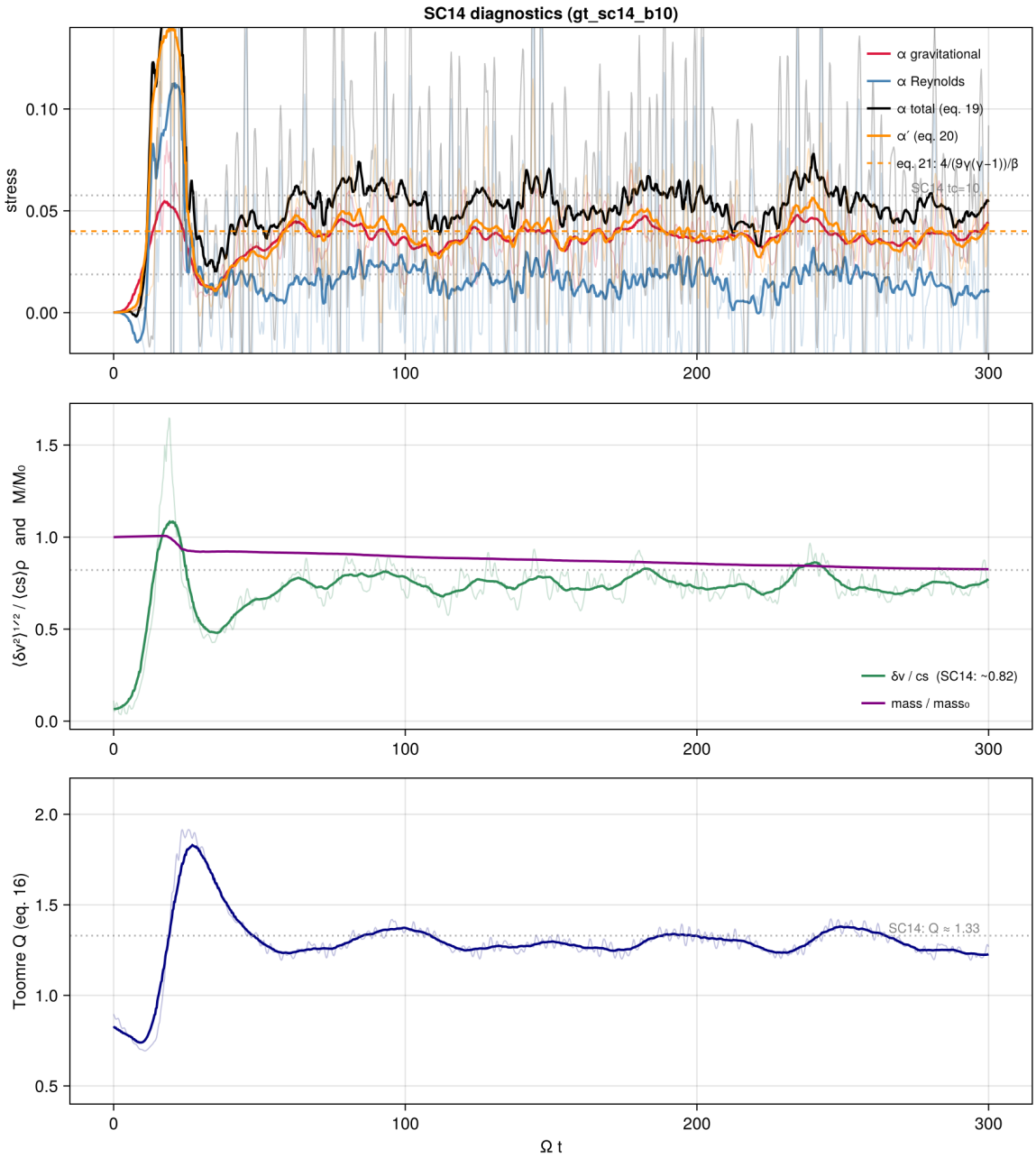


Figure 12: The full-width ($64H$) $\beta = 10$ run in SC14's Figure-2 form (thin: raw history; thick: $12\,\Omega^{-1}$ boxcar as in the paper; dotted gray: their tc=10 values; dashed: the eq.-21 energy-balance line). Top: stresses (gravitational, Reynolds, total, α'); middle: $\delta v/c_s$ and M/M_0; bottom: Toomre Q.

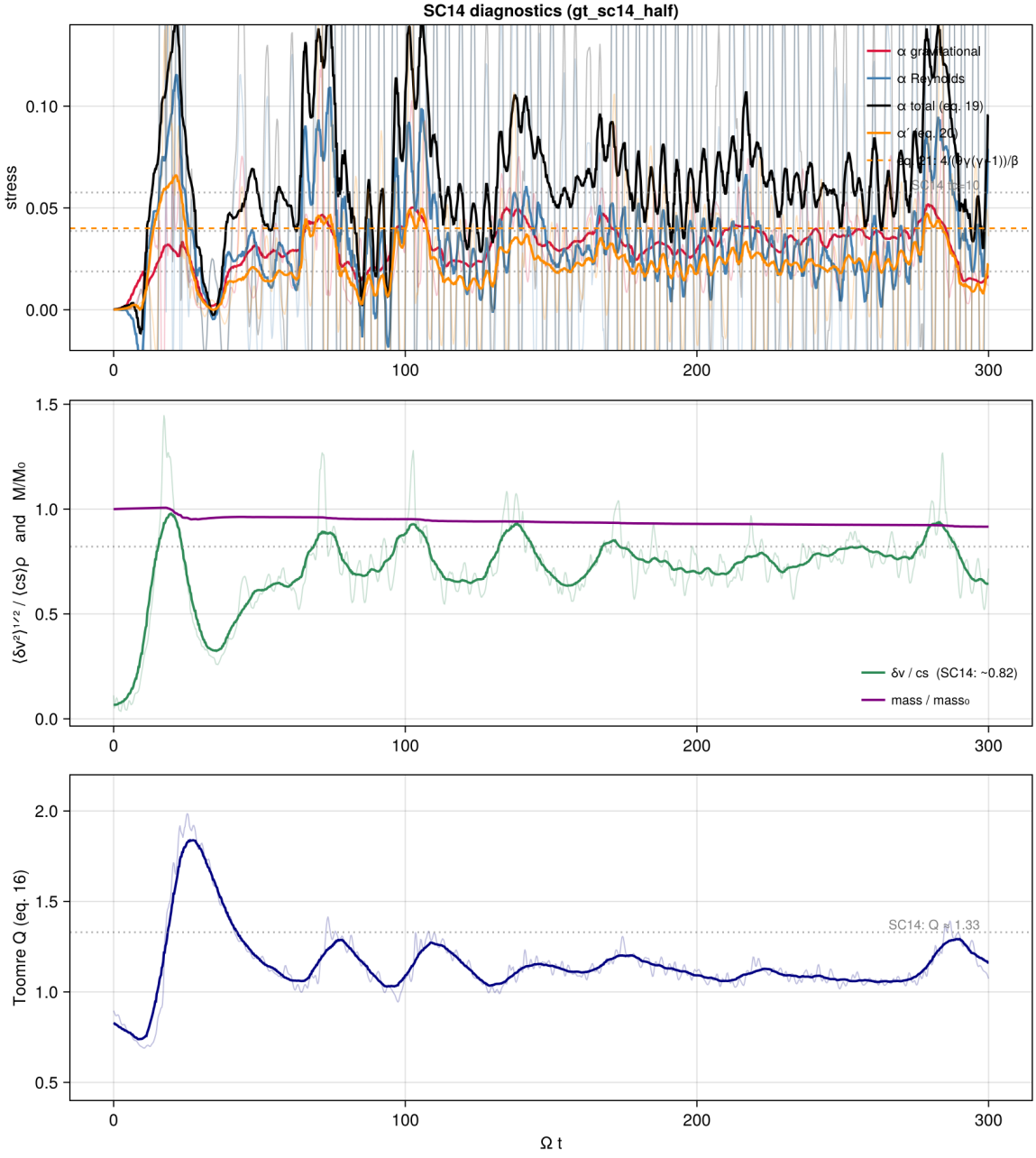


Figure 13: The half-width ($32H$) twin: same diagnostics. The quasi-periodic box-scale bursts — absent at $64H$ — are the half-width signature discussed in the text.

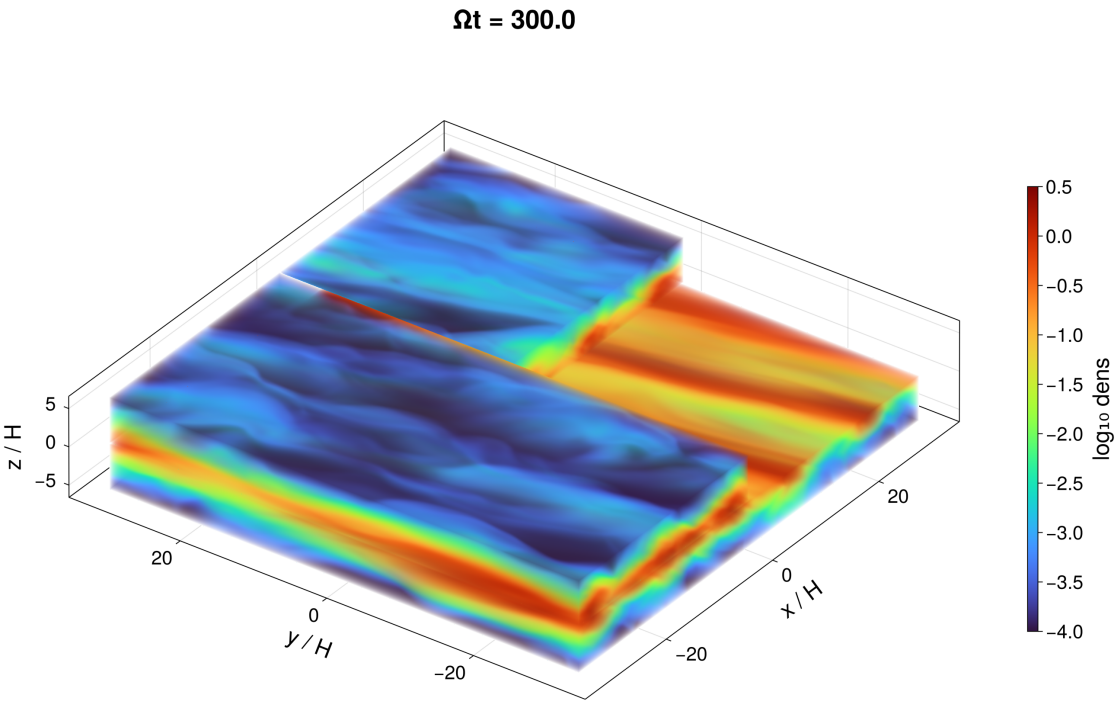


Figure 14: Volume rendering of $\log_{10}\rho$ in the full-width box at $\Omega t = 300$ (cutaway in the style of SC14's Fig. 2 right panel): the gravito-turbulent midplane with trailing wakes exposed at $z = 0$, under the 10^{-3}–10^{-4} halo whose striations are forced from below — the paper's central phenomenon, gas at altitude driven by midplane self-gravity.

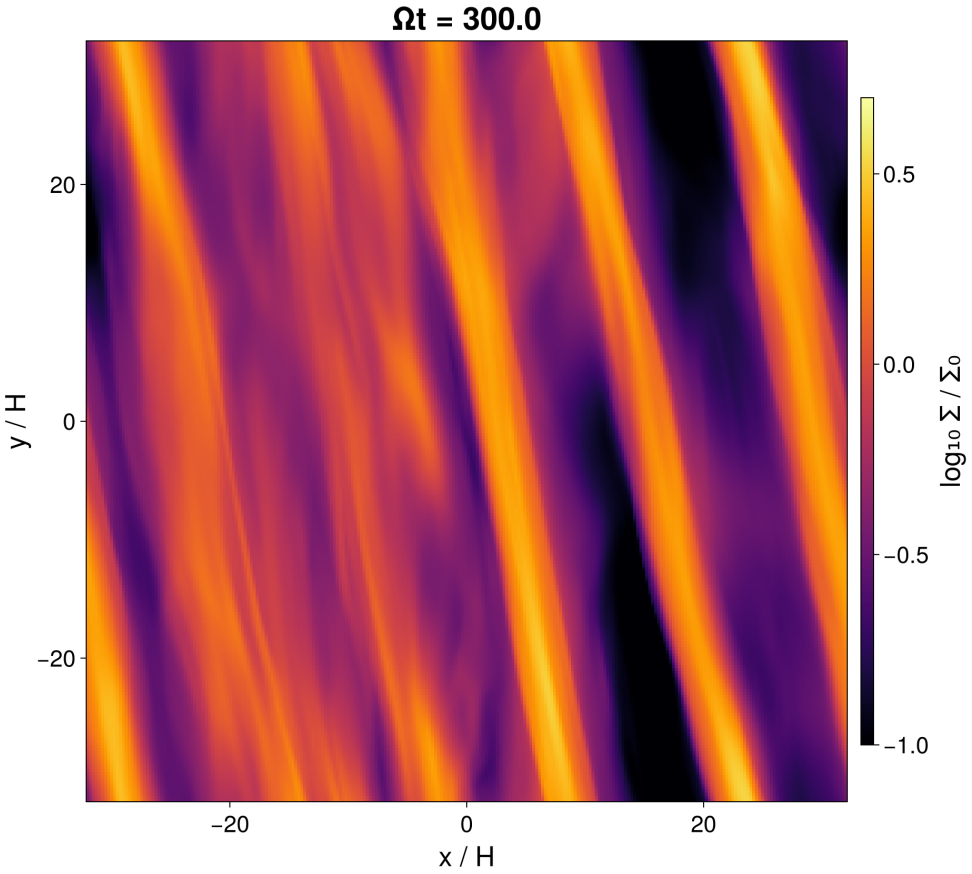


Figure 15: Vertically integrated surface density $\log_{10}\Sigma/\Sigma_0$ of the full-width box at $\Omega t = 300$: the trailing gravito-turbulent wake field, with Σ contrasts of $\sim \times 5$.

The lesson generalises past this solver. A library call that allocates internally is invisible in the source, and the allocation churn it creates surfaces first as a *network* error on a cluster with an RDMA registration cache — three days were spent on the interconnect before the arithmetic of the error message (4 MiB per plane $\times$ 32) identified it. Where a transform axis can be made the trailing one, it should be.

12.8 Summary

- The slab-open multigrid boundary reproduces the infinite-vacuum discrete solution to solver tolerance (10^{-12}, absolute comparison), matches the independent FFT-KO09 method to 3×10^{-13} statically and 8×10^{-5} through nonlinear evolution, and is bit-identical across MPI decompositions.
- Shear, SMR, and dynamic AMR all compose with it; the only interface cost is a constant, force-free $O(h^2)$ offset in enclosed refined regions.
- The stratified hydro side is complete (energy work term, cooling, diode boundaries) and the SC14 initial state holds hydrostatically.
- The SC14 $\beta = 10$ reproduction is complete at full size ($256 \times 256 \times 48$ on 64 NAS ranks): over their averaging window, $\alpha_{\rm grav}$ within 2%, Q within 3%, δv within 4%, α' within 5% of the eq.-21 energy balance — and the half-width twin isolates a clean box-width lesson (quasi-periodic box-scale bursts) for future economy runs.

13 Phase 4: ideal MHD with self-gravity

Phase 4 adds a magnetic field to the self-gravitating gas. The code pieces (the magneto-Jeans mode of `pgen/tests/jeans_wave.cpp`, and the gravito-turbulent box run with an `<mhd>` block) are described in Phase 4 of the implementation document; this section validates them, with linear problems that have exact answers. It starts without a field: the hydrodynamic Jeans wave (§13.1), stable and growing, on uniform and refined meshes, with the same box, integrator and meshes as the magneto-Jeans suite. The magneto-Jeans wave asks whether the MHD integrator (constrained transport, HLLD) coupled to the multigrid gravity source terms reproduces the linear theory of a magnetized, self-gravitating medium; on a uniform grid it does, for every field angle, both branches, stable and unstable wavelengths, and a field strong enough to stabilise a hydro-unstable wave (§§13.2–13.3), and on statically and adaptively refined meshes it does to within the bracket of the uniform runs at the root and the refined resolution (§13.5). The first pass of that suite found a numerical instability of AthenaK's *isothermal* HLLD solver in strong fields (§13.4); the suites were rerun adiabatic. Magnetized swing amplification (§13.6) then puts the field, self-gravity and the shear together, on uniform and refined shearing boxes.

13.1 The hydrodynamic Jeans test

Earlier phases checked the hydrodynamic Jeans wave only indirectly: FFT against multigrid on one stable wave (FFT document), and the error convergence of the Tomida & Stone (2023) §4.3 wave (§6), also stable. This suite measures the dispersion relation itself, on both sides of λ_J and on refined meshes. It is the hydro counterpart of the magneto-Jeans suite below (same box, integrator and meshes; the hydro module instead of MHD), so the two can be read together, and the $\theta = 0$ magneto-Jeans cases repeat it with a field along $\mathbf{k}$.

Theory. A uniform self-gravitating gas carries

$$\omega^2 = k^2 c_s^2 - 4\pi G\rho_0, \tag{17}$$

a sound wave for $\lambda < \lambda_J$ and pure growth, $\omega^2 = -s^2$, for $\lambda > \lambda_J$. The discrete operators change the gravity term. The solver inverts the 7-point Laplacian, whose eigenvalue is $D = \sum_d (2/\Delta x_d)^2 \sin^2(k_d \Delta x_d/2)$, and `SourceTerms::SelfGravity` applies the centred gradient $(\Phi_{i+1} - \Phi_{i-1})/2\Delta x$, whose symbol is $\sin(k\Delta x)/\Delta x$; so

$$\omega_{\rm disc}^2 = k^2 c_s^2 - 4\pi G\rho_0 \, \frac{\sum_d k_d \sin(k_d \Delta x_d)/\Delta x_d}{D} \approx k^2 c_s^2 - 4\pi G\rho_0 \big[1 - (k\Delta x)^2/12\big] \tag{18}$$

for a wave along an axis. Gravity is slightly weaker than in the continuum; the difference matters only near marginal stability, where the two terms of eq. (17) nearly cancel. (The hydro fluxes are fourth order in this suite, PPM4, and are not corrected.)

Setup. `inputs/tests/hydro_jeans.athinput` (with `_smr` and `_amr` variants): $\mathbf{k}$ along x_1 in a $1 \times \frac{1}{8} \times \frac{1}{8}$ periodic box of $64 \times 8 \times 8$ cells in 8^3 MeshBlocks; adiabatic hydro, $\gamma = 5/3$, $p_0 = 0.6$ ($c_s = 1$), HLLC, PPM4, RK3, CFL 0.3; full multigrid and eight V-cycles. The mode is the exact eigenmode with $\delta\rho = 10^{-4}\rho_0 \sin kx$: a standing wave from rest, or pure growth with the matching velocity $\delta v = (s/k)\, 10^{-4} c_s \cos kx$. Units as in §13.2: $c_s = \rho_0 = \lambda = 1$, $n = \lambda/\lambda_J$ fixes $4\pi G = k^2 n^2$. The cases:

- dispersion: $n = 0.25$, 0.5, 0.75, 0.9, 0.95, 1.05, 1.1, 1.25, 1.5, 2, 3 at 64 cells per wavelength;
- resolution: $n = 0.5$ and 2 at 16, 32 and 128 cells per wavelength;
- isothermal gas ($c_s = 1$, isothermal HLLC): $n = 0.5$, 0.95, 1.5, 2;
- 3-D: the wave along the diagonal of a $1 \times \frac{1}{2} \times \frac{1}{2}$ box, $64 \times 32 \times 32$ cells ($\lambda = 1/3$, 21 cells per wavelength), $n = 0.5$, 2;
- refined meshes, $n = 0.5$ and 2: the six one-dimensional meshes of the magneto-Jeans suite (§13.5: uniform 64 and 128, static level-1 and level-2 slabs, the moving window at one and two levels) and, in 3-D, a static central block and the moving window.

Growing modes are fitted with a straight line in $\ln|a|$ over the last 90% of three e-folds, stable ones with $\cos(\omega t)e^{-\gamma t}$ over two periods, as in the magneto-Jeans suite. The 35 cases take about 25 minutes, six at a time on a laptop. The runner is `validation/run/hydro_jeans/hydro_jeans.py`; the figure script, for Fig. 16, is `validation/figures/plot_hydro_jeans.jl`. The binary is a clean build of `dust-multigrid d0d5f564`.

Results.

- **Dispersion relation** (Fig. 16a,b; Table 1). The error in ω against eq. (17) is 3.7×10^{-4} to 5.9×10^{-3}, and $|\Delta\omega^2|/k^2 \le 6 \times 10^{-3}$. The largest errors are next to marginal stability ($n = 0.95$, 1.05), where $\omega \to 0$. Against eq. (18) they fall to 1.8×10^{-5}–1.6×10^{-3}: most of the near-marginal error is the weaker discrete gravity, not the hydro.
- **Convergence** (Fig. 16c). At 16, 32, 64 and 128 cells per wavelength the error is 1.5×10^{-2}, 3.4×10^{-3}, 5.9×10^{-4}, 1.1×10^{-4} for $n = 0.5$ (local orders 2.1, 2.5, 2.4) and 1.2×10^{-2}, 2.1×10^{-3}, 4.4×10^{-4}, 1.2×10^{-4} for $n = 2$ (orders 2.5, 2.3, 1.9). The stable wave damps at $\gamma \approx 5 \times 10^{-4}$ ($\gamma/\omega \approx 10^{-4}$) at 64 cells per wavelength.

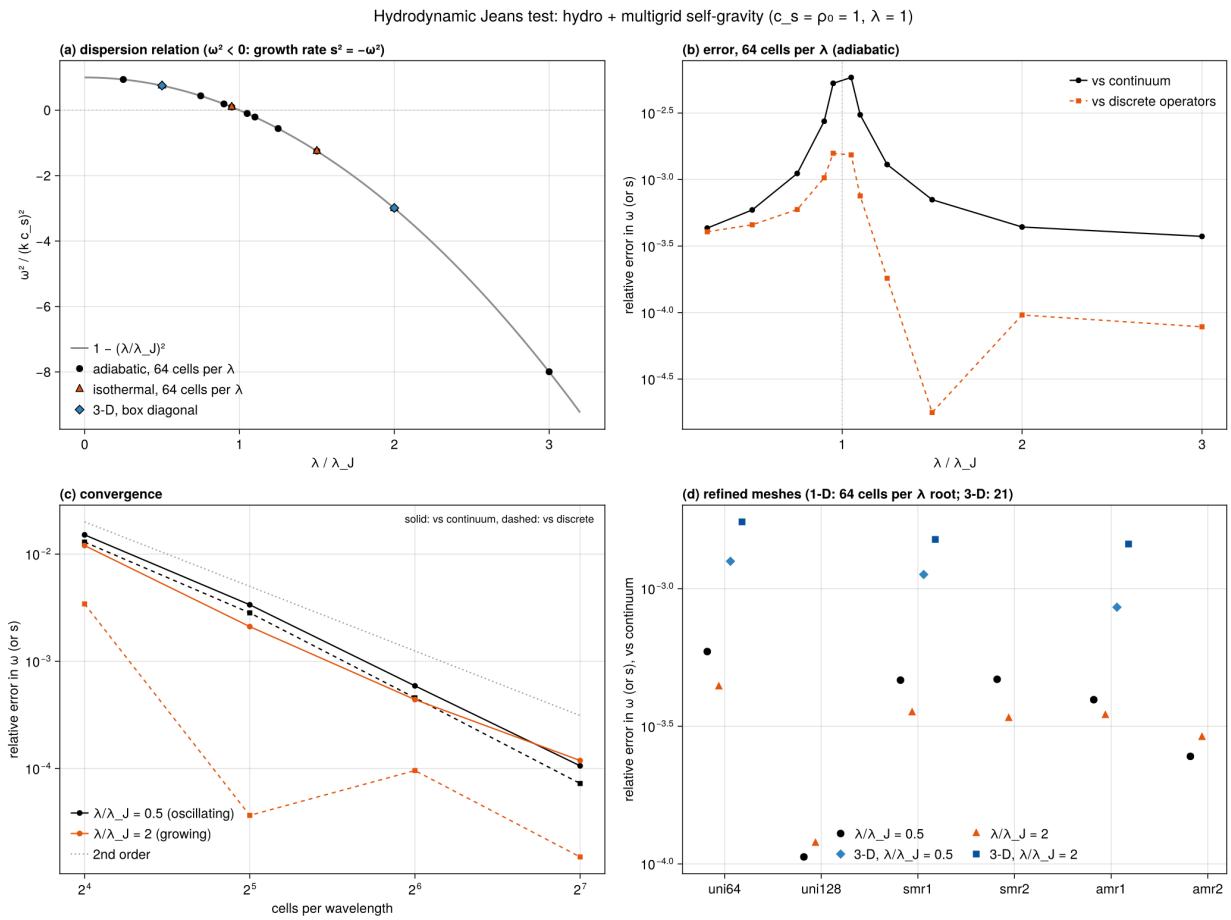


Figure 16: The hydrodynamic Jeans test. (a) Measured $\omega^2/(kc_s)^2$ (points) against eq. (17) (line) for adiabatic and isothermal gas at 64 cells per wavelength and the 3-D waves along the box diagonal. (b) Relative error in ω (or s) at 64 cells per wavelength against the continuum relation and against the discrete one, eq. (18). (c) Convergence for a stable ($n = 0.5$) and a growing ($n = 2$) mode. (d) Refined meshes, against the uniform bracket (64 and 128 cells per wavelength).

- **Equation of state.** The isothermal runs give the adiabatic errors to two or three digits (5.9×10^{-4}, 5.2×10^{-3}, 7.2×10^{-4}, 4.5×10^{-4} at $n = 0.5, 0.95, 1.5, 2$), as they should for an isentropic eigenmode.
- **3-D.** Along the box diagonal, at 21 cells per wavelength, the error is 1.3×10^{-3} ($n = 0.5$) and 1.7×10^{-3} ($n = 2$).
- **Refined meshes** (Fig. 16d; Table 1). Every refined run lies inside the bracket of the uniform runs at 64 and 128 cells per wavelength. The two-level moving window, which creates and destroys 1148 and 1008 MeshBlocks, comes closest to the fine run. In 3-D the central block and the moving window (2016 created, 1792 destroyed) both lower the error of the uniform root. The spurious $\cos kx$ density component, reflected at the level interfaces, is at most 7.8×10^{-4} of the mode in 1-D and 9×10^{-5} in 3-D (about 10^{-9} on the uniform grid).

Table 1: Hydrodynamic Jeans test: relative error in ω (or the growth rate s) against the continuum relation, eq. (17), and against the discrete one, eq. (18). Top: dispersion at 64 cells per wavelength (adiabatic). Bottom: refined meshes (continuum).

λ/λ_J	ω^2 (eq. 17)	ω^2 measured	error (continuum)	error (discrete)
0.25	37.011	37.043	4.3×10^{-4}	4.0×10^{-4}
0.5	29.609	29.644	5.9×10^{-4}	4.6×10^{-4}
0.75	17.272	17.310	1.1×10^{-3}	6.0×10^{-4}
0.9	7.501	7.542	2.7×10^{-3}	1.0×10^{-3}
0.95	3.849	3.890	5.3×10^{-3}	1.6×10^{-3}
1.05	−4.047	−3.999	5.9×10^{-3}	1.5×10^{-3}
1.1	−8.290	−8.240	3.1×10^{-3}	7.5×10^{-4}
1.25	−22.207	−22.149	1.3×10^{-3}	1.8×10^{-4}
1.5	−49.348	−49.278	7.1×10^{-4}	1.8×10^{-5}
2	−118.435	−118.331	4.4×10^{-4}	9.6×10^{-5}
3	−315.827	−315.591	3.7×10^{-4}	7.8×10^{-5}

mesh	$n=0.5$	$n=2$
uniform 64	5.9×10^{-4}	4.4×10^{-4}
uniform 128	1.1×10^{-4}	1.2×10^{-4}
static, 1 level	4.7×10^{-4}	3.5×10^{-4}
static, 2 levels	4.7×10^{-4}	3.4×10^{-4}
moving, 1 level	4.0×10^{-4}	3.5×10^{-4}
moving, 2 levels	2.5×10^{-4}	2.9×10^{-4}
3-D uniform 64	1.3×10^{-3}	1.7×10^{-3}
3-D central block	1.1×10^{-3}	1.5×10^{-3}
3-D moving, 1 level	8.6×10^{-4}	1.5×10^{-3}

13.2 The magneto-Jeans test

A uniform medium of density ρ_0, sound speed c_s, and field $\mathbf{B}_0$ at angle θ to the wavevector $\mathbf{k}$ carries compressive waves with the perturbed field along $\hat{\mathbf{e}}_\perp$ (in the plane of $\mathbf{k}$ and $\mathbf{B}_0$, perpendicular to $\mathbf{k}$). Their dispersion relation (Chandrasekhar 1961, ch. XIII) is

$$\left(\omega^2-\Omega_J^2\right)\left(\omega^2-k^2v_A^2\cos^2\theta\right)=k^2v_A^2\sin^2\theta\,\omega^2,\qquad \Omega_J^2=k^2c_s^2-4\pi G\rho_0,\quad v_A^2=B_0^2/\rho_0, \tag{19}$$

with a slow (smaller ω^2) and a fast root. At $\theta=0$ the compressive root is the hydro Jeans wave, $\omega^2=\Omega_J^2$, the field playing no part; at $\theta=90^\circ$ it is the magnetosonic Jeans wave, $\omega^2=k^2(c_s^2+v_A^2)-4\pi G\rho_0$, so a perpendicular field stabilises a wavelength $\lambda=n\lambda_J$ once $v_A>c_s\sqrt{n^2-1}$. At oblique angles the slow branch goes unstable at $\lambda>\lambda_J$, exactly as in hydro (the field along $\mathbf{k}$ cannot resist compression along it), while the fast branch stays stable.

For an ideal gas $c_s^2=\gamma p_0/\rho_0$ and the eigenmode is isentropic, $\delta p=c_s^2\,\delta\rho$; eq. (19) is then the same as for an isothermal gas. Units are $c_s=\rho_0=1$ and $\lambda=1$ ($k=2\pi$); $n=\lambda/\lambda_J$ fixes $4\pi G=k^2n^2$ and $\mathtt{b0}=v_A$. The mode is set up as an exact eigenvector, with $\delta\rho=10^{-4}\rho_0\sin k\xi$ and $\delta B_\perp=C\,\delta\rho$,

$$C=\frac{B_\perp\,\omega^2}{\rho_0\left(\omega^2-k^2v_{A\parallel}^2\right)}, \tag{20}$$

the field built from a vector potential so that $\nabla\cdot B=0$ to round-off: a standing wave from rest if $\omega^2>0$, pure growth e^{st} (with the matching velocities) if $\omega^2=-s^2<0$.

Setup. `inputs/tests/magneto_jeans.athinput`: $\mathbf{k}$ along x_1 in a $1 \times \frac{1}{8} \times \frac{1}{8}$ periodic box of $64 \times 8 \times 8$ cells (64 cells per wavelength) in 8^3 MeshBlocks, so the multigrid hierarchy spans eight blocks along the wave; adiabatic MHD ($\gamma = 5/3$, $p_0 = 0.6$, so $c_s = 1$), HLLD, PPM4, RK3, CFL 0.3. (The first pass was isothermal; §13.4 explains the change. The isothermal and adiabatic results agree to two or three digits wherever the isothermal run was not corrupted.) The gravity solve uses full multigrid and then a fixed eight V-cycles (`threshold` $= -1$): a fixed defect tolerance failed at 128 cells and in 3-D, because the round-off floor of the defect grows with the cell count and with $4\pi G$, and no single absolute value fits the whole suite; eight cycles reach the floor everywhere. Three runs instead put $\mathbf{k}$ along the diagonal of a $1 \times \frac{1}{2} \times \frac{1}{2}$ box ($64 \times 32 \times 32$ cells, $\mathbf{k} = 2\pi(1, 2, 2)$, $\lambda = 1/3$, i.e. about 21 grid spacings per wavelength), which exercises multi-dimensional constrained transport and a fully 3-D gravity solve.

Measurement. The user history projects the density onto the mode, $a(t) = (2/V) \int \delta\rho \sin k\xi \, dV$, and $\delta B_\perp$ likewise. A growing mode is fitted with a straight line in $\ln|a|$ over the last 90% of three e-folds; a stable one with $a/a_0 = \cos(\omega t)\, e^{-\gamma t}$ over two periods. One case needs more care. A stable *fast* mode at a wavelength where the *slow* branch is unstable ($\theta = 30, 60°$, $n > 1$) is not alone: the discretisation error of the initial state projects onto the slow eigenvector at the 10^{-4} level, and that component grows as e^{st} until it swamps the oscillation. These runs are fitted with $a/a_0 = A\cos\omega t + Be^{st} + C'e^{-st}$, ω and s both free, and stopped after six slow e-folds. The eigenvector check is the median of $\delta B_\perp/\delta\rho$ over the run (over the first quarter period in the mixed case) against eq. (20). `validation/run/magneto_jeans/magneto_jeans.py` runs the 42 cases and writes the fits; `validation/figures/plot_magneto_jeans.jl` makes Fig. 17. The suite takes 23 minutes on one laptop core.

13.3 Uniform-grid results

Dispersion relation. Over the 30 quasi-one-dimensional cases at $v_A = c_s$ (Fig. 17a) the relative error in ω (or s) lies between 9×10^{-5} and 4.8×10^{-3}, median 6.6×10^{-4}; the 19 stable cases stay below 1.8×10^{-3}. The largest errors sit near marginal stability, where $|\omega|$ is small: at $\theta = 90°$, $n = 1.5$, $\omega^2 = -9.87$ is measured as -9.78. The absolute error there, $|\Delta\omega^2| = 0.09$, is like that of the other cases ($\theta = 0$, $n = 1.5$: 0.07), so the method is not worse there; the quantity is smaller. The slow and fast branches are both reproduced at $\theta = 30$ and $60°$, including the slow branch's sign change at $\lambda = \lambda_J$, which the field does not move. The three oblique 3-D runs give 2.1×10^{-3} (slow, $\theta = 45°$, $n = 2$), 1.5×10^{-3} (fast, $\theta = 45°$, $n = 0.8$), and 1.4×10^{-2} ($\theta = 90°$, $n = 1.5$). The last is the near-marginal case again, now at about 21 grid spacings per wavelength: its absolute error, $|\Delta\omega^2|/k^2 = 6.7 \times 10^{-3}$, is 2.8 times the 1-D run's (2.4×10^{-3}) at a third of the resolution. (The run length of the 3-D cases is set from their own $k = 6\pi$; a first pass that used $k = 2\pi$ ran the growing cases nine e-folds, to $\delta\rho/\rho \sim 1$, and its fits are discarded.)

Magnetic support. At $\theta = 90°$ and $\lambda/\lambda_J = 1.5$ (Fig. 17b) the hydro wave is unstable. With $v_A/c_s = 0.5, 1.0$ the measured $\omega^2 = -39.40, -9.78$ (theory $-39.48, -9.87$); with $v_A/c_s = 1.5, 2.0$ the wave oscillates, $\omega^2 = 39.61, 108.76$ (39.48, 108.57). The threshold falls between, at $v_A = 1.12\, c_s$, as predicted: magnetic pressure enters the Jeans criterion with the coupling between the field, the gas and the multigrid potential right.

Convergence. At $\theta = 60°$ (Fig. 17c) the error falls from 1.9×10^{-2} at 16 cells per wavelength to 1.6×10^{-4} at 128 for the growing slow mode ($n = 2$; local orders 2.6, 2.3, 2.0), and from 1.3×10^{-2} to 7.2×10^{-5} for the oscillating fast mode ($n = 0.5$; orders 2.2, 2.6, 2.7): second order or better, as

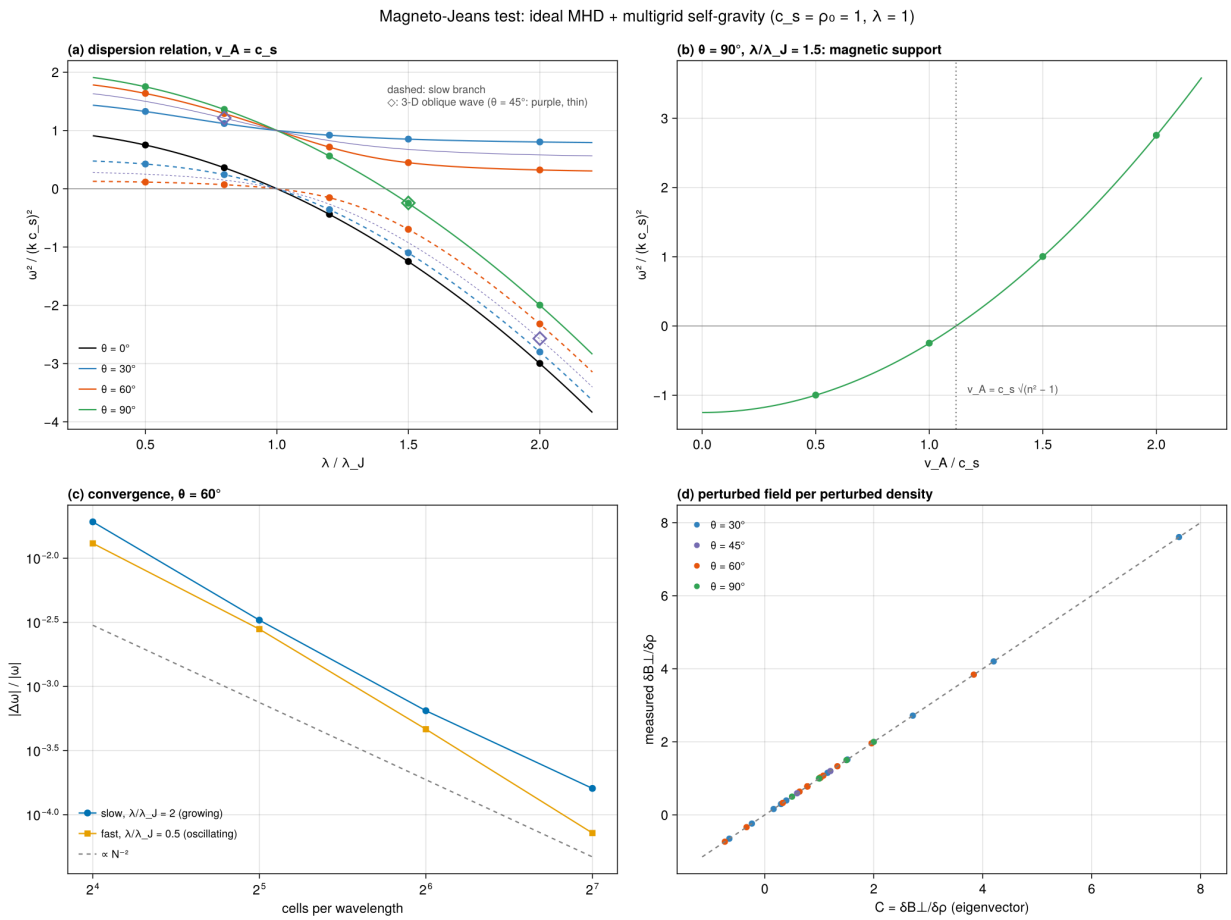


Figure 17: The magneto-Jeans test. (a) Measured ω^2 (points) against eq. (19) (lines) at $v_A = c_s$ and 64 cells per wavelength, for $\theta = 0, 30, 60, 90°$ (dashed: slow branch); open diamonds are the 3-D runs along the box diagonal, each at its own k and λ/λ_J, on the thin $\theta = 45°$ curves or the $90°$ one. (b) $\theta = 90°$, $\lambda/\lambda_J = 1.5$: the field stabilises the hydro-unstable wave above $v_A = c_s\sqrt{n^2 - 1} = 1.12\, c_s$. (c) Relative error in ω against resolution at $\theta = 60°$ for a growing slow mode and an oscillating fast one. (d) Measured $\delta B_\perp/\delta\rho$ against the eigenvector C for every case with a field perturbation.

expected from PPM4 reconstruction with the second-order multigrid Laplacian. The damping rate fitted to the standing waves converges with it, $\gamma = 7.8 \times 10^{-2}$, 7.4×10^{-3}, 8.0×10^{-4}, 1.0×10^{-4} at 16–128 cells for the fast mode ($\gamma/\omega \approx 10^{-4}$ at 64 cells); at 64 cells per wavelength no stable case exceeds $\gamma = 1.9 \times 10^{-3}$ (the 3-D diagonal fast wave, at about 21 spacings per wavelength, gives 8.7×10^{-3}, $\gamma/\omega = 4 \times 10^{-4}$).

Eigenvector. The measured $\delta B_\perp/\delta\rho$ matches C of eq. (20) in every case (Fig. 17d): across the 37 cases with $|C| > 0.05$ the relative difference has median 4×10^{-4} and maximum 4.9×10^{-3} (the 16-cell run). This checks the field and the gas moving together, which a frequency alone does not: C changes sign between the branches ($C < 0$ on stable slow modes, $C > 0$ on fast and on growing slow modes) and reaches 7.6 for the fast mode at $\theta = 30°$, $n = 2$, where nearly all the restoring force is magnetic.

The seeded slow mode. In the six mixed fits the free slow growth rate reproduces theory to 4% or better ($s = 3.61, 6.55, 10.60$ against $3.77, 6.59, 10.52$ at $\theta = 30°$; $2.39, 5.24, 9.44$ against $2.47, 5.25, 9.58$ at $60°$), with a seed amplitude of 0.7–4×10^{-4} of the fast mode's: the contamination is the expected slow eigenmode, excited by truncation error, not a defect of the coupling.

13.4 An instability of the isothermal HLLD solver

The first pass of the refined-mesh suite (§13.5) was isothermal, like the uniform one, and one mode failed: the magnetically supported wave at $\theta = 90°$, $\lambda/\lambda_J = 1.5$, $v_A = 1.5\,c_s$ ($\beta = 2c_s^2/v_A^2 = 0.89$) gave $\omega^2 = 23$–25 against 39.48 on the uniform 128-cell run and on both static refinements, while uniform 64 and the moving refinements were right. The failure was not the refinement. The standing wave followed theory until $t \approx 1.1$; then grid-scale structure with $k_y, k_z \neq 0$ (Fourier peaks at k_x of 30–100 out of 128) grew out of round-off at $s \approx 17$, faster than any physical rate in the problem ($\sqrt{4\pi G\rho_0} = 9.4$), and swamped it.

Seeding a uniform medium at rest with 10^{-8} white-noise velocity (`noise_amp`; the quasi-one-dimensional runs are otherwise one-dimensional to the last bit) isolates it ($128 \times 8 \times 8$ cells, $c_s = 1$, $\mathbf{B} = 1.5\,\hat{\mathbf{y}}$ unless noted; growth rate of the transverse kinetic-energy amplitude between $t = 1$ and 1.5):

Configuration	s
isothermal HLLD, PPM4, no gravity	14.8
the same with multigrid gravity / FFT gravity	15.4 / 15.5
PLM reconstruction	14.7
CFL 0.1	15.5
field along x_1 ($\theta = 0$)	12.7
$v_A = 0.5$ ($\beta = 8$)	stable
isothermal HLLE / LLF	stable / stable
adiabatic HLLD ($\gamma = 5/3$, $p_0 = 0.6$, $\beta = 0.53$)	stable

So the instability belongs to the isothermal HLLD solver in a strong field, independent of the reconstruction, the time step and self-gravity; gravity seemed involved at first only because, without the seed, multigrid round-off is what breaks the one-dimensional symmetry. It grows faster at finer resolution ($s \approx 7$ at 64 cells, from the multigrid seed), which is why the 64-cell runs of §13.3 escaped it within their two periods, and it did not show on the moving refinements, presumably because repeated restriction and prolongation damp grid-scale noise. The isothermal branch of `mhd/rsolvers/hlld_mhd.hpp` follows Mignone (2007) as written in Athena++ on reading; the cause is not yet found. Both magneto-Jeans suites were rerun adiabatic, which is also the equation of state of the production box; isothermal strong-field runs should use HLLE until the solver is fixed.

13.5 The magneto-Jeans wave on refined meshes

Setup. The eigenmodes of §13.2 on six meshes, all with a 64-cell root: uniform at 64 and 128 cells per wavelength (the bracket); a static level-1 slab, $x_1 \in [0.25, 0.5)$, two root blocks wide (`inputs/tests/magneto_jeans_smr.athinput`); the same with a level-2 slab $x_1 \in [0.3125, 0.4375)$ inside it; and a slab of the same width at level 1 or level 2 that moves through the box twice per run (`amr_mode = window`, `magneto_jeans_amr.athinput`; at level 2 the 2:1 rule adds a level-1 rim). The wave crosses two (or four) fine/coarse interfaces normal to $\mathbf{k}$. Four quasi-one-dimensional modes are run: the growing slow mode at $\theta = 60°$, $n = 2$; the oscillating fast mode at $\theta = 60°$, $n = 0.5$; the magnetically supported mode at $\theta = 90°$, $n = 1.5$, $v_A = 1.5$; and the field-carried

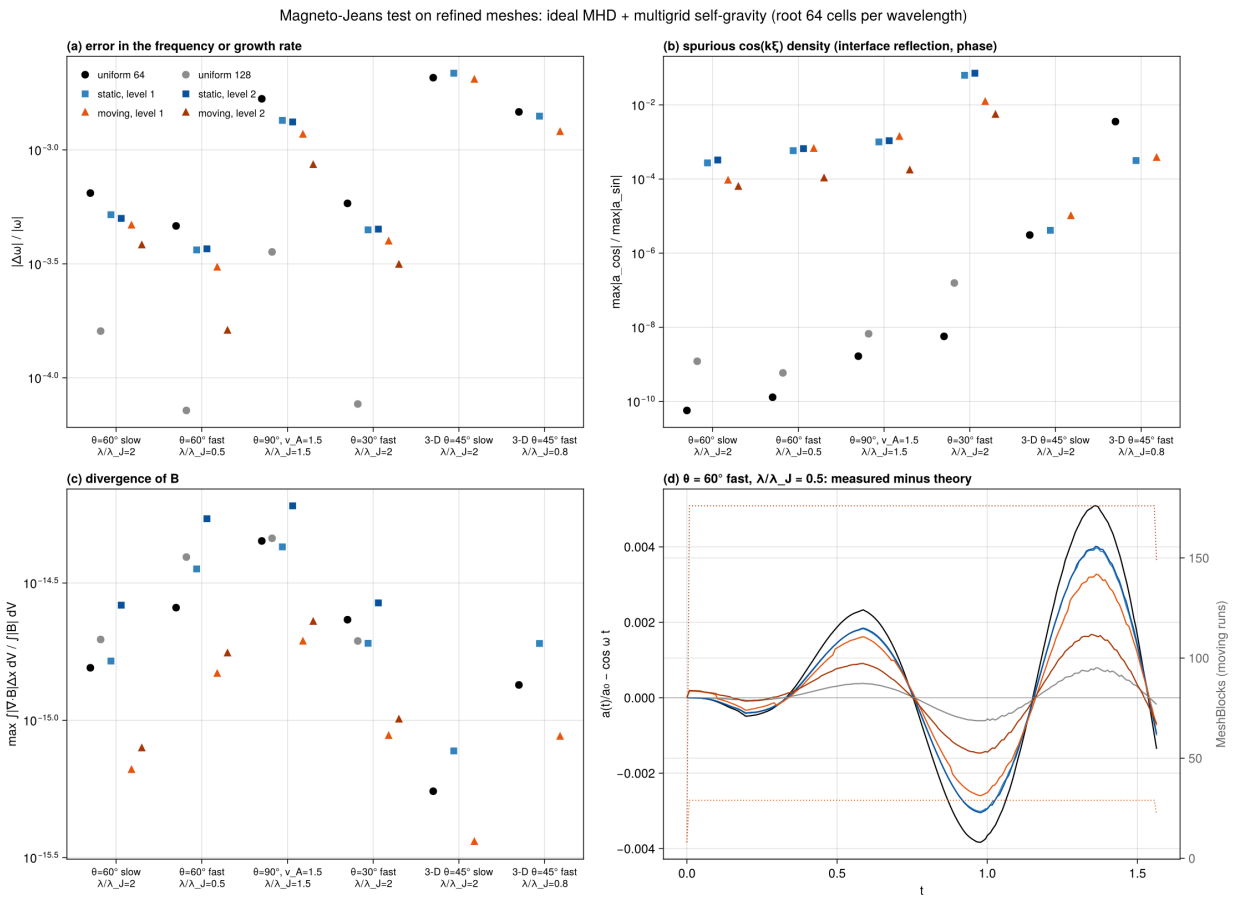


Figure 18: The magneto-Jeans test on refined meshes (root 64 cells per wavelength; colours and symbols as in the legend of panel a). (a) Relative error in the frequency or growth rate per mode and mesh. (b) The spurious $\cos k\xi$ density component, zero by symmetry on a uniform 1-D grid: reflection and phase error at the level interfaces. (c) The largest relative divergence of $\mathbf{B}$ over each run. (d) Measured minus theoretical density amplitude of the oscillating fast mode on each mesh (solid), and the MeshBlock count of the moving runs (dotted, right axis).

fast mode at $\theta = 30°$, $n = 2$ ($C = 7.6$, slow branch unstable). The 3-D diagonal waves (slow $n = 2$ and fast $n = 0.8$ at $\theta = 45°$) are run on the uniform 64 mesh, a static level-1 central block ($x_1 \in [0.25, 0.75)$, $x_{2,3} \in [0.125, 0.375)$; interfaces oblique to $\mathbf{k}$ on all six faces) and the moving slab. `validation/run/magneto_jeans_amr/magneto_jeans_amr.py` runs the 30 cases (25 minutes, five at a time) with the fits of §13.2.

Relative error in ω	uniform 64	uniform 128	static L1	static L2	moving L1	moving L2
$\theta = 60°$ slow, $n = 2$	6.5×10^{-4}	1.6×10^{-4}	5.2×10^{-4}	5.0×10^{-4}	4.6×10^{-4}	3.8×10^{-4}
$\theta = 60°$ fast, $n = 0.5$	4.6×10^{-4}	7.2×10^{-5}	3.6×10^{-4}	3.7×10^{-4}	3.0×10^{-4}	1.6×10^{-4}
$\theta = 90°$, $v_A = 1.5$	1.7×10^{-3}	3.6×10^{-4}	1.4×10^{-3}	1.3×10^{-3}	1.2×10^{-3}	8.6×10^{-4}
$\theta = 30°$ fast, $n = 2$	5.8×10^{-4}	7.7×10^{-5}	4.5×10^{-4}	4.5×10^{-4}	4.0×10^{-4}	3.1×10^{-4}
3-D $\theta = 45°$ slow	2.1×10^{-3}	—	2.2×10^{-3}	—	2.0×10^{-3}	—
3-D $\theta = 45°$ fast	1.5×10^{-3}	—	1.4×10^{-3}	—	1.2×10^{-3}	—

Results. Every refined run lies inside the bracket of the uniform runs at its root and finest resolutions, apart from the 3-D slow mode on the central block, which is 5% above uniform 64 (Fig. 18a). The static slabs cover a quarter of the wavelength and cut the error by 20–25%; the moving slab, which sweeps the whole wave, cuts it most. The level interfaces show up where they must, in the $\cos k\xi$ density that a uniform grid keeps at round-off (Fig. 18b): 6×10^{-5} to 1.3×10^{-3} of the mode on the refined meshes. The one large value, $1\text{–}7\times10^{-2}$ for the $\theta = 30°$ mode, is physical: the refined initial state projects on the unstable slow branch at about 2×10^{-4} of the amplitude, and that $\cos k\xi$ component then grows at $s = 10.25$ against the slow rate 10.52. (The 3-D fast wave's 3.5×10^{-3} on the uniform mesh is also physical: the $1\times\frac{1}{2}\times\frac{1}{2}$ box holds longer, Jeans-unstable modes, $\lambda = 1 > \lambda_J = 0.42$, which grow from round-off and couple to the projection nonlinearly.) The divergence of $\mathbf{B}$ stays at round-off, $\le 6\times10^{-15}$, through 1148 MeshBlocks created and 1008 destroyed in the 1-D level-2 runs and 2016/1792 in the 3-D moving runs (Fig. 18c): the face-field prolongation and restriction keep $\nabla\cdot B = 0$ on adaptive meshes. The eigenvector $\delta B_\perp/\delta\rho$ matches C to a median 1.5×10^{-4} (maximum 3.4×10^{-3}). The phase error of the standing wave (Fig. 18d) is ordered as the resolution: uniform 64 largest, uniform 128 smallest, the refined runs between.

13.6 Magnetized swing amplification

The test. The swing test of §7 — a leading shearing wave, $k_{x0} = -6$, $k_y = 1$, $k_z = 0$, in a $2\pi\times2\pi\times\pi/4$ box with $\kappa = k_y = 1$ and $4\pi G\rho_0 = 1.9$, swinging through $k_x = 0$ at $t = 4$ — threaded by a uniform azimuthal field $B_0\hat{\mathbf{y}}$ (the geometry of Kim & Ostriker 2001), the gas adiabatic with $c_s = 1$. The linear system (implementation document, Phase 4c) couples the wave to the field through magnetic tension in w_x and the shear of δB_x into δB_y; `validation/run/swing_mhd/swing_ode.py` integrates it with the 7-point Laplacian eigenvalue at each run's root resolution. Its $B_0 = 0$ limit gives the hydro peak 3.609 at $t = 7.81$ of §7 (documented there as 3.605). With $B_0 = 0.5$ ($\beta = 4.8$) and 1.0 ($\beta = 1.2$) the trajectory changes completely (Fig. 19a): the field-carried oscillation replaces the single swing peak, and the azimuthal field perturbation reaches $c_y \approx 2\text{–}4$ times the initial density amplitude (panel b). `inputs/tests/swing_mhd.athinput`: $128^2\times16$ cells in 16^3 MeshBlocks, HLLD, PLM, RK2, CFL 0.3, multigrid with full multigrid and ten V-cycles, $A = 10^{-4}$, $t \le 10$. MHD with orbital advection cannot refine, so the refined runs, and their uniform twins, integrate the full shear flow (`orbital_advection = false`). The refined meshes are a static level-1 ring over the central two of eight block columns ($|x| < \pi/4$) and the moving level-1 ring of §9 (centre $(\pi/4)\sin t$, half-width $\pi/16$; 448 MeshBlocks created and 392 destroyed). (The six-column ring of the hydro test puts refined cells where the shear flow is $3.5\,c_s$; without orbital advection that costs about five hours per run on a laptop.) The deviations quoted are $\max|\text{measured} - \text{reference}|/\max|\text{reference}|$ over the swing ($t \le 6$, $|k_x| \le 3$) and to $t = 8$; later, the wave is wound to $|k_x| \approx 9$, 14 cells per wavelength, and amplitude damping dominates every run (23–30% to $t = 10$ at 128^2, 7% at 256^2). `validation/run/swing_mhd/swing_mhd.py` runs the 19 cases and `validation/figures/plot_swing_mhd.jl` makes Fig. 19.

Uniform grid. Deviation from linear theory:

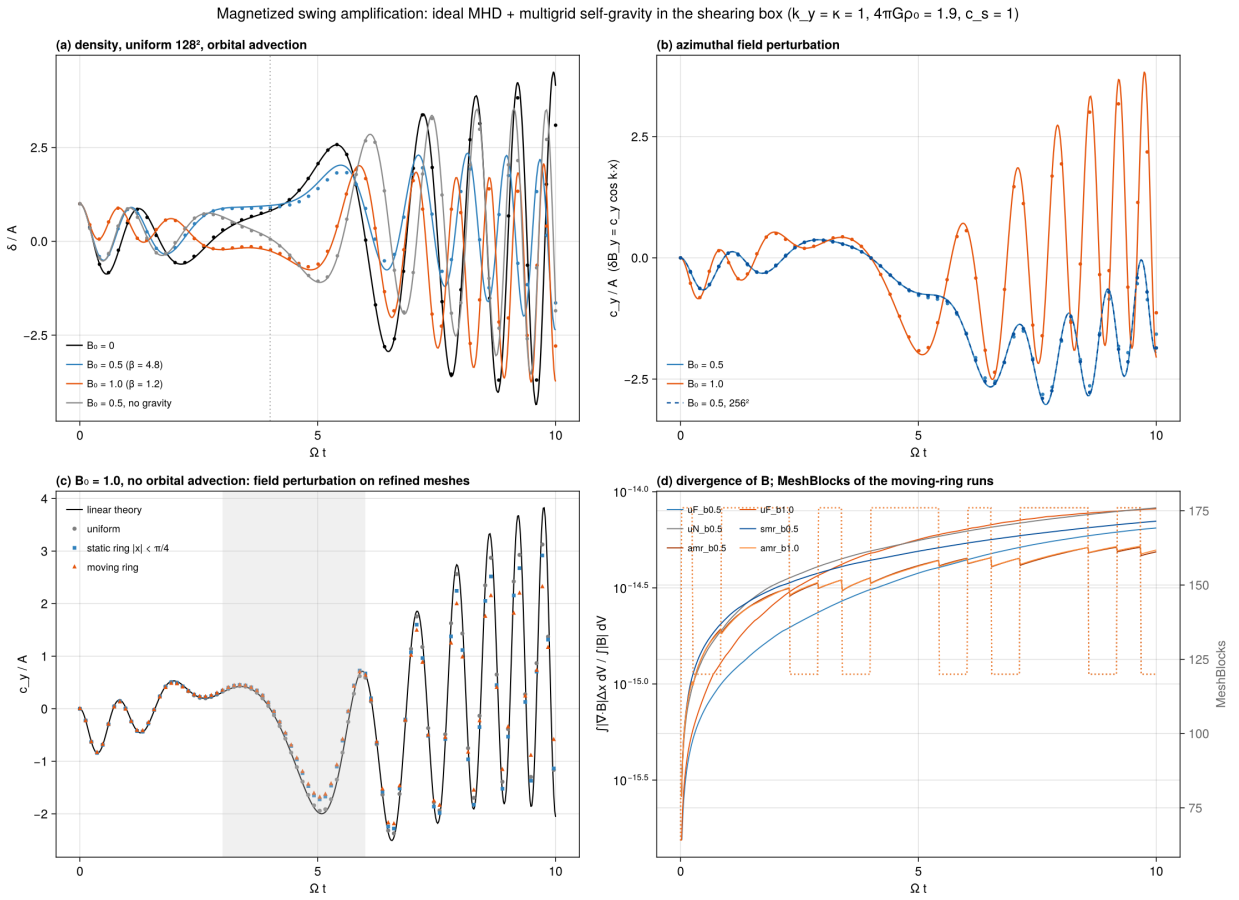


Figure 19: Magnetized swing amplification. (a) Density amplitude δ/A on the uniform 128^2 mesh with orbital advection (points) against linear theory (lines), for $B_0 = 0, 0.5, 1.0$ and for $B_0 = 0.5$ without self-gravity; dotted: the swing at $t = 4$. (b) The azimuthal field perturbation c_y/A for $B_0 = 0.5$ and 1.0, and $B_0 = 0.5$ at 256^2 (dashed theory). (c) $B_0 = 1.0$ without orbital advection: c_y/A on the uniform mesh, the static ring and the moving ring; the shaded interval is the swing, where the refined runs fall short of theory and of the uniform run. (d) The relative divergence of $\mathbf{B}$ and (dotted, right axis) the MeshBlock count of the moving-ring runs.

Case (uniform)	δ, $t \le 6$	c_y, $t \le 6$	δ, $t \le 8$
$B_0 = 0$, orbital advection	0.020	—	0.024
$B_0 = 0.5$, orbital advection	0.093	0.049	0.122
the same at 256^2	0.031	0.016	0.041
the same without self-gravity	0.016	0.012	0.048
$B_0 = 1.0$, orbital advection	0.041	0.040	0.079
$B_0 = 0.5$, full shear flow	0.060	0.029	0.079
$B_0 = 1.0$, full shear flow	0.030	0.030	0.078

Ideal MHD with multigrid self-gravity in the shearing box follows linear theory through the swing to a few per cent, and the error converges: the largest one, 9.3% in δ for $B_0 = 0.5$ with gravity (where the trajectory has a long, nearly cancelling plateau before the swing), falls to 3.1% at 256^2. The two frames agree with each other (orbital advection against the full shear flow, $B_0 = 0.5$: 3.5% in δ, 2.4% in c_y to $t = 6$), and $\nabla\cdot B$ stays at round-off, $\le 9 \times 10^{-15}$, in every run (panel d).

Refined meshes. Deviation of each refined run from its uniform twin (same field, same gravity, full shear flow) to $t = 6$:

	static ring		moving ring	
Case	δ	c_y	δ	c_y
$B_0 = 0$	0.018	—	0.030	—
$B_0 = 0.5$	0.025	0.071	0.019	0.087
$B_0 = 1.0$	0.039	0.133	0.040	0.150
$B_0 = 1.0$, no self-gravity	0.043	0.163	0.033	0.173
$B_0 = 1.0$, no self-gravity, 64-cell root	0.057	0.224	—	—

Without a field the refined runs match their twins as in the hydro tests (1.8% and 3.0%), so gravity on the refined, sheared mesh is right with the MHD integrator too. With a field they do not: the field perturbation of the refined runs is off by 7–9% at $B_0 = 0.5$ and 13–17% at $B_0 = 1.0$, against 3% for the uniform twins against theory, and the density follows at a few per cent (panel c: the refined runs undershoot the swing's c_y minimum, -1.7 against -2.0). Three controls locate it.

- *Not gravity*: without self-gravity the deviation is the same (16–17%).
- *Barely convergent*: from a 64-cell to a 128-cell root the static ring's field deviation falls only from 22% to 16%, while the uniform twin's own error falls from 7.3% to 1.7%; a truncation error at the level interfaces would fall at least twofold.
- *The shear flow*: with $q = 0$ (rotation, no shear flow; $B_0 = 1.0$, no gravity, to $t = 8$) the static ring is *closer* to theory than the uniform mesh, 19% against 25% in c_y (both dominated by phase error over eleven periods at 21 cells per wavelength), as refinement should be.

So the error needs a field, mesh refinement and the background shear flow carried across the level interfaces (here $|v_y| \leq 1.2$ at the ring edge); the divergence of $\mathbf{B}$ stays at round-off throughout, so it is not a loss of the constrained-transport constraint. It does not affect the production configuration, which uses orbital advection and, with MHD, cannot refine at all yet; but MHD shearing-box runs on refined meshes without orbital advection are *not* validated. §13.7 narrows it down.

13.7 Narrowing the refined-mesh field error

A minimal reproducer. `inputs/tests/swing_mhd_repro.athinput` keeps only what the error needs: a thin 3-D shearing box ($64 \times 64 \times 4$ cells, $8 \times 8 \times 4$ MeshBlocks), ideal MHD, $B_0 = 1$ along y, the leading shearing wave, no self-gravity, no orbital advection, and the static level-1 ring $|x| < \pi/4$. Run as a pair — as is, and with `mesh_refinement/refinement = none` — it takes about three minutes on one core, and the ring's c_y is 22.6% off linear theory over $t \leq 6$ against 7.5% for the uniform run, the error of the full-size runs at the same root resolution. `validation/run/swing_mhd_repro/repro.py` runs variants of the pair:

Variant	c_y against theory		
	uniform	ring	
HLLD, PLM, RK2 (base)	0.075	0.226	the defect
$B_0 = 0.5$	0.041	0.142	scales with B_0
PPM4	0.039	0.222	not the reconstruction
HLLE / LLF	0.27 / 0.27	0.36 / 0.34	not the Riemann solver
RK3; CFL 0.1	ring vs uniform 0.22, 0.22		not the integrator
$\Omega = 0.01$, $q = 150$ (pure shear)	0.10	0.18	rotation not needed
wave amplitude 10^{-5}, 10^{-3}	ring vs uniform 0.215, 0.224		linear in the wave
primitive prolongation	0.075	0.216	not the energy prolongation

The error is linear in the wave (the same fraction at amplitudes 10^{-5} to 10^{-3}, and round-off at zero amplitude), independent of the solver, the reconstruction, the integrator and the rotation, and needs the field and the velocity *shear*. Two simpler candidates do not reproduce it:

- *A uniform flow along the interface.* A 2-D MHD linear wave (slow, Alfvén, fast) crossing a level-1 slab with the flow ($V = 0, 1, 2$) and the field along the slab faces, or the flow across them, without a shearing box (input `linear_wave_mhd_tflow_smr.athinput`, runner in `validation/run/tflow_smr/`): the slab's error is 0.8× the uniform-64 one in every case, as it should be.
- *Winding of a uniform radial field.* With no wave and $B_x = 0.1$ (`<problem> bx0`), $B_y = 1 - q\Omega B_x t$ exactly. The uniform mesh keeps it to the last digit; the ring has an interface error of 4.3×10^{-4}, 1.1×10^{-4}, 2.6×10^{-5} at 64-, 128- and 256-cell roots: an ordinary second-order error.

The edge EMFs at the interface. A temporary dump (not committed) wrote the E_z edge EMFs on every MeshBlock x_1 face, before and after the fine/coarse correction of `bvals/flux_correct_fc.cpp`, for the ring, uniform 64 and uniform 128 runs; the wave part is the difference from a zero-amplitude run, compared with the linear-theory EMF $E_z = -q\Omega x\, A c_x \cos\psi - A w_x B_0 \sin\psi$.

- *The correction is right.* After it, every coarse edge on the interface equals the mean of its two fine edges to 10^{-20}, and the fine EMFs are unchanged.
- *The fine side's own interface EMFs are not.* At $t = 0.5$ they are 14% off theory, against 5.7% for uniform 64 at the same face and about 1% for uniform 128; the correction then imposes them on the coarse side, and by $t = 5$ the error fills the box (12–15% outside the ring). The fine block builds these EMFs from ghost cells prolongated from the coarse level, so the prolongation of the face fields and cell-centred variables in a sheared flow (`bvals/prolongation.cpp`) is the next suspect; not yet examined.

A second defect: the background shear flow. With no wave, the fine cells next to the interface develop a spurious radial velocity, $|v_x| = 2.8 \times 10^{-4}$ by $t = 0.03$, growing by about 7×10^{-5} per step — also in pure hydro. The ghost-cell total energy carries the shear flow's kinetic energy $\frac{1}{2}\rho v_y^2$, quadratic in x and large here (up to 11 against a thermal energy of 0.9), and linear prolongation of the *conserved* variables recovers the wrong ghost pressure. Prolongating primitives (`<mesh_refinement> prolong_primitives = true`) or an isothermal gas keeps the background exact ($|v_x| \sim 10^{-16}$). This error does not project on the wave and does not explain the field error (primitive prolongation leaves it unchanged), but adiabatic shearing boxes refined without orbital advection should prolongate primitives.

A caveat. AthenaK's 2-D shearing box is the x–z one only (`shearing_box_r_phi` is "not yet implemented"); a 2-D x–y input runs without complaint with plain periodic x_1 boundaries, which without orbital advection leaves a velocity jump of $q\Omega L_x$ at the boundary. All the tests here are 3-D.

13.8 The gravito-turbulent box with a field: consistency checks

The SC14 box with an `<mhd>` block is not a test with an analytic answer; these checks establish only that the MHD path reproduces the hydro one where it must and keeps the field solenoidal.

- With $B = 0$ (`b_config = none`) the MHD build reproduces the hydro run.
- A uniform vertical field at $\beta_p = 10$ with no Alfvén ceiling, laminar from the SC14 initial state on the $16H$-wide box at 4 cells per H, tracks the hydro run (mass to 0.08%, internal energy to 1%). This is the test of the `hse_outflow` boundary with a field: without excluding $B^2/2$ from the edge temperature the halo heats by $B^2/2\rho \sim 10^3$.
- The Alfvén ceiling $v_{A,\max} = 20$ is too aggressive at $\beta_p = 10$: its density floor $B^2/v_{A,\max}^2$ exceeds the halo density, and the run gained ~ 0.8 box masses through the floor by $t = 10/\Omega$ (most of it leaving through z). At the planned $\beta_p = 10^3$ the floor, 1.4×10^{-5}, starts below `dfloor` $= 10^{-4}$ and is inactive.
- Under constrained transport the history column `divb`/`babs`, the mean of $|\nabla{\cdot}B|\,\Delta x/|B|$, stays at $\sim 8 \times 10^{-16}$.
- On 512 NAS ranks the MHD step costs 1.12× the hydro one at $\beta_p = 10^3$, and adding constant-coefficient ambipolar diffusion ($\eta_A = B^2$) 1.13×.

13.9 Not yet validated

- **MHD shearing boxes on refined meshes.** Without orbital advection the field perturbation carries a 10–20%, barely convergent error that needs a field, refinement and the shear flow together (§13.6). It enters through the fine block's interface EMFs, built from prolongated ghost cells; the EMF correction is right (§13.7). With orbital advection MHD cannot refine until the face-field remap at fine/coarse interfaces exists (implementation document, Phase 4). The clump-following AMR of the gravito-turbulent box with a field depends on both.
- **The isothermal HLLD solver** at $\beta \lesssim 1$ (§13.4).
- **Conserved-variable prolongation in an adiabatic shear flow** does not keep the background equilibrium (§13.7); `prolong_primitives = true` does.
- **Ambipolar diffusion** with a density-dependent coefficient and super-time-stepping, which do not yet exist.

13.10 Summary

- Hydro with multigrid self-gravity reproduces the Jeans dispersion relation, stable and growing ($\lambda/\lambda_J = 0.25$–3): relative error in ω (or s) $\leq 5.9\times10^{-3}$ at 64 cells per wavelength, $\leq 1.6\times10^{-3}$ against the dispersion relation of the discrete operators (the rest is the weaker discrete gravity near marginal stability), second-order convergence, the same for an isothermal gas; every statically or adaptively refined run (1-D and 3-D) lies inside the uniform bracket (§13.1).
- Ideal MHD with multigrid self-gravity reproduces the magneto-Jeans dispersion relation on a uniform grid at every angle, on both branches, stable and unstable: relative error in ω $\leq 4.8\times10^{-3}$ (median 6.6×10^{-4}) at 64 cells per wavelength; 1.5–2×10^{-3} for 3-D waves along the box diagonal at about 21 spacings per wavelength (1.4×10^{-2} for the near-marginal one). The error converges at second order or better; a perpendicular field stabilises a hydro-unstable wavelength at the predicted $v_A = c_s\sqrt{n^2-1}$; and the perturbed field matches the eigenvector to a median 4×10^{-4}.
- On static and adaptive meshes (one and two levels, slabs normal to $\mathbf{k}$ and a 3-D block oblique to it) every magneto-Jeans run lies inside the bracket of the uniform runs at its root and

finest resolution (the 3-D slow wave on a central block 5% above it), the interfaces reflect $\lesssim 10^{-3}$ of the mode, and $\nabla \cdot B$ stays at round-off through thousands of MeshBlocks created and destroyed.

- AthenaK's isothermal HLLD solver amplifies grid-scale noise at $\beta \lesssim 1$ (no gravity needed; adiabatic HLLD, HLLE and LLF are stable); the MHD tests are therefore adiabatic, like the production box.
- Magnetized swing amplification on the uniform shearing box follows linear theory through the swing to a few per cent, converging with resolution, with or without orbital advection. On refined shearing boxes (without orbital advection) gravity is right ($B_0 = 0$ twins within 2–3%), but the field perturbation is not (7–17%, barely convergent, caused by the field, the refinement and the shear flow together): an open problem for refined MHD shearing boxes. A three-minute reproducer isolates it; the fine/coarse EMF correction is verified, and the error enters through the fine side's interface EMFs. Conserved-variable prolongation also breaks the adiabatic background shear flow at the interfaces; primitive prolongation fixes that.
- The gravito-turbulent box runs with a field, its vertical boundary is correct with MHD, and the field stays solenoidal to round-off.